Packings of Smoothed Polygons

A Self-Contained Guide to the Proof of Mahler's First Conjecture

Lecture notes for a reader with undergraduate mathematics

These notes are a pedagogical reconstruction of Thomas Hales and Koundinya Vajjha, *Packings of Smoothed Polygons*, arXiv:2405.04331v1 (2024). They reorganize the argument, supply undergraduate-level background, and distinguish conceptual reductions from computer-assisted verification. They do not replace the source paper for citation or formal auditing.

Prepared as an explanatory companion

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How to read these notes

The source paper is a 283-page research monograph. It combines discrete geometry, convexity, ordinary differential equations, optimal control, matrix Lie groups, Hamiltonian mechanics, hyperbolic geometry, singular perturbation, blowup, Poincaré maps, stable-manifold theory, and computer-assisted dynamical estimates. These notes begin from the geometric question and introduce each technical language only when it becomes useful. Almost every abstract construction is specialized immediately to explicit 2×2 matrices or elementary complex coordinates.

The main theorem and the open theorem

A centrally symmetric convex disk is a compact convex set $K \subset \mathbb{R}^2$ with nonempty interior and $K = -K$. The full Reinhardt conjecture predicts that the unique minimizer, up to affine transformation, is the smoothed octagon. The theorem proved by Hales and Vajjha is the structural statement that every global minimizer is a finite-sided smoothed polygon, with boundary alternating between straight segments and hyperbolic arcs of Reinhardt–Mahler type. It does not determine the number of sides.

The proof in one page

1. Classical packing theory reduces the density problem to minimizing the area of K after normalizing a smallest centrally symmetric hexagon circumscribing K.
2. A minimizing boundary is encoded by six synchronized curves $\sigma_0, \dots, \sigma_5$, hence by a curve $g(t) \in \mathrm{SL}_2(\mathbb{R})$.
3. Writing $g' = gX$ and normalizing $\det X = 1$, convexity produces a triangular curvature control. The area becomes an integral cost.
4. The Pontryagin maximum principle implies finite bang–bang motion away from a singular locus. Such motion gives a smoothed polygon.
5. The singular locus is the circle, and a second variation excludes a positive-length circular singular arc.
6. The only remaining obstruction is chattering: infinitely many switches accumulating at the singular locus in finite time.
7. A weighted blowup replaces the singular point by an exceptional divisor carrying a triangular Fuller system.

8. The Fuller Poincaré map has inward and outward self-similar fixed points. A computer-assisted cell decomposition proves a global basin theorem.
9. Stable-manifold theory transfers this structure to the exact Reinhardt map.
10. The required chattering stable curve cannot belong to a periodic minimizer. Chattering is impossible, so the minimizer has finitely many switches.

Prerequisites and proof audit

The main text assumes undergraduate linear algebra, multivariable calculus, elementary real analysis, ordinary differential equations, and complex numbers. Fejes Tóth's packing theorem, the Pontryagin maximum principle, the stable-manifold theorem, and Weierstrass preparation are stated as black boxes and specialized to this problem. The global basin theorem includes symbolic and numerical verification; the notes explain precisely where that computation enters.

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Part I

The geometric problem

Chapter 1

The Reinhardt minimax problem

1.1 Convex disks and packings

A convex disk is a compact convex subset $K \subset R2$ with nonempty interior. The word "disk" is topological: K need not be round. A convex disk is centrally symmetric if there is a point c such that $x \in K$ implies $2c - x \in K$. Translating the set does not affect packing density, so we place the center at the origin and write simply $K = -K$.

A family P of congruent copies of K is a packing if distinct copies have disjoint interiors. Let BR be the Euclidean ball of radius R centered at the origin. When the limit exists, the density of the packing is 1 δ(K, P) = lim X area(K′ ∩BR). R→∞ area(BR) K′∈P

The greatest packing density of K is

δ(K) = sup δ(K, P). P

This is the best fraction of the plane that can be covered by pairwise nonoverlapping congruent copies of K.

Geometric intuition

There are two competing optimizations. For a fixed shape K, the packer tries to make δ(K) as large as possible. Reinhardt's problem then asks the shape designer to make this best possible density as small as possible. Thus the objective is a minimax:

shape chooses K −→ packer chooses best arrangement .

A highly "unpackable" shape is one whose densest packing is still relatively sparse.

1.2 Lattice packings

A lattice in R2 is a discrete subgroup

$L = \mathbb{Z}v1 + \mathbb{Z}v2$

generated by linearly independent vectors v1, v2. Its determinant or covolume is

det(L) = |det(v1, v2)| .

The translates K + L = {K + l: l∈L} form a lattice packing when their interiors are disjoint. The density is then exactly area(K) δ(K, L) = . det(L)

The reason is elementary: a fundamental parallelogram of L has area det(L) and contains exactly one translate of K modulo the lattice.

Define the greatest lattice-packing density by

δL(K) = sup {δ(K, L) : K + L is a packing} .

A crucial two-dimensional theorem removes the apparent difference between arbitrary and lattice packings.

Standard theorem used as a black box

Fejes Tóth's theorem. If K ⊂R2 is centrally symmetric and convex, then

δ(K) = δL(K).

Thus the densest packing of such a disk can be realized by a lattice packing.

This theorem is special to the planar centrally symmetric setting. It is the first reason the Reinhardt problem is tractable.

1.3 The minimax quantity

Let Kccs denote all centrally symmetric convex disks centered at the origin. Set

δmin = inf δ(K). K∈Kccs

The Reinhardt problem asks for both the numerical value of δmin and a classification of all minimizing shapes.

The disk has density π δ(B) = √ ≈0.906899. 12 It is not the minimizer. Reinhardt found a slightly worse shape, the smoothed octagon, whose density is √ 8 − 32 −log 2 δoct = √ ≈0.902414. 8 −1

Figure note. *Figure 1.1: A schematic smoothed octagon. The dashed regular octagon is clipped at each vertex. The true rounded pieces are hyperbolic arcs; the drawing uses smooth surrogate arcs only for visualization.*

Definition

Definition 1.1 (Smoothed polygon). A smoothed polygon in the Reinhardt–Mahler sense is a centrally symmetric convex disk whose boundary is a finite cyclic concatenation of straight segments and arcs of hyperbolas. Each hyperbolic arc is tangent to the two adjacent straight segments, and its asymptotes are the lines extending the next pair of straight segments.

The full conjecture is the following.

Theorem

Theorem 1.2 (Reinhardt conjecture; open in the source paper). The smoothed octagon minimizes δ(K) over Kccs, and it is unique up to affine transformation.

The proved theorem is weaker but structural.

Theorem

Theorem 1.3 (Mahler's First conjecture; proved in [HV]). Every global minimizer is a finite-sided smoothed polygon.

1.4 Affine invariance

An invertible affine map has the form x 7→Ax + b with A ∈GL2(R). Translation is irrelevant, so consider A. Both the area of K and the covolume of a lattice are multiplied by |det A|. Hence

area(AK) |det A| area(K) δ(AK, AL) = = = δ(K, L). det(AL) |det A| det(L)

Taking suprema gives δ(AK) = δ(K).

Thus the problem naturally lives modulo affine transformations. This freedom will later be used to normalize six special boundary points to the sixth roots of unity.

1.5 Why a hexagon appears

Suppose K + L is a lattice packing. A fundamental domain of the lattice can be chosen as a parallelogram, but one can often crop its corners and obtain a smaller centrally symmetric hexagon still containing K. Every centrally symmetric hexagon tiles the plane by translations. Consequently, the optimal lattice packing of a non-parallelogram K can be represented by a least-area centrally symmetric hexagon HK circumscribing K.

The exact theorem is proved in the next chapter. Its consequence is the formula

area(K) δ(K) = area(HK).

Because the ratio is scale invariant, we can normalize $\surd\ \mathrm{area}(HK) = 12$.

After this normalization, minimizing packing density is equivalent to minimizing area(K).

Proof roadmap

The original minimax problem now starts to collapse:

arbitrary packings of K

$\Downarrow$ Fejes Tóth

lattice packings

$\Downarrow$ critical hexagon

$\delta(K) = \mathrm{area}(K)/\ \mathrm{area}(HK)\ \surd\ \Downarrow \text{normalize } \mathrm{area}(HK) = 12$

minimize the area of K.

The remaining difficulty is that K still ranges over an infinite-dimensional space of convex curves.

1.6 Historical orientation

Blaschke recorded the problem in 1923 as a conjecture attributed to Courant, who expected the disk to be worst. Reinhardt disproved this in 1934 by constructing the smoothed octagon and developed the critical-hexagon geometry. Mahler independently arrived at the same candidate in the 1940s. In 1947 he wrote that the boundary of an extremal domain should consist of line segments and hyperbolic arcs, but he could not prove this. That structural prediction is what the source paper establishes.

Chapter summary

The Reinhardt problem is not to pack one fixed shape; it is to choose the centrally symmetric convex shape whose best packing is least dense. Fejes Tóth reduces arbitrary packings to lattice packings. A least-area centrally symmetric circumscribed hexagon then converts the density into an area ratio. After normalization, the objective is simply to minimize the area of a convex disk satisfying a rigid critical-hexagon condition.

Chapter 2

Critical hexagons and the origin of hyperbolas

2.1 Support lines and circumscribed polygons

> Let K $\subset$R2 be a convex disk. A line lis a support line of K if K lies in one of the two closed half-planes bounded by land $l \cap K \neq$ empty. A circumscribed polygon is obtained by intersecting finitely many support half-planes.

For a centrally symmetric K, support lines naturally occur in opposite pairs. Hence a centrally symmetric circumscribed hexagon has three pairs of opposite edges.

Definition

Definition 2.1 (Critical hexagon). A critical hexagon HK of K is a centrally symmetric hexagon of least area among all centrally symmetric hexagons containing K.

When K is a parallelogram, the optimal density is 1, so such shapes are irrelevant to the minimization. We therefore exclude that degenerate case.

Theorem

Theorem 2.2 (Reinhardt–Fejes Tóth reduction). If K is a centrally symmetric convex disk that is not a parallelogram, then area(K) δ(K) = area(HK), where HK is any critical hexagon.

Explanation of the reduction. By Fejes Tóth, a densest packing can be taken to be a lattice packing. Starting from a fundamental parallelogram of the lattice, move supporting lines inward while preserving the tiling structure. At an area-minimal configuration one obtains either a parallelogram or a centrally symmetric hexagon. The non-parallelogram case gives HK. Since every centrally symmetric hexagon tiles by translations, its area is exactly the covolume of the corresponding tiling lattice. Thus the packing density is the occupied area divided by the tile area.

2.2 Why the edge midpoints touch K

Let the vertices of HK be $p_0, . . . , p_5$ in counterclockwise order, and let s_j be the midpoint of the edge $p_{j-1}p_j$. Minimality forces every s_j to lie on ∂K.

To see the mechanism, suppose an edge midpoint is separated from K. Then one can rotate the support line slightly about a nearby contact point, simultaneously adjusting its opposite line by central symmetry. To first order the area decreases, contradicting minimality. The midpoint condition is the stationary condition for the area of the hexagon.

HK

s_5 s_4 s_0 K s_3 s_1 s_2

Figure note. *Figure 2.1: A critical hexagon HK circumscribing K. Its six edge midpoints s_j lie on ∂K. The drawing is schematic.*

2.3 The algebra of midpoint hexagons

A centrally symmetric hexagon is determined conveniently by its edge midpoints. Label the midpoints cyclically by $s_0, . . . , s_5$. Central symmetry gives

$s_{j+3} = -s_j$.

The midpoint relations also imply $s_0 + s_2 + s_4 = 0$,

and hence, after shifting indices, $s_j + s_{j+2} + s_{j+4} = 0$.

The determinants of consecutive midpoints are equal:

$\det(s_j, s_{j+1}) = c$,

where $c > 0$ is proportional to the area of HK. Under the paper's normalization √
area(HK) = 12,

one has √ 3 $\det(s_j, s_{j+1})$ = . 2 Definition 2.3 (Multipoint). A six-tuple $(s_0, . . . , s_5)$
is a multipoint if √ 3 $s_{j+3} = -s_j$, $s_j + s_{j+2} + s_{j+4} = 0$, $\det(s_j, s_{j+1})$ = 2

for all indices modulo 6.

The standard example is the set of sixth roots of unity

cos(jπ/3)! s_j^* = . sin(jπ/3)

2.4 The hyperbolic envelope

The special hyperbolas arise before any optimal-control theory appears. Consider two intersecting lines, which after an affine change of coordinates we may take to be the coordinate axes. A moving line with intercepts (a, 0) and (0, b) cuts off a right triangle of area ab/2. If the area is fixed, then

ab = C.

The line is x y + = 1. a b Eliminating b = C/a gives a one-parameter family

x ay + = 1. a C

The envelope is obtained by solving the line equation together with its derivative with respect to a: −x + y = 0. a2 C

This gives a2 = Cx/y, and substitution yields

C xy = . 4

Thus the envelope is a hyperbola whose asymptotes are the coordinate axes.

y

moving support lines cut off constant area

xy = 1

x

Figure note. *Figure 2.2: A pencil of lines with constant product of intercepts has a hyperbola as its envelope.*

Now return to a critical hexagon. Keep four of its six edges fixed, so that they form a parallelogram. Move the remaining opposite pair of edges while keeping the total hexagon area constant. Each moving edge belongs to a constant-area pencil. Therefore its midpoint traces an arc of a hyperbola, and the asymptotes are the lines extending the neighboring fixed edges.

Lemma

Lemma 2.4 (Local hyperbola mechanism). If, during an interval of the boundary motion, the two neighboring midpoint curves are straight, then the intervening midpoint curve lies on a hyperbola whose asymptotes are those two straight lines.

This elementary envelope calculation will later translate a vertex of the control triangle into a rounded hyperbolic corner.

2.5 Critical lattices and the inscribed hexagon

There is a dual description through the geometry of numbers. A lattice L is K-admissible if no nonzero lattice point lies in the interior of K. Its minimal determinant is

Δ(K) = inf det(L). L K-admissible

A lattice achieving the infimum is critical.

Minkowski's theory produces three boundary points s0, s1, s2 such that s0, s1 form a lattice basis and 0, s0, s1, s2 form a parallelogram. Adding their negatives gives an inscribed centrally symmetric hexagon hK = conv {±s0, ±s1, ±s2} .

For the associated circumscribed critical hexagon HK,

1 1 Δ(K) = area(hK) = area(HK). 3 4

This is the precise bridge between admissible lattices and critical hexagons.

2.6 What is special about a global minimizer?

Reinhardt proved several regularity and coverage properties for a global minimizer Kmin.

Standard theorem used as a black box

For a normalized global minimizer Kmin: 1. a minimizer exists; 2. every boundary point lies on an edge of some critical hexagon; 3. Kmin has no corners: every boundary point has a unique support line; 4. every boundary point is the midpoint of a unique edge of a critical hexagon; 5. as one midpoint moves counterclockwise around the boundary, the other five move continuously and strictly counterclockwise.

The uniqueness of the critical hexagon through each boundary point is essential. It synchronizes six boundary points and turns one planar curve into a six-component moving frame.

2.7 Monotonicity triangles

Use affine invariance to send one multipoint to s∗j. The convex hull of these six points is a regular hexagon contained in K. The boundary between s∗j and s∗j+1 is confined to the triangle

n o Tj = conv s∗j, s∗j+1, s∗j + s∗j+1 .

Reinhardt's monotonicity theorem says that if a second critical multipoint differs from the first, then its jth point lies in the interior of Tj. These six triangles are the geometric origin of the later "star inequalities."

T0

hK

Figure note. *Figure 2.3: The six triangles Tj surrounding the normalized midpoint hexagon. A moving critical multipoint stays in the corresponding triangles.*

2.8 Normalization and the reduced variational problem

After fixing √ area(HK) = 12,

we have area(K) δ(K) = √ . 12

Therefore the Reinhardt minimizer is exactly the minimum-area disk among normalized disks satisfying the critical-hexagon synchronization described above.

The next chapter packages these geometric conditions into the notion of a balanced disk and derives the six synchronized boundary curves.

Chapter summary

A critical hexagon is a least-area centrally symmetric hexagon around K. Its edge midpoints lie on ∂K and satisfy simple determinant relations. For a global minimizer, every boundary point is the midpoint of a unique critical hexagon, so six boundary points move together. The hyperbolas are already present: a support line that moves while cutting off constant area has a hyperbolic envelope.

Chapter 3

Balanced disks and six synchronized boundary curves

3.1 From one boundary point to six

Let $K = K_{min}$ be a normalized global minimizer. Choose a counterclockwise regular parametrization of its boundary, $t \mapsto \sigma_0(t) \in \partial K$. For each t, the point $\sigma_0(t)$ is the midpoint of a unique critical hexagon. Let the other five edge midpoints be $\sigma_1(t), \ldots, \sigma_5(t)$.

At every time, the six points form a multipoint:

$\sigma_{j+3}(t) = -\sigma_j(t)$, (3.1) $\sigma_j(t) + \sigma_{j+2}(t) + \sigma_{j+4}(t) = 0$, (3.2) $\sqrt{3}$ $\det(\sigma_j(t), \sigma_{j+1}(t)) = .$ (3.3) 2 The strict monotonicity of critical hexagons implies that every σ_j runs counterclockwise around the same boundary curve.

Definition

Definition 3.1 (Multicurve). A family of curves

$\sigma_j : I \to \mathbb{R}^2$, $j \in \mathbb{Z}/6\mathbb{Z}$,

is a multicurve if $(\sigma_0(t), \ldots, \sigma_5(t))$ is a multipoint for every $t \in I$.

A multicurve should be visualized as six synchronized particles on the same convex boundary. The parameter t is not six unrelated time variables; it records one moving critical hexagon.

3.2 Regularity inherited by all six curves

The boundary of a general convex disk can have corners and flat pieces. Reinhardt's minimizer has no corners, so its tangent direction is continuous. If σ_0 is a regular C^1 parametrization, then all other σ_j are regular C^1 curves.

Here is the mechanism for σ_2. Differentiate the constant determinant relation $\sqrt{3}$ $\det(\sigma_0(t), \sigma_2(t)) = .$ 2 Writing the unit tangent of σ_2 as $u_2(t)$ and its speed as $v_2(t) > 0$, so that $\sigma'_2 = v_2 u_2$, gives $\det(\sigma'_0, \sigma_2) + v_2 \det(\sigma_0, u_2) = 0$.

The monotonicity triangles guarantee that neither determinant vanishes. Hence

$0(t), \sigma_2(t))\ v_2(t) = -\det(\sigma'\ \det(\sigma_0(t), u_2(t))$,

which is positive and continuous. The same argument cycles through all indices.

A further geometric estimate bounds the curvature from above by the curvature of the local hyperbolic envelopes. If σ_0 is parametrized by arclength, its tangent is Lipschitz. The determinant relations then transfer Lipschitz regularity to all σ'_j. Corollary 3.2. Each σ'_j is differentiable almost everywhere.

Proof. Proof. A Lipschitz function on an interval is differentiable almost everywhere by Rademacher's theorem. □

This almost-everywhere second derivative is sufficient for curvature controls and optimal-control equations.

3.3 The global star condition

Fix a time t and abbreviate $s_j = \sigma_j(t)$. The tangent $\sigma'_j(t)$ cannot point arbitrarily. Convexity confines it to the open cone based at s_j whose boundary rays point toward

s_{j+1} and $s_j + s_{j+1}$.

After translating the apex to the origin, this says that σ'_j is a positive linear combination of the two edge directions $s_{j+1} - s_j$, s_{j+1}.

The source calls this the star condition.

Why are the inequalities strict? If a tangent lay exactly on a bounding ray, convexity and the multipoint relations would force another curve to have zero velocity or would create a corner. Both are impossible for a regular minimizing multicurve.

3.4 The local curvature condition

For a regular plane curve $\gamma(t)$, the signed curvature is

$$\kappa_{signed}(t) = \frac{\det(\gamma'(t), \gamma''(t))}{|\gamma'(t)|^3}.$$

ray through $s_j + s_{j+1}$

$\sigma'_j(t)$ ray through s_{j+1}

s_j

Figure note. *Figure 3.1: The star condition: the tangent lies strictly inside a convex cone determined by the current critical multipoint.*

A counterclockwise simple closed curve bounds a convex disk if and only if its signed curvature is nonnegative, interpreted almost everywhere when the derivative is Lipschitz. Hence

$$\det(\sigma'_j(t), \sigma''_j(t)) \geq 0$$

for almost every t.

There are thus two distinct convexity constraints:

• the global star condition, which controls tangent directions relative to the synchronized hexagon; • the local curvature condition, which forbids negative turning.

Both are needed. Local nonnegative curvature alone would not ensure the six curves remain synchronized in the critical-hexagon geometry. The star condition alone would not prevent smallscale nonconvex bending.

3.5 Balanced disks

The paper packages the preceding properties into a class Kbal.

Definition

Definition 3.3 (Balanced disk). A centrally symmetric convex disk K is balanced if its boundary admits six regular C1 curves σ_j such that:

1. the derivatives σ'_j are Lipschitz; 2. at each time the six points form a normalized multipoint; 3. the signed curvature is nonnegative almost everywhere; 4. the star condition holds at every time; 5. the six curves have the same image and fit together periodically.

Every normalized global minimizer is balanced. The balanced class may contain nonminimizers; it is a convenient admissible class for the variational problem.

3.6 Circle representation

By affine invariance, choose one critical multipoint and send it to the sixth roots of unity $s*_j$. Reparametrize time so that this multipoint occurs at t = 0:

$$\sigma_j(0) = s*_j.$$

This normalization is called the circle representation. The term does not mean that K is a circle. It means only that one synchronized hexagon is the regular one inscribed in the unit circle.

The balanced Reinhardt problem is now:

minimize area(K) over balanced disks in circle representation.

3.7 A first preview of the matrix reduction

The multipoint equations are linear and preserve determinants. This strongly suggests that every current multipoint is obtained from the standard one by a determinant-one linear map. Indeed, the next chapter proves that there is a unique matrix $g(t) \in SL_2(\mathbb{R})$ such that

$$\sigma_j(t) = g(t)s_j^* \quad (j = 0, \ldots, 5).$$

Thus six plane curves are generated by one curve of matrices. The tangent directions are encoded by a traceless matrix $X(t) = g(t)^{-1}g'(t)$.

The star condition becomes three elementary linear inequalities on the entries of X. The curvature condition becomes a control lying in a triangle.

Chapter summary

A minimizing boundary is not treated as one free curve. Every boundary point determines a unique critical hexagon, so six points move together. Their multipoint identities, star inequalities, and nonnegative curvature define a balanced disk. Affine normalization fixes one multipoint as the sixth roots of unity. This makes a matrix representation unavoidable and prepares the passage from convex geometry to a finite-dimensional control system.

Part II

From planar geometry to a matrix control system

Chapter 4

The matrix language: $SL_2(\mathbb{R})$ without prior Lie theory

The matrix language: without $SL_2(\mathbb{R})$ prior Lie theory

4.1 Area-preserving matrices

Let $SL_2(\mathbb{R}) = \{g \in M_2(\mathbb{R}) : \det g = 1\}$. This is a group under matrix multiplication. Geometrically, $g \in SL_2(\mathbb{R})$ is an orientation-preserving linear transformation that preserves signed area:

$$\det(gu, gv) = \det(g)\det(u, v) = \det(u, v).$$

The corresponding space of infinitesimal matrices is

$$\mathfrak{sl}_2(\mathbb{R}) = \{X \in M_2(\mathbb{R}) : \operatorname{tr} X = 0\} = \left\{ \begin{pmatrix} a & b \\ c & -a \end{pmatrix} : a, b, c \in \mathbb{R} \right\}.$$

The terminology "Lie group" and "Lie algebra" will be explained in Appendix A; for now these are concrete matrix sets.

The matrix $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

generates rotations because $\exp(tJ) = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}$.

We write $R = \exp\left(\frac{\pi}{3} J\right)$ for counterclockwise rotation by 60°.

4.2 Every multipoint is an $SL_2(\mathbb{R})$-image of the standard one

Let (s_j) and (σ_j) be two normalized multipoints. Since s_0 and s_2 are linearly independent, there is a unique linear map g satisfying $gs_0 = \sigma_0$, $gs_2 = \sigma_2$.

The sum relations express every other sj as a linear combination of s0, s2, and the same relations hold for σj. Therefore gsj = σj

for all j. Finally, det(gs0, gs2) det(σ0, σ2) det(g) = = = 1. det(s0, s2) det(s0, s2) Thus g ∈SL2(R).

Applying this at every time gives the basic representation.

Theorem

Theorem 4.1 (Matrix representation of a multicurve). For a normalized multicurve there is a unique curve g : I →SL2(R) such that

σj(t) = g(t)s∗j for all j, t.

The differentiability of g is the same as that of the multicurve.

Six synchronized curves have now been replaced by one curve in a three-dimensional matrix manifold.

4.3 Body velocity

Differentiate σj(t) = g(t)s∗j: σ′j(t) = g′(t)s∗j.

Since g(t) is invertible, define

X(t) = g(t)−1g′(t), equivalently g′(t) = g(t)X(t).

This X is the velocity measured in the moving frame determined by g. It belongs to sl2(R).

To prove tracelessness, differentiate det g(t) = 1. Jacobi's formula gives

d 0 = det g = det(g) tr(g−1g′) = tr X. dt

Thus a(t) b(t) ! X(t) = . c(t) −a(t)

The boundary tangents are σ′j(t) = g(t)X(t)s∗j. Equivalently, because σj = gs∗j, σ′j = gXg−1σj. The conjugate gXg−1 is the same infinitesimal motion written in the fixed laboratory coordinates.

Background interlude

The map X 7→gXg−1 is called the adjoint action and is denoted Adg X. Nothing abstract is hidden here: it is ordinary matrix conjugation. The commutator

[X, Y] = XY −Y X

is the corresponding infinitesimal operation.

4.4 Translating the star condition into inequalities

The standard points are cos(jπ/3)! s∗j = . sin(jπ/3)

Because multiplication by g preserves open cones, the star condition for σ′j = gXs∗j is equivalent to a condition on Xs∗j before applying g. For each j, the vector Xs∗j must lie in the cone spanned by s∗j+1 and s∗j+2. Solving the corresponding linear systems yields three linear forms √ c + 3a ρ0(X) = √ , (4.1) 3 √ c − 3a ρ1(X) = √ , (4.2) 3 + c ρ2(X) = −3b√ . (4.3) 2 3

The six cone conditions reduce, by central symmetry, to

ρ0(X) > 0, ρ1(X) > 0, ρ2(X) > 0.

These are the matrix form of the star inequalities.

They imply several useful signs: √ c > 0, 3 |a| < c, 3b + c < 0, tr(JX) = b −c < 0.

The final inequality will guarantee that the area integrand is positive.

4.5 A determinant identity

For a b ! X = , c −a

we have det X = −a2 −bc.

A direct calculation using Equations (4.1) to (4.3) gives

det X = ρ0ρ1 + ρ1ρ2 + ρ2ρ0.

Since all ρj are positive, det X > 0.

This positivity permits a convenient reparametrization.

4.6 Normalizing time by det X = 1

If we replace the parameter t by an increasing parameter τ, then

dg dt = gX . dτ dτ Thus X is multiplied by the positive scalar dt/ dτ, and its determinant is multiplied by the square of that scalar. Choose dt 1 = dτ p det X(t). Then the reparametrized matrix, again denoted X, satisfies

det X(t) = 1.

This is analogous to arclength parametrization, but it is adapted to the matrix geometry rather than Euclidean speed.

For a traceless 2×2 matrix, the characteristic polynomial is

$\lambda^2 + \det X$.

$$\text{Under } \det X = 1, \text{ Cayley–Hamilton gives } X^2 = -I,\ X^{-1} = -X.$$

This simple identity is used repeatedly.

4.7 The trace form

$$\text{Define } \langle X, Y \rangle = \operatorname{tr}(XY),\quad X, Y \in \mathfrak{sl}_2(\mathbb{R}).$$

This symmetric bilinear form is nondegenerate. It has the invariance property $\left\langle gXg^{-1}, gYg^{-1} \right\rangle = \langle X, Y \rangle$,

and the commutator identity $\langle X, [Y, Z]\rangle = \langle [X, Y], Z\rangle$. For $X \in \mathfrak{sl}_2(\mathbb{R})$, $\langle X, X\rangle = \operatorname{tr}(X^2) = -2 \det X$.

Hence the determinant-one surface is the quadratic surface

$\langle X, X\rangle = -2$.

The sign is not a mistake: the trace form is indefinite, not an ordinary Euclidean inner product.

4.8 Reconstructing the critical hexagon from X

The directions of the six support edges at the standard multipoint are Xs^*_j. The intersection of the lines through s^*_j and s^*_{j+1} in these directions gives a vertex of the current critical hexagon. Solving the two linear equations yields

$$p_j = s^*_j + \frac{\rho_{j+2}(X)}{\det X} Xs^*_j = s^*_{j+1} - \frac{\rho_j(X)}{\det X} Xs^*_{j+1}.$$

When $\det X = 1$, this simplifies further. Thus X does not merely encode velocity; together with the current multipoint it reconstructs the support hexagon.

Geometric intuition

The three-dimensional matrix X simultaneously records:

• the six tangent directions of the synchronized boundary curves; • the shape of the current critical hexagon; • the position of the system in a hyperbolic state space introduced later.

The star inequalities select one connected region of the determinant-one quadratic surface.

4.9 Endpoint conditions

In circle representation, $g(0) = I$.

After one sixth of the boundary has been traversed, the jth curve must arrive where the $(j + 1)$st curve started. Therefore $g(t_f)s^*_j = s^*_{j+1} = Rs^*_j$,

so $g(t_f) = R$.

Smooth matching of tangents gives

$X(t_f) = R^{-1}X(0)R$.

Extending the solution by this symmetry produces a periodic orbit:

$X(t + 3t_f) = X(t)$, $g(t + 6t_f) = g(t)$.

The periodicity of a minimizer will be decisive at the end of the proof.

Chapter summary

The six-point geometry is encoded by one area-preserving matrix $g(t)$, and its body velocity $X = g^{-1}g'$ is traceless. Convexity becomes three linear star inequalities $\rho_j(X) > 0$. These imply $\det X > 0$, so time can be normalized by $\det X = 1$. The endpoint conditions are rotational, making every admissible minimizer periodic after extension.

Chapter 5

Curvature becomes a control variable

5.1 State-dependent curvatures

Recall $\sigma_j(t) = g(t)s_{*j}$, $g' = gX$, $\det X = 1$.

Differentiate twice: $\sigma'_j = gXs_{*j}$, $\sigma''_j = g(X^2 + X')s_{*j}$.

Since $\det g = 1$, signed determinants are unchanged by g. For the three even-indexed curves define

$\kappa_j = \det(\sigma'_{2j}, \sigma''_{2j}) = \det Xs_{*2j}, (X^2 + X')s_{*2j}$, $j = 0, 1, 2$.

Using $X^{-1} = -X$, one can rewrite this as

$\kappa_j = \det s_{*2j}, (X + X^{-1}X')s_{*2j}$.

Convexity gives $\kappa_j \geq 0$

almost everywhere.

At least one of the three κ_j is positive at almost every time. Geometrically, if two vanish then two boundary curves are straight, and the third is a genuine hyperbolic arc with positive curvature. Algebraically, the star inequalities force the sum to be positive.

5.2 Normalized curvature coordinates

Set $\kappa = \kappa_0 + \kappa_1 + \kappa_2 > 0$

and define κj uj = . κ Then uj ≥0, u0 + u1 + u2 = 1.

e2 = (0, 0, 1)

UT

13, 13, 13

e0 = (1, 0, 0) e1 = (0, 1, 0)

each vertex means two curvatures vanish

Figure note. *Figure 5.1: The triangular control set. Interior points distribute curvature among all three synchronized arcs; vertices concentrate it in one arc.*

Thus the control lies in the standard two-simplex

n o UT = (u0, u1, u2) ∈R3 : uj ≥0, u0 + u1 + u2 = 1 .

The normalization is crucial. The unnormalized curvatures depend on the speed of parametrization; the ratios uj do not.

5.3 Encoding the control by a traceless matrix

For each u ∈UT , define

(u1 −u2 u0 −2u1 −2u2) √ 3 3 | Zu = | | u2 −u1 | (u0 √)∈sl2(R). 3

This affine map is chosen so that

uj = det(s∗2j, Zus∗2j), j = 0, 1, 2.

Because three such determinant values determine a traceless 2 × 2 matrix, Zu is unique.

The vertices give three particularly simple matrices. We write

Z0 = Ze0, Z1 = Ze1, Z2 = Ze2.

They are cyclically conjugate by R: RZjR−1 = Zj+1

with indices modulo 3.

5.4 The star inequalities in control form

A direct calculation gives

$$\langle Z_u, X\rangle = -2\sqrt{3}\,\rho_2(X)u_0 + \rho_1(X)u_1 + \rho_0(X)u_2 .$$

Since u ranges over the whole simplex,

$$\rho_0, \rho_1, \rho_2 > 0 \iff \langle Z_u, X\rangle < 0 \text{ for every } u \in UT .$$

The denominator appearing below is therefore never zero on an admissible trajectory.

5.5 Deriving the differential equation for X

From the definition of the curvature controls,

$$\det\left(s^*_{2j}, (X + X^{-1}X')s^*_{2j}\right) = \kappa u_j = \det(s^*_{2j}, \kappa Z_u s^*_{2j}).$$

A traceless matrix is determined by these three determinant values, so

$$X + X^{-1}X' = \kappa Z_u.$$

Multiply by X: $X' = \kappa X Z_u - X^2$.

Since $X^2 = -I$, this is $X' = \kappa X Z_u + I$. Take traces. Because $\operatorname{tr} X' = 0$, $0 = \kappa \operatorname{tr}(XZ_u) + 2$,

after using $\operatorname{tr} I = 2$. Therefore $\kappa = -\frac{2}{\langle X, Z_u\rangle}$.

Substitute this back. The identity

$XZ_u + Z_uX = \langle X, Z_u\rangle I$

for traceless 2 × 2 matrices yields $X' = \frac{[Z_u, X]}{\langle Z_u, X\rangle}$.

This is the central state equation.

Theorem

Theorem 5.1 (Lie-algebra state equation). The normalized body velocity satisfies

---

$$X' = \frac{[Z_u, X]}{\langle Z_u, X\rangle},\quad u(t) \in UT .$$

Together with $g' = gX$, this determines the boundary multicurve.

5.6 Why the equation is isospectral

Set $Z_u P = \langle Z_u, X\rangle$.

Then $X' = [P, X]$.

> This is a Lax equation. Its basic property is that the characteristic polynomial of X is constant. For example, $\frac{d}{dt}\operatorname{tr}(X^2) = 2\operatorname{tr}(X[P, X]) = 2\operatorname{tr}(XPX - X^2P) = 0$. Thus det X is preserved, consistent with our normalization.

If P were constant, the solution would be

$X(t) = e^{tP} X(0)e^{-tP}$.

Even when P varies, the equation says that X(t) moves by infinitesimal conjugation on the determinant-one surface.

5.7 What the vertices mean geometrically

> Suppose $u = e_0 = (1, 0, 0)$. Then
>
> $\kappa_1 = \kappa_2 = 0, \kappa_0 > 0.$

Two of the three even-indexed curves have zero curvature and hence move along straight lines. By the hyperbolic-envelope lemma, the remaining curve moves along a hyperbola with those lines as asymptotes. The same holds cyclically at the other vertices.

Therefore: vertex control $\Longleftrightarrow$ two straight arcs and one hyperbolic arc .

A control that jumps among vertices produces a boundary that alternates among straight and hyperbolic pieces. This is why bang-bang control will imply a smoothed polygon.

5.8 Interior and edge controls

If u lies in the interior of U_T, all three curvature components are positive. The circle corresponds to the center $u = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$.

If u lies on an edge, exactly one curvature component vanishes. Such edge controls require special treatment because the Pontryagin maximization may fail to choose a unique endpoint of the edge. Chapter 10 develops an auxiliary edge-control problem to prove that true minimizers cannot slide arbitrarily along an edge.

5.9 Reconstruction of the boundary

Given measurable $u(t) \in UT$ and initial data $g(0) = I$, $X(0) = X0$ satisfying the star inequalities and $\det X0 = 1$, solve $[Zu, X]\, X' = \langle Zu, X\rangle$, $g' = gX$.

Then define $\sigma j(t) = g(t)s{*}j$.

The state equation was derived precisely so that the normalized curvatures of these curves are uj. As long as the star inequalities and endpoint conditions hold, the multicurve reconstructs a balanced disk.

Geometric intuition

The control does not directly push the point g in the plane. It chooses how the total curvature is distributed among three synchronized boundary arcs. That curvature distribution changes the moving-frame velocity X; X then moves the frame g; and g generates all six boundary curves. $u \longrightarrow X' \longrightarrow g' \longrightarrow \sigma j$.

Chapter summary

Convexity supplies three nonnegative curvature quantities. Dividing by their sum places the control in a triangle. The affine matrix Zu packages these curvature ratios. The complete normalized state dynamics is

$$g' = gX,\ X' = [Zu, X]/\langle Zu, X\rangle.$$

At a vertex of the control triangle, two boundary curves are straight and the third is hyperbolic. Consequently, finite vertex switching is exactly the geometric pattern of a smoothed polygon.

Chapter 6

The area functional and the Reinhardt control problem

6.1 Green's theorem

Let $\gamma(t) = (x(t), y(t))$ be a positively oriented simple closed curve. Green's theorem gives

$$1\ I\ 1\ Z\ \ \mathrm{area}(\gamma) = (x\,dy - y\,dx) = \det(\gamma, \gamma')\,dt.\ \ 2\ \gamma\ 2$$

For a curve generated by $g(t)$, write

$$\gamma(t) = g(t)v,\ g' = gX.$$

Then $\gamma' = gXv$,

and, because $\det g = 1$, $\det(\gamma, \gamma') = \det(v, Xv)$.

The identity $\det(v, Xv) = -vTJXv$

turns area into a matrix expression.

6.2 Adding three synchronized arcs

One sixth of the boundary is represented by three even-indexed curves

$$\sigma 0(t) = g(t)s{*}0,\ \sigma 2(t) = g(t)s{*}2,\ \sigma 4(t) = g(t)s{*}4,$$

for $0 \le t \le tf$. Their centrally symmetric copies complete the boundary. A direct matrix identity for any $Y \in sl2(R)$ is

3 JY + (R2)TJY R2 + (R4)TJY R4 = tr(JY)I. 2

Applying Green's theorem to the three arcs gives

Z tf area(K) = −3 tr(JX(t)) dt. 2 0

Since the star inequalities imply tr(JX) < 0, the integrand is positive.

Theorem

Theorem 6.1 (Area formula). For a normalized balanced disk in circle representation,

---

Z tf area(K) = −3 ⟨J, X⟩dt. 2 0

Example

Example 6.2 (The unit disk). For the unit disk,

π g(t) = etJ, X(t) = J, tf = . 3

Because J2 = −I and tr(J2) = −2,

Z π/3 area(K) = −3 (−2) dt = π. 2 0

6.3 The complete control problem

We can now state the geometric problem as an explicit optimal-control problem.

Theorem

Theorem 6.3 (Reinhardt control problem). Find a measurable control u : [0, tf] →UT , a terminal time tf > 0, and state curves

---

g : [0, tf] →SL2(R), X : [0, tf] →sl2(R)

that minimize −3 Z tf ⟨J, X⟩dt 2 0 subject to

g′ = gX, (6.1) [Zu, X] X′ = ⟨Zu, X⟩, (6.2) det X = 1, ρj(X) > 0, (6.3) g(0) = I, g(tf) = R, (6.4) X(tf) = R−1X(0)R. (6.5)

The terminal time is free.

The state space is the tangent bundle of SL2(R), written concretely as

SL2(R) × sl2(R).

No abstract tangent-bundle machinery is needed to solve the equations: the pair (g, X) is simply a matrix and its body velocity.

6.4 Why the free terminal time matters

> The parameter t was chosen by det X = 1, not by prescribing how long one sixth of the boundary must take. Different shapes can have different tf. Thus the endpoint g(tf) = R must be reached at a time chosen by the optimizer.

For free-terminal-time problems, the Pontryagin Hamiltonian will vanish along every extremal. This additional equation is used repeatedly in the singular-locus and blowup analysis.

6.5 Existence of an optimizer

The original geometric problem has a minimizer by Reinhardt's compactness argument. The control formulation also fits a standard existence theorem due to Filippov, once the relevant state region is made compact.

A simplified version of Filippov's theorem says:

Standard theorem used as a black box

If the control set is compact, the velocity set is convex at each state, and the relevant vector fields are confined to a compact region, then the attainable sets are compact and an optimal measurable control exists.

The control simplex UT is compact. The velocity set for the upper-half-plane version of the X equation is the convex hull of the three vertex velocities. The only subtle point is that the star domain is noncompact. The paper proves that a global minimizer never approaches its ideal boundary.

6.6 Why the boundary of the star domain is nonoptimal

A state X satisfying det X = 1 and the star inequalities determines a critical hexagon with fixed midpoints s∗j. As one star inequality approaches equality, the hexagon degenerates toward a parallelogram. A convex disk whose critical hexagon is nearly a parallelogram has packing density close to 1, far above the density of the smoothed octagon. Such a state cannot belong to a global minimizer. The source makes this quantitative. It constructs a compact subset H∗∗of the star domain by deleting six neighborhoods of the three ideal vertices and three boundary sides. If a trajectory leaves H∗∗, a geometric area estimate implies

> δ(K) > δoct.

Therefore every global minimizer remains in H∗∗.

Relation to the source paper

This is the compactification theorem of source Chapter 5. The explicit constants, such as the horocycle height 4.5 and slope 15, are chosen for a convenient rigorous exclusion, not because

they are optimal.

6.7 A useful time bound

In upper-half-plane coordinates, developed in the next chapter, the cost integrand is at least 3. Since a minimizer has area less than the disk's area π, one obtains

π tf < . 3

This uniform time bound, together with compactness of the X-state, controls g by Gronwall's inequality.

6.8 The conceptual achievement of the reduction

The original unknown is a convex boundary curve. In the control problem the unknowns are:

• six scalar functions: three coordinates of g and three of X, subject to algebraic constraints; • one measurable control u(t) in a fixed triangle; • one scalar terminal time tf.

The equations are explicit rational matrix ODEs. The price is that the control can be discontinuous and may switch infinitely often. This is exactly where optimal-control theory becomes necessary.

Proof roadmap

From now on the geometric theorem will follow from a statement about controls:

global minimizer is bang-bang with finitely many switches

Indeed, each constant vertex-control segment produces straight lines and a hyperbola, and finitely many segments produce a finite smoothed polygon.

Chapter summary

Green's theorem turns area into the linear cost −3 R ⟨J, X⟩. Together with the curvature 2 state equation and rotational endpoint conditions, this gives a free-time optimal-control problem on explicit 2 × 2 matrices. A geometric compactification excludes states near the degenerating boundary, ensuring that an optimal control exists in a compact region.

Chapter 7

The hidden hyperbolic plane

7.1 The determinant-one surface

The normalized state matrix X satisfies

$X \in sl2(R)$, det $X = 1$, $\langle J, X\rangle < 0$.

Write a b ! X = . c −a

Then $-a2 - bc = 1$. This is a two-dimensional quadratic surface inside the three-dimensional vector space sl2(R). The component selected by $\langle J, X\rangle = b - c < 0$ is naturally a copy of the hyperbolic plane.

Define the upper half-plane $H = \{z = x + iy \in C : y > 0\}$.

For $z = x + iy$, set x/y −(x2 + y2)/y ! Φ(z) = . 1/y −x/y

A direct computation gives

+ y2 + 1 tr Φ(z) = 0, det Φ(z) = 1, ⟨J, Φ(z)⟩= −x2 < 0. y

Conversely, every normalized state matrix in the selected component has this form.

Theorem

Theorem 7.1 (Upper-half-plane parametrization). The map

$\Phi : H \longrightarrow \{X \in sl2(R) : \det X = 1, \langle J, X\rangle < 0\}$

is a diffeomorphism.

7.2 Why matrix conjugation becomes a Möbius transformation

A matrix α β ! g = ∈SL2(R) γ δ

acts on H by αz + β g · z = . γz + δ The denominator never vanishes in H, and the imaginary part remains positive. The map Φ is equivariant: gΦ(z)g−1 = Φ(g · z). Thus the conjugation motion of X is exactly the usual action of SL2(R) on the Poincaré upper half-plane. The stabilizer of J = Φ(i) is the rotation group SO2(R). Hence

H ~= SL2(R)/SO2(R).

This is the geometric meaning of the adjoint orbit of J.

Background interlude

The Poincaré metric is dx2 + dy2 ds2 = . y2

Its geodesics are vertical lines and semicircles orthogonal to the real axis. Möbius transformations by SL2(R) preserve this metric. The source also uses the symplectic form dx ∧dy/y2, but the metric picture is enough for most of the proof.

7.3 The star domain is an ideal triangle

Substitute X = Φ(x + iy) into the three star inequalities. They become

−1√ < x < √1 x2 + y2 > 1 3 3, 3.

Therefore the admissible X-states form √ √ H∗= nx + iy ∈H : −1/ 3 < x < 1/ 3, x2 + y2 > 1/3 o .

Its boundary consists of two vertical geodesics and a semicircle orthogonal to the real axis. Its three vertices lie at the ideal boundary: −1√ √1 ∞. 3, 3, Thus H∗is an ideal hyperbolic triangle.

y

H∗

i

√ √ x −1/ 3 1/ 3

Figure note. *Figure 7.1: The star domain H∗in the upper half-plane. The point z = i corresponds to the circle.*

7.4 The cost in hyperbolic coordinates

Because + y2 + 1 ⟨J, Φ(x + iy)⟩= −x2 , y the area becomes 3 Z tf x(t)2 + y(t)2 + 1 area(K) = dt. 2 0 y(t) The level sets of x2 + y2 + 1 y are hyperbolic circles centered at i. Indeed,

x2 + y2 + 1 cosh dH(x + iy, i) = . 2y

Thus the cost density is 3 cosh dH(z, i).

It is smallest at the circular state z = i and increases as the state moves hyperbolically away from the center.

Point requiring care

A pointwise smaller cost density does not imply a globally better shape. The circle stays at z = i but requires terminal time π/3. Other trajectories can reach the rotational endpoint in shorter time. The optimization balances path location and travel time.

The elementary inequality x2 + y2 + 1 1 ≥y + ≥2 y y gives area(K) ≥3tf.

Since a minimizer has area less than π, π tf < . 3

7.5 State equations for (x, y)

Write au bu ! Zu = , cu −au

where u1 −u2 u0 −2u1 −2u2 au = √ , bu = , cu = u0. 3 3 Transferring X′ = [Zu, X]/ ⟨Zu, X⟩through Φ gives

y(2aux + bu −cux2 + cuy2) x′ = , (7.1) 2aux + bu −cux2 −cuy2 2y2(au −cux) y′ = . (7.2) 2aux + bu −cux2 −cuy2

Although these formulas look complicated, the vertex trajectories are simple generalized circles.

7.6 Constant controls

Fix u. The quantity ⟨X, Zu⟩is constant because

d ⟨[Zu, X], Zu⟩ ⟨X, Zu⟩= = 0. dt ⟨Zu, X⟩

Hence Zu P0 = ⟨Zu, X(0)⟩

is constant and X(t) = etP0X(0)e−tP0.

Equivariance implies z(t) = etP0 · z(0).

For a vertex control, P0 has real eigenvalues, and the Möbius orbit is a generalized Euclidean circle joining two ideal vertices of H∗.

> Up to cyclic symmetry: √ • one vertex gives straight lines from −1/ 3 toward ∞; √ • a second gives straight lines from ∞toward 1/ 3; √ √ • the third gives circular arcs from 1/ 3 toward −1/ 3.

e2-flow e1-flow

e0-flow

Figure note. *Figure 7.2: Schematic vertex-control trajectories in the ideal triangle. Exact trajectories are generalized circles determined by the corresponding Möbius one-parameter groups.*

7.7 Dihedral symmetry

The equilateral control triangle has the dihedral symmetry group of order 6 after the central sign in SL2(R) is factored out. Rotation by R cyclically permutes

e0, e1, e2

and the three ideal vertices of H∗. Reflection across the imaginary axis reverses time and exchanges two controls. These symmetries greatly reduce case analysis. A fundamental domain for the action on H∗is one of six sectors bounded by three geodesics through i. Later, a related time-reversal symmetry will exchange the inward and outward chattering fixed points.

7.8 Smoothed polygon trajectories

> A bang-bang control cycles through the three vertices. In H∗, the state follows a concatenation of generalized-circle arcs. For the known smoothed (6k + 2)-gons, the orbit repeatedly traces the sides of a small equilateral curvilinear triangle centered at i. As k →∞, the triangle shrinks to i, and the smoothed polygons converge to the circle.

This hyperbolic picture is not required to define the control problem, but it reveals the geometry of its explicit extremals and identifies the circle as the singular center.

Chapter summary

The determinant-one state surface is the hyperbolic upper half-plane. The star inequalities cut out an ideal triangle, and the circle is the central point z = i. The area density is 3 cosh dH(z, i). Constant vertex controls move along generalized circles joining ideal vertices, so bang-bang trajectories become curvilinear polygonal paths in the hyperbolic state space.

Part III

Optimal control without prerequisites

Chapter 8

Optimal control from first principles

8.1 From calculus of variations to control

In the classical calculus of variations one chooses a curve q(t) to minimize

Z tf L(q(t), q′(t)) dt. 0

Optimal control separates the velocity choice from the state. One prescribes

q′(t) = f(q(t), u(t)),

where u(t) is a control taking values in a fixed set U, and minimizes

Z tf l(q(t), u(t)) dt. 0

The control may be only measurable. Discontinuous controls are not pathologies to be removed; they are often the natural optimizers.

In the Reinhardt problem, q = (g, X), U = UT ,

and the control distributes curvature among the synchronized boundary arcs.

8.2 Why a multiplier is needed

The state equation is a constraint. Introduce a time-dependent row vector or covector p(t) and consider the augmented integral

Z tf λcostl(q, u) + p · (f(q, u) −q′) dt, 0

where λcost ≤0 is a constant multiplier. Integrating the term −p · q′ by parts and varying q suggests

p′ = −∂H , ∂q

where $H(q, p, u) = p \cdot f(q, u) + \lambda\text{cost}\, l(q, u)$

is the control-dependent Hamiltonian. Varying p recovers

$\partial H \quad q' = = f(q, u). \quad \partial p$

The remaining variation in u leads not to an ordinary derivative equation when U has corners, but to a pointwise maximization rule.

8.3 The Pontryagin maximum principle

Standard theorem used as a black box

Pontryagin maximum principle, simplified free-time form. Let $q*(t)$, $u*(t)$ be an optimal trajectory for a smooth control system with compact control set. Then there exist a nonzero multiplier pair $(\lambda\text{cost}, p(t))$, $\lambda\text{cost} \leq 0$,

such that, almost everywhere,

$\partial H \quad q' = , \quad \partial p \quad p' = -\partial H , \quad \partial q \quad H(q, p, u*) = \max H(q, p, u). \quad u \in U$

For a free-terminal-time problem with no terminal cost,

$H(q(t), p(t), u*(t)) = 0$

along the extremal. Endpoint constraints give transversality conditions on p.

A state–costate trajectory satisfying these necessary conditions is a Pontryagin extremal. Not every extremal is a minimizer, just as not every critical point in calculus is a minimum.

8.4 Normal and abnormal extremals

If $\lambda\text{cost} \neq 0$, scale the multiplier so that

$\lambda\text{cost} = -1.$

The extremal is then normal. If $\lambda\text{cost} = 0$,

the cost disappears from the first-order equations; the extremal is abnormal. Abnormal extremals reflect the geometry of the constraints rather than the cost itself. They can be genuine, but they can also be artifacts of a redundant formulation.

The Reinhardt problem has anomalous abnormal edge extremals in its full encoding. The source therefore introduces a reduced edge-control problem to remove them when proving finite switching along an edge.

8.5 Cotangent vectors without differential geometry

At a point $q \in \mathbb{R}^n$, a tangent vector is an ordinary vector $v \in \mathbb{R}^n$. A cotangent vector is a linear functional $p : v \mapsto p \cdot v$.

For a matrix group, left translation identifies every tangent space with the same matrix space. Thus the costates for $g \in SL_2(\mathbb{R})$ and $X \in \mathfrak{sl}_2(\mathbb{R})$ can both be represented by traceless matrices, using the nondegenerate trace pairing $\langle A, B\rangle = \mathrm{tr}(AB)$.

This is why the source writes the costates as matrices Λ_1 and Λ_2.

8.6 Maximized Hamiltonian

Define $H^+(q, p) = \max_{u\in U} H(q, p, u)$.

If the maximizer is unique and depends smoothly on (q, p), then the extremal follows the ordinary Hamiltonian flow of H^+. When the maximizer jumps from one extreme point of U to another, H^+ is not differentiable across the switching surface, but the state and costate remain continuous.

Definition

Definition 8.1 (Bang-bang control). A control is bang-bang if it takes values only at extreme points of the control set, except possibly on a set of measure zero.

For an interval control $U = [-1, 1]$, a bang-bang control takes the values -1 and 1. For the Reinhardt triangle, it takes the three vertex values e_0, e_1, e_2.

8.7 Switching functions

Suppose the Hamiltonian evaluated at two candidate controls is $H_i(t)$ and $H_j(t)$. Their difference

$$\chi_{ij}(t) = H_i(t) - H_j(t)$$

is a switching function. The maximization rule chooses i when $\chi_{ij} > 0$ and j when $\chi_{ij} < 0$. A switch can occur only at a zero of χ_{ij}.

If a zero is simple, $\chi_{ij}(t_0) = 0$, $\chi'_{ij}(t_0) \neq 0$,

the switching time varies smoothly with initial conditions by the implicit-function theorem. Multiple zeros are the source of discontinuities in Poincaré maps and require discriminants and resultants later.

8.8 Faces of a convex control set

A subset F of a convex set U is a face if whenever an interior point of a segment in U lies in F, both endpoints lie in F. The faces of a triangle are its vertices, edges, and the whole triangle.

In the Reinhardt Hamiltonian, the control-dependent term is a ratio of two affine functions of u. Along a line segment $u(t) = tu_1 + (1 - t)u_0$, such a ratio has the form $\frac{at + b}{ct + d}$. Its derivative has constant sign unless it is identically constant. Therefore the set of maximizers is always a face of U_T.

Consequently, at each time exactly one of the following occurs:

1. a unique vertex maximizes: regular bang-bang motion; 2. an entire edge maximizes: an edge singularity; 3. the whole triangle maximizes: the fully singular locus.

8.9 Singular arcs

A singular arc is a time interval on which the maximization rule fails to determine a unique control. The name refers to singularity of the first-order necessary conditions, not necessarily to a geometric singularity of the state curve.

In the Reinhardt problem, full-triangle singularity forces the circular state. Edge singularities are handled by the auxiliary edge-control problem. Outside the full singular locus, edge extremality will imply finite bang-bang switching.

8.10 Chattering

A bang-bang control may have infinitely many switches in a finite interval. If the switching times

$t_1 < t_2 < \cdots < t_n < \cdots$

converge to a finite limit t_*, the control chatters. The state can converge smoothly even though the control oscillates infinitely rapidly.

Chattering cannot be excluded by compactness or first-order optimality alone. Fuller's classical example shows that it can be the genuine optimal behavior. The central difficulty of the source paper is to prove that a periodic Reinhardt minimizer does not chatter into the circular singular locus.

8.11 Poincaré maps

For a bang-bang flow, record only the state at switching times. If q_n is the state at the nth switch, define $q_{n+1} = F(q_n)$.

This F is a Poincaré first-recurrence map. It converts a continuous-time control system into a discrete dynamical system.

Near a chattering singularity, the time intervals shrink, but after a suitable blowup the angular switching data can converge to a nontrivial fixed point of F. This is the key idea used in Part V.

Chapter summary

The Pontryagin maximum principle augments the state with a costate and replaces minimization by Hamiltonian evolution plus pointwise control maximization. For the triangular control set, maximizers are faces. Unique vertex maximizers give bang-bang motion; edges and the whole triangle give singular behavior. Infinite vertex switching in finite time is chattering, the only obstruction that survives the first-order analysis.

Chapter 9

The Pontryagin system for Reinhardt's problem

9.1 State and costate variables

The state is $(g, X) \in SL2(R) \times sl2(R)$.

We represent the two costates by matrices

$\Lambda 1, \Lambda 2 \in sl2(R)$

through the trace pairing. The costate $\Lambda 1$ corresponds to the group variable g, while $\Lambda 2$ corresponds to X.

The dynamics are Zu $g' = gX$, $X' = [P, X]$, $P = \langle Zu, X\rangle$. The running cost is $l(X) = -3$ $\langle J, X\rangle$. 2

9.2 The Hamiltonian

The group part contributes $\langle \Lambda 1, X\rangle$.

The X-state part contributes $\langle \Lambda 2, [P, X]\rangle = -\langle [\Lambda 2, X], Zu\rangle$. $\langle X, Zu\rangle$ It is convenient to define the reduced costate

$\Lambda R = [\Lambda 2, X]$.

Then the full Hamiltonian is

Zu⟩ $H(\Lambda 1, \Lambda R, X; u) = \Lambda 1 -3\ X -\langle \Lambda R$, $2\lambda cost J$, $\langle X, Zu\rangle$.

For normal extremals we take $\lambda cost = -1$, so the first term becomes

3 Λ_1 + 2J, X .

The source often keeps λ_{cost} symbolic until normality is established.

Point requiring care

The Pontryagin convention maximizes H and uses $\lambda_{cost} \leq 0$. Other textbooks minimize the Hamiltonian or use a positive cost multiplier. Signs must be translated before comparing formulas.

9.3 The state–costate equations

Let Zu P = ⟨Zu, X⟩.

The complete equations are

$g' = gX$, (9.1) $X' = [P, X]$, (9.2) $\Lambda_1' = [\Lambda_1, X]$, (9.3) 3 $\Lambda_R' = [P, \Lambda_R] - \langle \Lambda_R, P\rangle [P, X] +$ $-\Lambda_1 + 2\lambda_{cost} J, X$. (9.4)

The control u(t) maximizes H over UT almost everywhere, and

$H = 0$

along a free-time extremal.

The equation for Λ_1 is another Lax equation. Its solution is explicit:

$$\Lambda_1(t) = g(t)^{-1}\Lambda_1(0)g(t).$$

Thus $\det \Lambda_1(t) = \det \Lambda_1(0)$

is conserved.

9.4 Why Λ_R is the right reduced variable

By construction, $\langle \Lambda_R, X\rangle = \langle [\Lambda_2, X], X\rangle = 0$.

Hence Λ_R lies in the two-dimensional plane orthogonal to X under the trace form. This is exactly the cotangent space to the determinant-one orbit at X.

The control-dependent part of H depends only on (X, Λ_R):

Zu⟩ $H_2(X, \Lambda_R; u) = -\langle \Lambda_R, \langle X, Z_u\rangle$.

The variables g and Λ_1 matter for endpoint conditions and the inhomogeneous term in Equation (9.4), but the switching geometry is controlled by (X, Λ_R).

9.5 Hamiltonian maximizers are faces

Fix (X, ΛR). Along a segment in the control simplex, Zu is affine in u, so H2 is a ratio of affine functions. As explained in Chapter 8, the set of maximizers is a face. Thus the cotangent space is partitioned into seven strata indexed by nonempty subsets I ⊂{0, 1, 2}:

arg max H2 = (UT)I, u∈UT

where (UT)I is the face spanned by the vertices in I.

The three open vertex regions are separated by three codimension-one walls. The walls meet at the full singular set where the Hamiltonian is independent of u.

9.6 Switching functions

Let Zj Pj(X) = ⟨Zj, X⟩

for the three vertex controls. The difference between the control-dependent Hamiltonians at vertices i and j is, up to a harmless sign convention,

χij(X, ΛR) = ⟨ΛR, Pj(X) −Pi(X)⟩.

A switch between i and j occurs on the wall

χij = 0.

Away from multiple zeros, the first switching time is a smooth function of the initial state and costate.

9.7 Transversality and periodicity

The geometric endpoint conditions are rotational:

g(tf) = R, X(tf) = R−1X(0)R.

The costates satisfy matching conditions of the same form:

Λ1(tf) = R−1Λ1(0)R, ΛR(tf) = R−1ΛR(0)R.

Extending by symmetry yields

X(t + 3tf) = X(t), Λ1(t + 3tf) = Λ1(t), ΛR(t + 3tf) = ΛR(t).

The group variable has period $6t_f$ because $R^6 = I$.

Geometric intuition

The final contradiction will use only one global feature of the boundary conditions: a minimizer extends to a periodic Pontryagin trajectory. The local chattering analysis will show that every trajectory entering the singular locus lies on a stable curve that cannot close periodically.

9.8 Left invariance

The equations and cost depend on g only through $g' = gX$, and replacing $g(t)$ by $hg(t)$ with constant $h \in SL_2(\mathbb{R})$ changes neither X nor the cost. This is left invariance. It explains why the group costate satisfies a conjugation equation and why much of the dynamical analysis can be performed solely in the Lie algebra variables.

For the proof of Mahler's First conjecture, one often temporarily ignores the endpoint condition on g, analyzes $(X, \Lambda_1, \Lambda_R)$, and restores $g(t_f) = R$ at the end. This separation is legitimate because g is reconstructed by integrating $g' = gX$ once X is known.

9.9 Normality near the singular locus

At a point where $\Lambda_R = 0$, $\Lambda_R' = 0$,

the Hamiltonian equation and Equation (9.4) force

$3\ \Lambda_1 = 2\lambda_{cost} J$.

If $\lambda_{cost} = 0$, then both costates vanish, contradicting the nontriviality condition in the maximum principle. Therefore $\lambda_{cost} \neq 0$.

Any extremal that reaches the full singular locus is normal, and after scaling $\lambda_{cost} = -1$, $\Lambda_1 = -\frac{3}{2}J$.

These fixed parameter values are used in the blowup analysis.

9.10 Constant vertex segments

When $u = e_j$ is constant, $P = P_j$ is constant because $\langle X, Z_j \rangle$ is conserved. Hence X and g have explicit solutions:

$$X(t) = e^{tP} X_0 e^{-tP},$$

$$g(t) = e^{t(X_0+P)} e^{-tP}$$

when $g(0) = I$. The costate Λ_1 follows by conjugation, and Λ_R is obtained by elementary quadratures of matrix exponentials.

These explicit formulas make the Poincaré maps and switching functions computable. The eventual computer-assisted part of the proof depends heavily on the fact that constant-control solutions are analytic and explicit.

Chapter summary

The maximum principle lifts the Reinhardt state to the matrix system $(g, X, \Lambda_1, \Lambda_R)$. The Hamiltonian is Zu⟩ $H = \Lambda_1 - 3$ X $-\langle \Lambda_R, 2\lambda J, \langle X, Zu\rangle$,

and the control maximizers are faces of the triangle. Rotational transversality makes every minimizing extremal periodic. The full singular locus is necessarily normal and has fixed costate parameters.

Chapter 10

Bang-bang extremals and smoothed polygons

10.1 Finite vertex switching gives the desired geometry

Suppose the control is piecewise constant and takes values only in the three vertices e0, e1, e2. On each constant segment, two state-dependent curvatures vanish. Therefore two synchronized boundary arcs are straight and the third is a hyperbolic arc. At a switching time, the state g, X is continuous, so the plane curves and their tangent lines match continuously.

Proposition

Proposition 10.1. A bang-bang state trajectory with finitely many switches reconstructs a finitesided smoothed polygon of Reinhardt–Mahler type.

Conversely, a smoothed polygon whose corners are rounded by the prescribed hyperbolas yields a finite vertex-control sequence. Thus the geometric theorem is equivalent to finite bang-bang behavior of the global minimizer.

10.2 Explicit constant-control splines

Let u0 be one vertex and let g0(z, t) be the constant-control solution with initial upper-half-plane state z. Cyclic symmetry gives gk(z, t) = Rkg0(z, t)R−k.

A finite control word I = ((k1, t1), ... , (kN, tN)) means: use control Rk1u0 for time t1, then Rk2u0 for time t2, and so on. Multiplying the corresponding segment solutions constructs a spline

g(I, z0, t).

Every finite bang-bang state trajectory arises in this way. The total area is the sum of the elementary segment costs.

This explicit parametrization turns the search for special extremals into algebraic endpoint and switching equations.

10.3 The known family of smoothed (6k + 2)-gons

Choose the cyclic control sequence

e2, e1, e0, e2, e1, e0, . . .

with equal switching time tsw. In H∗, the state travels counterclockwise around a curvilinear equilateral triangle centered at i. Let the starting point be

1 z0 = iy0, √ < y0 < 1. 3

The requirement that one segment moves z0 to R−1 · z0 gives log 4/(3y20 + 1) tsw = √ . 3y0

The group endpoint condition after 3k + 1 segments imposes

4 πk = 2 cos . 3y20 + 1 3k + 1

For each positive integer k, these equations determine a smoothed (6k + 2)-gon that is a normal Pontryagin extremal.

For k = 1, one obtains the smoothed octagon. Its parameters are s√ 8 −1 log 2 y0 = , tsw = . 3 q√ 2 8 −1

Its area and density are √ √ 8 − 32 −log 2 area(Koct) = 12 √ , 8 −1 √ 8 − 32 −log 2 δoct = √ . 8 −1 As k →∞, the triangular state orbit contracts to z = i, and the smoothed polygons converge to the circle.

Relation to the source paper

The source proves the Hamiltonian maximization inequalities on each segment by reducing a switching-function inequality to an elementary but lengthy inequality on a triangle in two real variables. This establishes that the constructed splines are genuine Pontryagin extremals, not merely state trajectories satisfying endpoint conditions.

10.4 Why the maximum principle alone is not yet enough

At a generic cotangent state, the Hamiltonian has a unique maximizing vertex. But on a wall, an entire edge of UT maximizes. A priori the optimal control might move through interior points of that edge for a positive time or switch infinitely often between its endpoints.

The full Reinhardt encoding admits abnormal edge extremals with arbitrary measurable edge controls. To obtain a useful conclusion, the source re-encodes the same state motion as a smaller three-dimensional edge-control problem.

10.5 The edge-control problem

Consider the edge

1 1 u0 = 0, u1 = + v, u2 = −v, −1 ≤v ≤1 2 2 2 2. In upper-half-plane coordinates the state equations simplify to √ 2 3y2v x′ = y, y′ = √ −1 + 2 3xv.

In particular, x′ = y > 0,

so x is strictly increasing. Introduce a third state variable

1 s′ = y.

The group variable g can then be reconstructed explicitly from (s, x, y). After subtracting an endpoint-dependent constant, the cost reduces to Z x2 dt. y

This is the edge-control problem.

Let (λ1, λ2, λ3) be its costates. The control-dependent part of the Hamiltonian is λ2y′. Since y′ is strictly monotone in v, the maximizing control is

1 v = −1 or v = 2 2 according to the sign of λ2. Thus λ2 is the edge switching function.

10.6 Zeros of the edge switching function are isolated

When λ2(t0) = 0, λ′2(t0) ≠ 0, the zero is plainly isolated. The delicate case is λ2(t0) = λ′2(t0) = 0.

The Hamiltonian equation H = 0 and the costate equation give

λ1(t0) = 0, λ3 = −x(t0)2λcost, λcost ≠ 0.

Because x is strictly increasing, this exceptional relation can occur at at most two times.

A local expansion then shows that the zero remains isolated. After shifting t0 to 0 and normalizing λcost = −1:

- if x(0) ≠ 0, then λ2(t) = Ct2 + O(t3), C ≠ 0;

- if x(0) = 0, then λ2(t) = −2 + O(t4). 3t3

The estimates are uniform over all measurable edge controls. They are obtained by constructing explicit polynomial approximations to the costates and controlling the error with Gronwall's inequality.

Theorem

Theorem 10.2 (Finite switching on an edge; source Theorem 8.1.4). Every Pontryagin extremal of the edge-control problem is bang-bang with finitely many switches on a compact time interval.

Logical core. The edge control is determined by the sign of the continuous switching function $\lambda 2$. Every zero is isolated, including the possible double or triple zeros described above. A compact interval cannot contain infinitely many isolated zeros without an accumulation point, and every accumulation point would be another zero, contradicting its isolatedness. Hence only finitely many switches occur.

10.7 Edge extremality of a global minimizer

A globally minimizing Reinhardt trajectory is automatically optimal among all variations whose control remains on the same edge during any subinterval. Therefore it is also an extremal of the corresponding edge-control problem.

Definition

Definition 10.3 (Edge-extremal trajectory). A Reinhardt trajectory is edge-extremal if, on every interval where its control lies in one edge of UT , it satisfies the Pontryagin conditions for the corresponding reduced edge-control problem.

Every global minimizer is both Reinhardt extremal and edge-extremal.

10.8 The full singular locus

Let us temporarily denote by S the set where the Hamiltonian is independent of the entire triangle. We will identify it precisely in the next chapter. The following local result is the bridge from edge analysis to global bang-bang behavior.

Theorem

Theorem 10.4 (Local edge confinement). Let a Reinhardt extremal be edge-extremal. At any time when it does not lie in S, there is a neighborhood on which all Hamiltonian-maximizing controls lie in one fixed edge of UT .

> The proof is by contrapositive. If arbitrarily near t0 the maximizer is not confined to one edge, then all three walls accumulate at t0. Continuity forces $\Lambda R(t0) = 0$. Estimates on the costate equation then force $3\ \ \Lambda 1(t0) = 2\lambda \mathrm{cost} J,\ X(t0) = J$, which is exactly the singular locus.

Combining local confinement with finite edge switching gives the decisive regular-region theorem.

Theorem

Theorem 10.5 (Finite bang-bang switching away from the singular locus). Let a Reinhardt extremal be edge-extremal and avoid the singular locus on a compact interval. Then its control is bang-bang with finitely many switches on that interval.

Proof. Proof. Every time has a neighborhood in which the maximizers lie in one edge. On that neighborhood, the edge-control theorem gives finitely many switches between the endpoints. A finite subcover of the compact interval gives global finiteness. □

10.9 What remains to prove

For a global minimizer, the preceding theorem already gives a smoothed polygon unless the trajectory meets or accumulates at the singular locus. The next chapter shows:

• a trajectory cannot remain on the singular locus for positive time; • the singular locus is the circle; • the circle is not locally minimizing.

After that, the only unresolved possibility is chattering into the singular locus in finite time.

Chapter summary

Finite vertex switching is exactly the geometry of a smoothed polygon. The source constructs explicit smoothed (6k +2)-gon extremals, including the octagon. More importantly, a reduced edge-control problem proves that every edge-extremal trajectory has only finitely many edge switches. Hence any global minimizer that stays away from the full singular locus is already a finite bang-bang solution.

Chapter 11

The singular locus is the circle

11.1 When the Hamiltonian forgets the control

The control-dependent Hamiltonian is

Zu⟩ $H2(X, \Lambda R; u) = -\langle \Lambda R, \langle X, Zu\rangle$.

Suppose this is independent of u on the whole triangle. The velocity set generated by UT spans the entire tangent plane to the determinant-one orbit at X. A covector that takes the same value on all these velocities must vanish. Hence $\Lambda R = 0$.

Conversely, if $\Lambda R = 0$, the Hamiltonian has no explicit control dependence.

Lemma

Lemma 11.1. The Hamiltonian is independent of all $u \in UT$ if and only if $\Lambda R = 0$.

11.2 Determining the state on a singular interval

Assume $\Lambda R(t) = 0$ on an open interval. Then also $\Lambda' R(t) = 0$. The Hamiltonian equation gives $\Lambda 1 - 3$ X = 0. 2λcostJ,

The costate equation gives 3 $-\Lambda 1 + 2\lambda \mathrm{cost} J, X = 0$. The first relation says the bracketed matrix is orthogonal to X; the second says it commutes with X. For a regular determinant-one traceless matrix, the centralizer is the line RX. The only matrix both parallel and orthogonal to X is zero because $\langle X, X\rangle = -2$. Therefore

3 $\Lambda 1 = 2\lambda \mathrm{cost} J$.

The nontriviality condition forces $\lambda \mathrm{cost} \neq 0$, so the extremal is normal.

Now use $\Lambda' 1 = [\Lambda 1, X] = 0$.

Since $\Lambda 1$ is a nonzero multiple of J, [J, X] = 0.

The only X in the normalized orbit that commutes with J and satisfies $\langle J, X\rangle < 0$ is

X = J.

Then $X' = 0$ forces the control matrix to commute with J. The unique control is the center of the triangle: 1 1 1 1 u = 3, 3, 3 , Zu = 3J.

Finally, $g' = gJ \Longrightarrow g(t) = g0etJ$. The curves $g(t)s*j$ are circular arcs, up to the fixed affine transformation g0.

Definition

Definition 11.2 (Singular locus). For normal scaling λcost = −1, the singular locus is

Ssing = (g, X, Λ1, ΛR) : X = J, Λ1 = −3 ΛR = 0 . 2J,

The group coordinate $g \in SL2(R)$ is arbitrary.

In upper-half-plane coordinates, X = J is the point

z = i.

Theorem

Theorem 11.3 (Classification of full singular arcs). A Pontryagin extremal on which the Hamiltonian is independent of the entire control triangle is, up to affine transformation, a multicurve of circular arcs. Conversely, the circular multicurve gives such a singular extremal.

11.3 Why the circle is still a Pontryagin extremal

The circle satisfies all first-order equations:

X = J, u = (1/3, 1/3, 1/3), ΛR = 0, Λ1 = −3J/2.

This illustrates a central limitation of first-order necessary conditions. The circle is stationary under first-order variations but is not a local minimum. One needs a second-order variation.

11.4 A second variation that lowers the area

Consider the circular segment g0(t) = etJ.

Choose smooth compactly supported functions ψ1, ψ2 on an interval (t1, t2) and set

ψ1(t) ψ2(t) ! A(t) = . ψ2(t) −ψ1(t)

For small s, perturb by gs(t) = esA(t)etJ. Because A(t) is traceless, det esA(t) = 1, so gs(t) ∈SL2(R). The compact support keeps the endpoints fixed.

The area cost can be written in a parameterization-invariant form using the Maurer–Cartan matrix g−1 dg: Z D cost(g) = −3 J, g−1g′E dt. 2 Expanding in s gives

Z t2 cost(gs) = cost(g0) −3s2 ψ′1ψ2 −ψ1ψ′2 dt + O(s3). t1

Choose nonnegative functions whose supports overlap and such that

ψ′1 > 0, ψ′2 < 0

on the overlap. Then ψ′1ψ2 −ψ1ψ′2 > 0,

so for sufficiently small s > 0, cost(gs) < cost(g0).

Because the circular curvature is strictly positive, sufficiently small C2 perturbations remain convex. Therefore the circular arc is not locally minimizing.

Theorem

Theorem 11.4 (No singular arc in a global minimizer). A global Reinhardt minimizer cannot remain in Ssing on any interval of positive length.

Proof. Proof. A singular interval is an affine image of a circular arc. The preceding compactly supported second variation lowers the area while preserving endpoint data and convexity, contradicting local optimality. □

11.5 A single vertex segment cannot hit the singular locus

Suppose an extremal uses one constant vertex control and reaches the singular locus at t = 0. The constant-control solution is analytic. At the singular point,

ΛR(0) = 0, Λ′R(0) = Λ′′R(0) = 0,

but the third derivative is nonzero and points in a direction for which one of the other two vertex controls gives larger Hamiltonian. Hence the assumed vertex fails the maximum condition immediately. Therefore:

Lemma

Lemma 11.5. An extremal segment with constant vertex control does not meet the singular locus.

This does not rule out an infinite sequence of successively shorter vertex segments whose switching points converge to the singular locus. That is the chattering possibility.

11.6 The trichotomy for a global minimizer

A global minimizer is edge-extremal. We now know:

1. if it stays a positive distance from the singular locus, then it is finite bang-bang and hence a smoothed polygon; 2. it cannot remain on the singular locus for positive time; 3. no single constant-control segment reaches the singular locus.

Therefore the only unresolved behavior is an infinite switching sequence accumulating at the singular locus in finite time.

Proof roadmap

The rest of the proof is entirely devoted to excluding the following scenario:

$$t1 < t2 < \cdots < tn \nearrow t*, (X(tn), \Lambda1(tn), \Lambda R(tn)) \longrightarrow J, -3\ 0\ .\ 2J,$$

The control cycles among the vertices faster and faster while the state approaches the circular singularity.

Chapter summary

Full-triangle singularity forces $\Lambda R = 0$, $X = J$, and the central control $u = (1/3, 1/3, 1/3)$; geometrically this is the circle. The circle satisfies Pontryagin's first-order conditions but a compactly supported second variation lowers its area, so no minimizer contains a singular circular arc. The only remaining obstruction is chattering into the singular locus.

Part IV

The chattering obstruction and its model

Chapter 12

Chattering and the Fuller phenomenon

12.1 Why infinitely many switches can be optimal

It is tempting to argue that an optimal piecewise-constant control should have only finitely many pieces. This is false. Fuller's classical problem is

$$x' = y,\quad y' = u,\quad -1 \le u \le 1,$$

with the objective $\int_0^\infty x(t)^2\,dt \longrightarrow \min .$ For suitable initial data, the optimal control alternates between -1 and 1 infinitely many times before reaching the origin in finite time. The switching intervals form a geometric progression.

The phenomenon is not an isolated curiosity. In sufficiently high-dimensional optimal-control systems, singularities of the maximized Hamiltonian generically produce Fuller-type chattering.

12.2 What chattering means in the Reinhardt geometry

At a vertex control, the boundary is locally made of two straight arcs and one hyperbolic arc. During chattering, the identity of the curved arc changes faster and faster. The boundary itself can remain C^1 and converge to a smooth circular state even though the curvature control jumps infinitely often.

In the upper-half-plane state space, the switching points form a shrinking triangular spiral around $z = i$. The control cycles through the three vertices of U_T. The challenge is not merely to show that some chattering trajectories exist; it is to classify every possible trajectory that reaches the singular locus.

12.3 Coordinates centered at the singular locus

The normal singular point in state–costate space is

X = J, Λ1 = −3 ΛR = 0. 2J,

The paper uses the Cayley transform to move from real matrices in sl2(R) to matrices in su(1, 1) and then writes the deviations in three complex coordinates

w, b, c ∈C.

The singular locus becomes simply w = b = c = 0.

The exact coordinate formulas are given in the next chapter. For now, their orders have a transparent interpretation: w = O(t), b = O(t2), c = O(t3)

for a trajectory reaching the singularity at t = 0.

This hierarchy follows from a chain of differential equations:

w′ ∼control direction, b′ ∼w, c′ ∼b.

Thus c is effectively the third integral of the control.

12.4 Leading-order truncation

Near the singular locus, the exact Reinhardt equations have the form

c w′ = −iρ + O(|w|), |c| b′ = 2iw + O(|b|), c′ = −3ib + O(|c| + |b| |w|2 + |b|2 |w|).

Discarding higher-order terms gives

cF w′F = −iρ |cF |, b′F = 2iwF , c′F = −3ibF .

Introduce rescaled variables

wF cF z1 = , z2 = −ibF , z3 = ρ 2ρ 6ρ.

Then z3 z′3 = z2, z′2 = z1, z′1 = −i |z3|.

This is the complex length-three Fuller system with circular control.

12.5 Scaling symmetry

The chain is homogeneous under

$$(z_1(t), z_2(t), z_3(t)) \longmapsto \left(r z_1(t/r), r^2 z_2(t/r), r^3 z_3(t/r)\right).$$

It is also invariant under simultaneous complex rotation

$$z_j \longmapsto e^{i\theta} z_j.$$

Combining the two gives a two-dimensional virial group. Self-similar chattering trajectories are fixed by a one-parameter subgroup of this scaling-rotation action; they appear as logarithmic spirals.

12.6 An explicit logarithmic spiral

For $t > 0$, the circular Fuller system has the outward solution

$$z_3(t) = \frac{1}{10} t^{3-i}, \quad z_2(t) = \frac{3-i}{10} t^{2-i}, \quad z_1(t) = \frac{(2-i)(3-i)}{10} t^{1-i}.$$

Because $t^{-i} = e^{-i \log t}$,

the phase rotates logarithmically while the magnitudes scale as t, t^2, t^3. Time reversal and complex conjugation produce an inward spiral reaching the origin in finite time.

Geometric intuition

The exponent $3 - i$ has two jobs. Its real part 3 gives the cubic scaling expected from three integrations. Its imaginary part -1 makes the control direction rotate by an angle proportional to $-\log t$, hence infinitely many turns as $t \to 0^+$.

12.7 Why a blowup is natural

At the singular point, every direction of approach is collapsed to $(0, 0, 0)$. To distinguish them, write

$$z_1 = r\xi_1, \; z_2 = r^2\xi_2, \; z_3 = r^3\xi_3,$$

where $r \geq 0$ is a weighted radius and ξ lies on a weighted unit sphere. The set $r = 0$ is the exceptional divisor. It records the angular profile of a chattering trajectory after factoring out its shrinking scale.

A self-similar spiral becomes a fixed point on the exceptional divisor. The infinite switching sequence in physical coordinates becomes an ordinary orbit of a discrete Poincaré map on a compact angular space.

exceptional divisor r = 0

qout

blow up qin singular point

Figure note. *Figure 12.1: A blowup replaces one singular point by a space of approach directions. Inward and outward self-similar spirals become fixed points on the exceptional divisor.*

12.8 Why the circular model is not the final proof

> The exact Reinhardt control set is a triangle, not a circle. Circular control is smoother and has continuous rotational symmetry, making the first model easier to analyze. But the true chattering sequence uses only three discrete directions. The final proof therefore constructs a triangular Fuller system with control n u ∈ 1, ζ, ζ2o , ζ = e2πi/3.

Its self-similar solutions are triangular spirals rather than smooth logarithmic spirals.

12.9 The remaining proof strategy

The chattering exclusion will proceed as follows.

1. Derive the triangular Fuller system as the leading term on the exceptional divisor. 2. Define its Poincaré map from one switching time to the next. 3. Find the inward and outward fixed points qin, qout. 4. Prove that qout has a global basin of attraction on the divisor, except for qin. 5. Show that the true Reinhardt Poincaré map analytically extends to the divisor and has corresponding stable and unstable manifolds. 6. Use periodicity to rule out the only possible approach to qin.

Chapter summary

Chattering is a genuine optimal-control phenomenon, so it cannot be dismissed abstractly. Near the circular singularity, the Reinhardt equations have a three-step chain with weighted orders 1, 2, 3. Their leading term is a Fuller system with a scaling symmetry. Weighted blowup separates shrinking scale from angular profile, converting self-similar chattering into fixed points of a Poincaré map on an exceptional divisor.

Chapter 13

Optional model: circular control and a conserved angular momentum

13.1 Why enlarge the triangle to a disk?

The triangular control set has only discrete rotational symmetry. Replace it temporarily by a circular disk in the affine plane u0 + u1 + u2 = 1.

The modified problem is no longer exactly the Reinhardt packing problem, because some controls outside the triangle correspond to negative boundary curvature. It is a toy model. Its advantage is continuous rotational symmetry.

Continuous symmetries often produce conserved quantities. In elementary mechanics, invariance under spatial translation gives linear momentum, and invariance under rotation gives angular momentum. The same principle holds for optimal control.

13.2 Noether's principle in the present setting

Let a one-parameter rotation act simultaneously on

X, Λ1, ΛR, Zu

by conjugation. When the control set is a disk, the rotated control is still admissible, and the Hamiltonian is invariant. The control-theoretic Noether theorem then yields the conserved scalar

A = ⟨J, Λ1 + ΛR⟩.

The source calls this angular momentum.

Theorem

Theorem 13.1 (Angular momentum for circular control). Along every Pontryagin extremal of the circular-control problem, d ⟨J, Λ1 + ΛR⟩= 0. dt

One can verify the conservation directly by differentiating and substituting the costate equations; the control-dependent terms cancel exactly because of rotational symmetry.

13.3 The Cayley transform and hyperboloid coordinates

The real algebra sl2(R) is isomorphic to (it z !) su(1, 1) = : t ∈R, z ∈C . ¯z −it

The isomorphism is a fixed change of basis called the Cayley transform. Its main advantage is that rotations act by multiplication of the complex coordinate by a phase.

For a determinant-one state, write

(q) 1 + |w|2 w X = (−i q). ¯w i 1 + |w|2

The pair q 1 + |w|2, w

lies on the upper sheet of the hyperboloid t2 −|w|2 = 1.

This is another standard model of the hyperbolic plane.

The other costates can be encoded by complex variables b, c. Near the singular locus,

w = b = c = 0.

This is why these coordinates are adapted to the local analysis.

13.4 Optimal control on a circle

The boundary of the circular control set can be parametrized by a phase z with |z| = 1. In su(1, 1) coordinates, the control matrix has the form −iα βz ! Z∗(z) = , β/z iα

after fixing positive constants α, β. The ratio

β ρ = α records the disk radius. The circumscribed disk of the triangle has ρ = 2; the inscribed disk has ρ = 1.

Hamiltonian maximization and angular-momentum conservation reduce the optimal phase to a quadratic equation. In the source notation, if Q = [ΛR, X], then αQ21z2 −2iβQ11z + αQ12 = 0.

Near the singular locus, the maximizing root satisfies c z∗= + O(|w|). |c|

Thus the optimal control direction is, to leading order, the phase of the third-chain variable c.

13.5 Exact equations near the singular locus

With fixed normal parameters

$$3 \quad \lambda_{cost} = -1, \ d_1 = 2, \ A = 3,$$

the hyperboloid-coordinate equations imply

$$c \quad w' = -i\rho + O(|w|), \quad |c| \quad b' = 2iw + O(|b|), \quad c' = -3ib + O(|c| + |b|\,|w|^2 + |b|^2\,|w|).$$

If a trajectory reaches the singularity at $t = 0$, Gronwall estimates give

$$|w| = O(t), \ |b| = O(t^2), \ |c| = O(t^3).$$

This justifies the Fuller truncation rigorously.

13.6 Hamiltonian structure of the truncated system

$$\text{For} \quad z_3 \quad z_3' = z_2, \ z_2' = z_1, \ z_1' = -i \ |z_3|,$$

$$\text{define} \quad i \quad H_F = 2(z_2\bar{z}_1 - \bar{z}_2 z_1) + |z_3| \ , \quad A_F = |z_2|^2 - (z_1\bar{z}_3 + \bar{z}_1 z_3).$$

Both are conserved. The system is Hamiltonian for the nonstandard symplectic form

$$-1 \ 1 \ \omega_F = dz_1 \wedge d\bar{z}_3 + dz_2 \wedge d\bar{z}_2 - 1 \ dz_3 \wedge d\bar{z}_1. \quad 2i \ 2i \ 2i$$

The unusual pairing of z_1 with z_3 reflects the third-order chain.

This Hamiltonian structure is conceptually valuable: the Fuller system is not an ad hoc approximation but retains the conservation laws and symplectic geometry of the original problem at leading order.

13.7 Reducing by the virial symmetry

$$\text{On the zero set} \quad H_F = A_F = 0,$$

introduce scale-invariant quantities

$$|z_2| \ |z_3| \quad x_2 = \ , \ x_3 = \ . \quad |z_1|^2 \ |z_1|^3$$

The conservation laws imply

1 2 ≤x3 ≤x2. x2 > 0, x3 > 0, 2x2

Thus (x2, x3) lies in a curved planar region Ω. Phase signs produce four copies of Ω, glued along their boundary curves to form a topological plane.

The Fuller flow descends, after a positive time rescaling, to an explicit vector field on this plane. It has two nontrivial equilibrium points:

2 r2 ! 2 r2 ! q∗+ = √ 10, 5, +, + , q∗−= √ 10, 5, −, − .

They are the outward and inward logarithmic spirals. The Jacobian eigenvalues at q∗+ are √ √ − 2 ± i 3,

so q∗+ is asymptotically stable; time reversal makes q∗−unstable.

13.8 Global behavior of the circular Fuller model

The source analyzes the vector field on the four glued copies of Ω. Apart from one special timereversal-symmetric trajectory passing through the corner (2, 2), the flow crosses the boundary curves in a controlled sequence. A Lyapunov function near q∗+ and monotonic winding around it show:

Theorem

Theorem 13.2 (Global circular Fuller dynamics). Every trajectory other than the inward logarithmic spiral converges, after quotienting by scaling and rotation, to the outward logarithmic spiral as t →+∞. In reverse time, every trajectory other than the outward spiral converges to the inward spiral.

This is a complete classification for the smooth circular model. It suggests the geometry later proved for the triangular Poincaré map: one inward fixed point, one outward fixed point, and a global basin for the outward one.

13.9 What transfers to the triangular problem

The following structural features survive when the circle is replaced by the triangle:

• weighted scaling of orders 1, 2, 3; • a time-reversal involution; • inward and outward self-similar spirals; • a stable outward angular profile and unstable inward profile; • global attraction on the blown-up angular space.

What changes is regularity. Circular control gives a smooth phase z∗. Triangular control jumps among three phases, so the continuous-time angular flow is replaced by a Poincaré map whose switching times are roots of cubic polynomials.

Chapter summary

Circularizing the control set creates continuous rotational symmetry and a conserved angular momentum. Hyperboloid coordinates center the singularity at $w = b = c = 0$ and rigorously produce a complex length-three Fuller system. After quotienting by scaling and rotation, that model has one stable outward spiral and one unstable inward spiral, with a global basin. This chapter supplies the conceptual template for the true triangular analysis, but the final proof does not depend logically on it.

Chapter 14

The triangular Fuller system

14.1 The correct leading-order control

The true Reinhardt control uses the three vertices of a triangle. After the Cayley-transform and rescaling used near the singular locus, these three directions become the cube roots of unity

n $V_T = 1, \zeta, \zeta^2$o , $\zeta = e^{2\pi i/3}$.

The triangular Fuller system is

$z'_3 = z_2$, (14.1) $z'_2 = z_1$, (14.2) $z'_1 = -iu$, $u(t) \in V_T$. (14.3) For constant u, the solution with initial data $z_j(0) = z_j^0$ is $z_1(t) = -iut + z_1^0$, (14.4) $z_2(t) = -iu\ t^2 + z_1^0 t + z_2^0$, (14.5) 2 z_1^0 $z_3(t) = -iu\ t^3 + t^2 + z_2^0 t + z_3^0$. (14.6) 6 2 Thus every switching function is a cubic polynomial in time.

14.2 Hamiltonian and maximum rule

Define the real bilinear form on $\mathbb{C}$ by

$\mathrm{Re}(a, b) = \mathrm{Re}(\bar{a}b)$.

The Fuller Hamiltonian is $H_F(z, u) = \mathrm{Re}(z_1, iz_2) + \mathrm{Re}(u, z_3)$.

The first term is independent of u. Therefore the maximizing control is the vertex $u \in V_T$ whose direction has the largest projection onto z_3:

$u(t) \in \arg\max\ \mathrm{Re}(v, z_3(t))$. $v \in V_T$

For the free-time extremals relevant to the blowup,

$H_F(z(t), u(t)) = 0$.

The maximization partitions the complex z_3-plane into three sectors. Switching occurs on the three rays where two projections tie: $W = \mathbb{R}_{\le 0} \cup \mathbb{R}_{\le 0}\zeta \cup \mathbb{R}_{\le 0}\zeta^2$.

These rays are the walls.

Imz3

u = 1

W0 u = ζ Rez3

u = ζ2

Figure note. *Figure 14.1: The maximizing sectors in the z3-plane. Their boundaries are the three switching walls W.*

14.3 Switching functions are cubic polynomials

Suppose the current control is uk = ζk and compare it with uj = ζj. Define

1 χkj(t) = √ 3Re(uk −uj, z3(t)).

Substituting Equation (14.6) shows

t2 t3 χkj(t) = c0 + c1t + c2 + c3 , 2 6 with coefficients determined by z03, z02, z01, uk. The leading coefficient is nonzero because uk ≠ uj.

The first positive switching time is the least positive root of one of two competing cubics. If a root is simple, it depends smoothly on the initial data. Discontinuities occur only when:

• a switching cubic has a multiple root, detected by its discriminant; • two candidate switching cubics share a root, detected by their resultant.

This algebraic observation drives the later cell decomposition.

14.4 Choosing the first control lexicographically

At a switching point, two or even three control projections may tie at t = 0. To determine which control is optimal for small positive time, compare successively the coefficients of

1 z3(t) = z03 + z02t + 1t2 −iu t3. 2z0 6 For each candidate u, form the real vector

v(z0, u) = Re(z03, u), Re(z02, u), Re(z01, u) .

Choose the lexicographically largest vector. If two controls remain tied through these three entries, the cubic control term breaks the tie in a prescribed cyclic direction.

This rule guarantees that every nonzero switching state has a well-defined first control. It also gives a uniform estimate: if z0j ≤rj,

then the first switching time is less than 10r. Thus switching times shrink linearly with the weighted radius near the singularity.

14.5 Symmetries

The triangular Fuller system has three important symmetries.

Weighted scaling

For $r > 0$, $z_j(t) \mapsto r^j z_j(t/r)$

preserves the equations and Hamiltonian condition.

Discrete rotation

$$\text{Multiplication by } \zeta \text{ sends } (z_1, z_2, z_3, u) \mapsto (\zeta z_1, \zeta z_2, \zeta z_3, \zeta u).$$

Time reversal

$$\text{Define } \tau(z_1, z_2, z_3) = (\bar{z}_1, -\bar{z}_2, \bar{z}_3).$$

If $z(t)$ is a trajectory with control $u(t)$, then

$\tau(z(-t))$

is a trajectory with control $\bar{u}(-t)$.

Scaling and discrete rotation form the triangular virial group. Time reversal exchanges inward and outward chattering.

14.6 No nontrivial singular arcs

Allow temporarily controls in the convex triangle conv(V_T). If the Hamiltonian is maximized by a positive-dimensional face throughout an interval, then the source proves that

$$z_1 = z_2 = z_3 = 0, u = 0$$

almost everywhere. Thus the origin is the only singular Fuller arc.

A finite constant-control segment starting at a nonzero wall point cannot reach the origin. Therefore any approach to the origin requires infinitely many switches, just as in the Reinhardt system.

14.7 Weighted blowup

Define the weighted norm

$1/6\ \phi(z_1, z_2, z_3) = |z_1|^6 + |z_2|^3 + |z_3|^2$.

It satisfies $\phi(rz_1, r^2z_2, r^3z_3) = r\phi(z_1, z_2, z_3)$.

Let $\Xi = \left\{ \xi \in \mathbb{C}^3 : \phi(\xi) = 1 \right\}$.

Every nonzero state has unique weighted polar coordinates

$$z_1 = r\xi_1, \; z_2 = r^2\xi_2, \; z_3 = r^3\xi_3, \; r = \phi(z) > 0.$$

The oriented weighted blowup is $[0, \infty) \times \Xi$.

The exceptional divisor is $\{0\} \times \Xi$.

Restrict to switching points $\Xi_W = \{\xi \in \Xi : \xi_3 \in W\}$ and quotient by the discrete rotation V_T.

14.8 The Poincaré map

Given a switching state $q = (r, \xi)$, follow the triangular Fuller flow with the first control prescribed by the maximum principle until the next switching time. Rewrite the result in weighted polar coordinates. This defines

$$F : (0, \infty) \times \Xi_W / V_T \longrightarrow (0, \infty) \times \Xi_W / V_T .$$

Scaling equivariance implies that the angular component is independent of r:

$F_{ang} : \Xi_W / V_T \to \Xi_W / V_T$.

The Hamiltonian is conserved, so we restrict to the compact three-dimensional section

$$\Xi_{W,0}/V_T = \{\xi \in \Xi_W : H_F(\xi) = 0\} / V_T .$$

14.9 Self-similar fixed points

A fixed angular point means that after one switch the state is obtained from the initial state by weighted scaling and discrete rotation:

$$z_j(t_{sw}) = r^j \zeta_1 z_j(0).$$

Solving the constant-control equations and the maximum conditions leads to a palindromic polynomial $p(r) = 1 - 5r - 7r^2 - 5r^3 - 7r^4 - 5r^5 + r^6$.

It has exactly two positive real roots, reciprocal to each other:

$$r_{scale} \approx 6.2741673995, \; r^{-1}_{scale} \approx 0.1593836977.$$

They give the outward and inward fixed points qout, qin = τ(qout).

After one convenient virial normalization, the outward point begins with approximately z1 = −1 + 0.418610i, z2 = −1√ + 0.131590i, z3 = −0.216292, 3

and the first switching time is tsw ≈7.40352

in the normalized Fuller scale.

Theorem

Theorem 14.1 (Fixed points of the angular Fuller map). On the zero-Hamiltonian Poincaré section, Fang has exactly two fixed points: the outward and inward triangular spirals qout and qin.

The iterates at the outward point satisfy

F n(qout) = (rscale, ζ2)n · qout,

so the physical switching points move outward. For qin, the reciprocal scaling makes them move inward toward the singularity.

14.10 Local stability

In local real coordinates on the three-dimensional angular section, the Jacobian of Fang at qout has all eigenvalues of absolute value less than 0.1 according to the source computation. Therefore qout is a strongly asymptotically stable fixed point. Time reversal makes qin unstable in forward time.

Point requiring care

Local stability does not imply a global basin. The Poincaré map is only piecewise continuous because the active cubic switching polynomial can change. The hardest discrete-dynamical theorem is to prove that every other point eventually enters the local basin of qout.

Chapter summary

The true leading-order system has three discrete control phases. Its switching functions are cubic, its singular locus is only the origin, and weighted blowup produces a compact three-dimensional Poincaré section. The angular recurrence map has exactly two self-similar fixed points, an outward stable triangular spiral and its inward time reverse. The next task is to prove that the outward point attracts the entire section except for the inward point.

Part V

Blowup, recurrence, and the global basin theorem

Chapter 15

The blown-up Poincaré section and its cells

15.1 Why the angular section is three-dimensional

The triangular Fuller state (z_1, z_2, z_3) has six real coordinates. The zero-Hamiltonian equation removes one. Quotienting by positive weighted scaling removes another, and quotienting by the discrete rotation only identifies finitely many points. Restricting to a switching wall removes one more real dimension. Therefore the Poincaré section

$$\Xi_{W,0}/V_T$$

is three-dimensional.

To analyze a piecewise-defined recurrence map globally, we need explicit coordinates and a finite partition into regions where one switching polynomial determines the next switch continuously.

15.2 Coordinates on a wall

$$\text{Write } z_j = r_j e^{i\theta_j}, \; j = 1, 2, 3.$$

$$\text{Using the discrete rotation, assume } z_3 = x_3 \leq 0$$

is real. The zero-Hamiltonian relation becomes

$$x_3 = 2r_1r_2 \sin(\theta_1 - \theta_2) = -2r_1r_2 \sin\psi, \quad \psi = \theta_2 - \theta_1.$$

When $x_3 < 0$, one has $r_1r_2 > 0$, $0 < \psi < \pi$.

$$\text{Weighted scaling acts by } r_1 \mapsto r_1/s, \; r_2 \mapsto r_2/s^2.$$

There is a unique $s > 0$ for which $r1/s + r2/s2 = 1$.

After this normalization, rename the scaled radii and write

$$r1 = 1 - r2, \ 0 \le r2 \le 1.$$

Thus a typical point is described by $(r2, \psi, \theta 2)$.

15.3 The two big cells

For $x3 \neq 0$, the sign of Imz2 determines the first control.

Definition

Definition 15.1 (Big cells). The first big open cell is

$$C3(\zeta)^{\circ} = \{0 < r2 < 1, 0 < \psi < \pi, 0 < \theta 2 < \pi\} .$$

Its first control is ζ. The second is

$$C3(\zeta 2)^{\circ} = \{0 < r2 < 1, 0 < \psi < \pi, -\pi < \theta 2 < 0\} ,$$

with first control $\zeta 2$.

Each is an open rectangular box in $(r2, \psi, \theta 2)$ coordinates. Their complement consists of lowerdimensional boundary strata.

$$\text{Time reversal acts by } \ \tau(r2, \psi, \theta 2) = (r2, \pi - \psi, \pi - \theta 2)$$

on the first cell, with the corresponding $-\pi - \theta 2$ branch on the second.

15.4 Two competing switching polynomials

Within a big cell, the current control u can switch to either neighboring control. Denote the corresponding switching cubics by

$$\chi A = \chi u,u/\zeta, \ \chi B = \chi u,\bar{\ }u.$$

At the initial wall, one of these has a zero at $t = 0$. Remove its known factor and write

$$\chi A \ \chi B \ \ \chi A,mA = , \ \chi B,mB = . \ tmA \ tmB$$

The next switching time is the least positive root among the two reduced polynomials.

The identity of the active polynomial can change only on algebraic loci:

• $\Delta A = 0$, where χA has a multiple root; • $\Delta B = 0$, where χB has a multiple root; • $\mathrm{Res}(\chi A, \chi B) = 0$, where they share a root.

These loci are surfaces inside the big cells. Crossing one can make the least positive root jump from one branch to another.

Background interlude

For a polynomial p, the discriminant vanishes exactly when p and p′ share a root. The resultant of p and q vanishes exactly when they share a complex root. Both are polynomial functions of the coefficients, hence analytic functions of the cell coordinates.

15.5 Boundary strata and zero switching time

On parts of the boundary, the limiting first switching time is 0. One must not simply identify the two sides, because the first controls can differ and the Poincaré map has a nontrivial limiting action. The source attaches lower-dimensional cells to the faces so that the dynamics extends continuously wherever possible.

> The two-dimensional cells with $x_3 \neq 0$ have $\mathrm{Im} z_2 = 0$ and a double root at $t = 0$ for one switching polynomial. Further two-cells occur when $z_3 = 0$. Together they cap the unfilled faces of the big cells.

Topologically, every point of the Poincaré section is represented in the union of the closures of the two big cells, with specified identifications on their faces.

15.6 The time-reversal involution

The recurrence map F and time reversal satisfy

$$F^{-1} = \tau \circ F \circ \tau.$$

Therefore $\iota = \tau \circ F$

is an involution: $\iota^2 = \mathrm{id}$.

This converts information about images into information about preimages and makes the discontinuity surfaces geometrically manageable.

For example, a boundary two-cell on which a switching polynomial has a multiple zero is mapped by ι to the corresponding discriminant surface inside a big cell. A boundary where both candidate polynomials share a switching time maps to a resultant surface.

15.7 The five main parts D0, . . . , D4

The discriminant and resultant surfaces partition the two big cells into five regular closed pieces.

• D1 lies in the first big cell and contains both fixed points qin and qout. The active switching polynomial is χA. The piece is invariant under ι. • D2 lies in the second big cell, also with active polynomial χA, and is invariant under ι. • D3 is the remaining part of the second big cell, with active polynomial χB.

Figure note. *Figure 15.1: Boundary regions of the two big cells where the limiting first switching time is zero. This is the source paper's computational visualization (source Figure 15.1.1), included to orient the cell attachment.*

• D4 = ι(D3) lies in the first big cell. • D0 is a very small residual bubble in the first big cell, bounded by pieces of the discriminant surfaces. It is invariant under ι.

On the interior of each Di, the first switching time is given by one continuous root branch. The map extends continuously to a function

Fi : Di →ΞW,0/VT .

(a) Partition of the first big cell. (b) Partition of the second big cell.

Figure note. *Figure 15.2: Computational visualizations of the geometric partition, adapted directly from source Figures 15.3.1 and 15.3.2. The surfaces are discriminant and resultant loci for cubic switching polynomials.*

15.8 Where the fixed points lie

In the (r2, ψ, θ2) coordinates, the inward fixed point is approximately

qin ≈(0.267949, 0.170594, 2.91574).

The outward point is its time reverse:

qout ≈(0.267949, π −0.170594, π −2.91574).

Both lie in D1. The inward point is extremely close to the triple juncture of D0, D1, D4, while the outward point lies well inside the regular part of D1. This asymmetry explains why the global basin proof is much easier near qout than near qin.

15.9 Continuity is enough

The Poincaré map is not globally continuous on the unpartitioned section. The proof does not require it to be. What is needed is:

1. a finite cover by closed pieces Di; 2. a continuous extension Fi on each piece; 3. explicit information about which pieces can contain Fi(Di).

This is analogous to a piecewise-smooth map in one-dimensional dynamics, except that the pieces are three-dimensional and the boundaries are algebraic surfaces.

Chapter summary

After normalizing scaling, the switching section is covered by two three-dimensional boxes. The next switch is the least positive root of one of two cubic polynomials. Discriminants and resultants locate every possible discontinuity. Time reversal turns the recurrence into an involution and organizes the section into five closed pieces D0, . . . , D4, on each of which the Poincaré map extends continuously. This finite partition is the platform for the global basin theorem.

Chapter 16

Inward and outward fixed points

16.1 Fixed angular profile versus physical motion

A fixed point of the angular Poincaré map does not mean the physical Fuller state is stationary. It means that one switching step changes only the scale and the discrete phase. If q is fixed angularly, then $F(r, q) = (\lambda r, q)$

for some scale factor $\lambda > 0$, after quotienting by the cyclic rotation.

Thus:

• $\lambda > 1$ gives an outward self-similar spiral; • $0 < \lambda < 1$ gives an inward self-similar spiral.

For the triangular Fuller system, $\lambda_{out} = r_{scale} \approx 6.27417$, $\lambda_{in} = r^{-1}_{scale} \approx 0.159384$.

16.2 Solving the self-similarity equations

Choose a first control $u \in V_T$ and a switching time $T > 0$. The constant-control solution is polynomial in T. Self-similarity requires $z_j(T) = r^j \zeta_1 z_j(0)$, $j = 1, 2, 3$,

for some $r > 0$ and $\zeta_1 \in V_T$. These are linear equations in the initial data once (r, ζ_1, T, u) are fixed.

Solving the chain equations for $z_3(0)$ gives a rational expression. Imposing:

• the switching-wall condition at time 0; • the switching-wall condition at time T; • Hamiltonian maximization at both endpoints; • the zero-Hamiltonian equation;

reduces the scale to the roots of

$1 - 5r - 7r^2 - 5r^3 - 7r^4 - 5r^5 + r^6 = 0.$

The palindromic coefficients imply that if r is a root, so is 1/r. The two positive roots give q_{out} and q_{in}.

A third formal fixed point with r = 1 appears before the zero-Hamiltonian condition is imposed. Its Hamiltonian is nonzero, so it is irrelevant to the singular extremal problem. It nevertheless arises as a limiting angular profile of the known smoothed (6k + 2)-gon extremals as they converge to the circle. This is a useful warning: taking a geometric limit of Reinhardt extremals need not preserve the leading-order Fuller Hamiltonian normalization.

16.3 Local contraction at qout

Choose local coordinates in D1 centered at qout. Differentiate the analytic branch of the next-switch map. The source computes the 3 × 3 Jacobian and finds all three eigenvalues to have modulus less than 0.1. Hence there is a neighborhood Uout such that

F n(q) →qout

for every q ∈Uout.

The small eigenvalues mean the contraction is strong. This numerical fact is later useful when tracking the unstable manifold of the full Reinhardt map: errors transverse to the one expanding direction decay rapidly.

16.4 Time reversal and qin

The involution $\tau(z1, z2, z3) = (\bar{z}1, -\bar{z}2, \bar{z}3)$

conjugates forward and backward dynamics. It exchanges the two fixed points:

$\tau(qout) = qin.$

Therefore qin is unstable in forward angular time and stable in backward angular time.

For a physical trajectory, however, inward motion means the radial scale decreases under forward iteration. The terminology can therefore be confusing:

• qin is radially contracting but angularly unstable in forward time; • qout is radially expanding but angularly stable in forward time.

The full four-dimensional Reinhardt map will combine these radial and angular directions.

16.5 Stable and unstable directions after adding the radius

At qout, the angular map has three contracting eigenvalues, while the radial multiplier is

rscale > 1.

Thus the full blown-up map has:

• a three-dimensional stable space tangent to the exceptional divisor; • a one-dimensional unstable space transverse to the divisor.

At qin, time reversal exchanges them:

• a one-dimensional stable direction transverse to the divisor; • a three-dimensional unstable space tangent to the divisor.

This dimension count is the heart of the final rigidity argument. A true trajectory entering the singularity must lie on the one-dimensional stable manifold of qin.

16.6 What the global basin theorem must add

> Local hyperbolicity says nothing about an angular point far from the fixed points. The global basin theorem will show $q \neq$ qin $\Longrightarrow F^n(q) \to$ qout.

This implies a complete classification of Fuller trajectories emanating from or arriving at the singularity: an inward trajectory can have only the angular profile qin, and an outward one only qout.

Chapter summary

The two fixed points are self-similar switching profiles with reciprocal scale factors. The outward profile is a strong angular attractor, while the inward profile is its time reverse. After restoring the radial coordinate, qout has three stable angular directions and one unstable radial direction; qin has the opposite splitting. The remaining global task is to show that no other angular recurrent behavior exists.

Chapter 17

Extending the Reinhardt map across the exceptional divisor

17.1 The Fuller system is only the leading term

The global basin theorem concerns the truncated Fuller dynamics on the exceptional divisor. The actual minimizer follows the exact Reinhardt state–costate equations. To transfer the Fuller conclusions, we need to show that the true switching map has a regular extension to the blowup and that its restriction to the exceptional divisor is exactly the Fuller Poincaré map.

Fix the parameter values forced by a trajectory meeting the singular locus:

9 λcost = −1, det Λ1 = 4, ρ = 2.

The singular point is X = J, Λ1 = −3 ΛR = 0. 2J,

17.2 Weighted real coordinates

Near the singular point, write

X = J + r eX, (17.1) Λ1 = −3 + r2 eΛ1, 2J (17.2) ΛR = r3 eΛR. (17.3)

The powers 1, 2, 3 match the orders found in the Fuller truncation. To make this representation unique, normalize ex11 = 1. Then r itself is the (1, 1) coefficient of X −J.

Let ex = (ex21, el11, el21) ∈R3.

The constraints 9 det X = 1, det Λ1 = 4 determine the remaining entries of eX and eΛ1. The three linear equations

⟨X, ΛR⟩= 0, H = 0, χ23 = 0

determine eΛR. Therefore (r, ex) ∈R4 is a local coordinate system on the switching section near the outward fixed point.

The exceptional divisor is simply r = 0.

17.3 Matching to Fuller angular coordinates

Apply the Cayley transform and hyperboloid coordinates to the triple (X, Λ1, ΛR), obtaining complex variables (wr, br, cr). Define the Fuller-scaled quantities

wr cr = , = −ibr , = bz1 ρ bz2 2ρ bz3 6ρ.

Their expansions are

r bz1(r, ex) = 2(ex21 + i) + O(r2), r2 bz2(r, ex) = 3! (el11 −iel21) + O(r3), r3 bz3(r, ex) = 4! (elR,12 + elR,21) + O(r4).

Hence the limit = lim bz1 , bz2 , bz3 z(ex) r→0 r r2 r3

exists and is nonzero. Its weighted angular component is a point

ξ(ex) ∈ΞW,0. At the parameter exout, this is the Fuller fixed point qout modulo virial symmetry. The exact Reinhardt Hamiltonian has the asymptotic expansion

HRein(r, ex; u) = 24r3HF (z(ex); eu) + O(r4), where the three Reinhardt vertices correspond to 1, ζ, ζ2.

17.4 Matching the constant-control flows

Use rescaled time t s = . r0 Let (r(s), ex(s)) be the exact Reinhardt solution with a fixed vertex control and initial condition (r0, ex0). Let z(s) be the corresponding Fuller solution starting at z(ex0). Then for bounded s, bz1(r(s), ex(s)) = r0z1(s) + O(r20), bz2(r(s), ex(s)) = r20z2(s) + O(r30), bz3(r(s), ex(s)) = r30z3(s) + O(r40).

The exact switching functions satisfy, for example,

χRein21 (sr0) = 24r30Re(ζ −1, z3(s)) + O(r40).

Thus after dividing by r30, the exact switching equation converges to the Fuller cubic.

17.5 Analytic continuation of the switching time

Near qout, the relevant Fuller switching cubic has a least positive root

ssw,out ≈8.84

that is simple and is separated from 0. The exact normalized switching function

χRein(sr0, r0, ex0) r0, = eχ(s, ex0) r30

extends analytically to r0 = 0. At

(s, r0, ex0) = (ssw,out, 0, exout),

one has = 0, ∂eχ ̸= 0. eχ ∂s The analytic implicit-function theorem therefore supplies an analytic switching-time function

ssw = ssw(r0, ex0)

for positive, zero, and even small negative r0.

Evaluate the exact constant-control solution at

tsw = r0ssw(r0, ex0)

and rotate by the appropriate power of R so that the next wall is represented in the same coordinate chart. Converting back to (r, ex) gives an analytic map FRein : (r, ex) 7−→(r′, ex′).

Theorem

Theorem 17.1 (Analytic extension; source Lemma 14.3.1). The Reinhardt Poincaré map extends to an analytic local diffeomorphism in a neighborhood of (0, exout) ∈R4. Its restriction to r = 0 is the Fuller angular Poincaré map.

The inverse is obtained by repeating the construction backward in time. Since forward and backward extensions agree as inverses for r > 0, analytic continuation makes them inverse near r = 0 as well.

17.6 Hyperbolic fixed points and invariant manifolds

At qout, the three angular eigenvalues have modulus less than 1, while the radial multiplier is

rscale > 1.

Thus the fixed point is hyperbolic. The stable-manifold theorem gives:

Theorem

Theorem 17.2 (Local invariant manifolds at qout). The local stable manifold is the three-dimensional exceptional divisor r = 0. The local unstable manifold is a one-dimensional smooth curve transverse to the divisor.

By time reversal:

Theorem

Theorem 17.3 (Local invariant manifolds at qin). The local stable manifold of qin is a onedimensional curve transverse to the exceptional divisor, and its local unstable manifold is the three-dimensional divisor.

17.7 The global unstable curve from qout

The local one-dimensional unstable manifold can be continued by iterating the exact Reinhardt map. The source uses r itself as a parameter and numerically follows

> r 7−→(r, ex(r)). The curve begins at ex21 ≈−2.39 and reaches the boundary of the star domain near

(r, ex21) ≈(0.21, −3.03). For r above roughly 0.065, the continuous Reinhardt trajectory hits the star-domain boundary before another switch occurs. Thus the unstable curve does not turn back toward the exceptional divisor.

Geometrically, the corresponding upper-half-plane trajectories form outward triangular spirals that chatter near z = i and then expand until they strike the star-domain boundary.

Theorem

Theorem 17.4 (No return along the unstable curve; source Theorem 14.4.1). A Reinhardt trajectory that emanates from qout on the exceptional divisor does not return to the divisor. By time reversal, a trajectory tending to qin did not emanate from the divisor at an earlier time.

Figure note. *Figure 17.1: The source paper's schematic of the blown-up Reinhardt dynamics. The exceptional divisor carries the Fuller dynamics from qin toward qout; a one-dimensional curve enters at qin and another exits at qout. Source Figure 14.0.1.*

Point requiring care

The continuation of the global unstable curve in the source uses high-resolution Mathematica numerics rather than a fully presented interval-arithmetic certificate. The dynamical conclusion is used as an input to the final proof. Chapter 21 separates this numerical component from the analytic local theory.

17.8 Why this matters for periodicity

If a periodic trajectory were to chatter into the singular locus along the stable curve of qin, periodicity would require it to have emerged from the singular locus at an earlier time. Time reversal of the no-return theorem says that this is impossible. The only missing step is to prove that every finite-time cluster on the divisor is indeed qin and lies on its stable curve. That conclusion will use the global basin theorem and the cluster-point argument.

Chapter summary

Weighted Lie-algebra coordinates make the exact Reinhardt Poincaré map analytic across the blowup. After rescaling time and dividing the switching function by r3, the map restricts at r = 0 to the triangular Fuller recurrence. The outward fixed point has a one-dimensional unstable curve leaving the divisor; the inward point has a one-dimensional stable curve entering it. Numerical continuation shows the outward curve hits the star-domain boundary and never returns, a fact that will contradict periodic chattering.

Chapter 18

The global basin theorem

18.1 Statement

The central discrete-dynamical result is:

Theorem

Theorem 18.1 (Global basin; source Theorem 16.1.1). Let $q \in \Xi_{W,0}/V_T$. If $q \neq q_{in}$, then

---

$$F^n(q) \longrightarrow q_{out} \ (n \to \infty).$$

The theorem rules out periodic, quasiperiodic, or chaotic angular switching patterns on the exceptional divisor. There is one exceptional backward attractor q_{in}; everything else is drawn to q_{out}.

18.2 Why local stability is insufficient

The map is three-dimensional and piecewise continuous. Even a strongly attracting fixed point can coexist with other invariant sets far away. A global proof must force every orbit through a finite sequence of regions into the local contraction neighborhood.

The source accomplishes this by a finite containment graph. The geometric partition $D_0, \ldots, D_4$ makes each branch continuous. Then D_1, the piece containing both fixed points, is subdivided into smaller rectangular boxes.

18.3 The coarse containment table

Let $D_{in} \subset D_1$

be a small rectangular box containing q_{in}, and

$D_{out} \subset D_1$

be a box containing q_{out}. The verified coarse inclusions are summarized by the following table. A star means that the row-set may map into the column-set.

Din Dout D4 D2 D0 D3 D1 \ Din

Din ∗ ∗ ∗ ∗ ∗ ∗ ∗ Dout ∗ D4 ∗ ∗ D2 ∗ D0 ∗ D3 ∗ D1 \ Din ∗

The first row imposes no restriction near the unstable inward point. Every other row is upper triangular in the displayed ordering.

The inclusions have conceptual explanations from the involution:

$$F(D_4) = \tau(D_3) \subset D_2 \cup D_3,\quad F(D_2) = \tau(D_2) \subset D_3,\quad F(D_0) = \tau(D_0) \subset D_1 \setminus D_{in},\quad F(D_3) = \tau(D_4) \subset D_1 \setminus D_{in}.$$

The contraction inclusion $F(D_{out}) \subset D_{out}$

is checked directly.

18.4 The boundary containment method

On each D_i, the selected switching root and the resulting map F_i are continuous. The sets used for containment are boxes or regular closed cells described by analytic inequalities. The source evaluates the images of their boundary faces and encloses those images in convex target regions.

The computational principle is:

1. express the constant-control solution and switching time analytically; 2. restrict the initial data to one face of a cell; 3. calculate or sample the resulting image surface; 4. verify that each supporting half-space defining the target contains the image; 5. use the branch's topological regularity to conclude containment of the full cell.

The source calls this the boundary method. It is a finite-dimensional analogue of interval-arithmetic containment, though the presentation uses Mathematica rather than a full interval package.

18.5 A strict order inside D1

The coarse table leaves $D_1 \setminus D_{in}$ invariant, so a finer descent is needed. Divide the coordinate ranges into thirds:

$$0 = a_0 < a_1 = \frac{\pi}{3} < a_2 = \frac{2\pi}{3} < a_3 = \pi,$$

Figure note. *Figure 18.1: The image of the box Dout lies strictly inside it and contracts toward qout. Source Figure 16.1.1.*

$$\pi = b_0 > b_1 = \pi - 1.1 > b_2 = 1.1 > b_3 = 0.$$

Define boxes $D_{ij} = \{0 \le r_2 \le 1,\ a_i \le \psi \le a_{i+1},\ b_{j+1} \le \theta_2 \le b_j\}$, and $D_{1,ij} = D_1 \cap D_{ij}$, $i, j \in \{0, 1, 2\}$.

Then $q_{out} \in D_{1,00} = D_{out}$, $q_{in} \in D_{1,22} = D_{in}$.

Order the index pairs lexicographically:

$(k, l) < (i, j) \iff k < i$ or ($k = i$ and $l < j$).

The verified refinement is $F(D_{1,ij}) \subset \bigcup_{(k,l)<(i,j)} D_{kl}$

after excluding $(i, j) = (0, 0), (2, 2)$. Therefore every orbit in $D_1 \setminus D_{in}$ strictly descends through a finite ordered list of boxes until it enters D_{out}.

Geometric intuition

The lexicographic pair (i, j) is a discrete Lyapunov function. It need not decrease at every point of the original continuous flow, but it decreases at every Poincaré step once the orbit is in $D_1 \setminus D_{in}$.

18.6 Representative computed images

The source displays full three-dimensional images of two difficult boxes. Their projections are contained in unions of earlier boxes, providing the strict descent.

18.7 Escaping the inward box

The inward fixed point is unstable in forward angular time. However, an arbitrary point in D_{in} could in principle remain there for many iterates. Time reversal resolves this.

Suppose $q \in D_{in}$ and $q \neq q_{in}$. Apply the time-reversal involution. The point $\tau(q)$ lies near q_{out} for the inverse dynamics. Because q_{out} is locally attracting forward, it is locally repelling backward. Therefore the backward orbit of $\tau(q)$ eventually leaves D_{out}, which translates into the forward orbit of q eventually leaving D_{in}.

Once the orbit exits D_{in}, the coarse upper-triangular structure sends it after finitely many steps into $D_1 \setminus D_{in}$. The refined lexicographic structure then sends it into D_{out}.

(a) A computed image corresponding to one refined box. (b) A second computed image.

Figure note. *Figure 18.2: Representative images used in the containment verification, from source Figure 16.1.4. The purpose is not visual aesthetics but certified range control for the recurrence map.*

18.8 Convergence inside Dout

The inclusion $F(D_{out}) \subset D_{out}$

keeps all subsequent iterates inside the contraction box. The Jacobian at qout has spectral radius less than 0.1, and the verified image is already substantially smaller than the box. Standard contraction or local asymptotic-stability theory gives

$F^n(q) \to q_{out}$.

This completes the proof of the global basin theorem.

18.9 Classification of Fuller trajectories reaching the singularity

Restore the radial coordinate. Suppose a Fuller trajectory reaches the singularity in forward time. Let q be any of its angular switching points. If $q \neq q_{in}$, the global basin theorem says its forward angular iterates approach qout. But qout has radial multiplier greater than 1, so the physical trajectory eventually moves outward, not inward. Contradiction.

Therefore every inward Fuller trajectory has all switching points on the ray over qin. By time reversal, every outward trajectory has all switching points over qout.

Theorem

Theorem 18.2 (Classification of singular Fuller trajectories). Every triangular Fuller trajectory emanating from the singularity is the outward self-similar triangular spiral, up to virial symmetry. Every trajectory arriving at the singularity is the inward self-similar triangular spiral.

18.10 What has been achieved

The complicated angular Poincaré map has no hidden recurrent set. Its entire global dynamics is organized by two fixed points and a finite transient graph. This rigidity is what makes the cluster-point theorem for the exact Reinhardt map possible.

Point requiring care

The global basin theorem is the most substantial computer-assisted step. Its logical implication from the containment relations is elementary and fully explained above. The burden of computation is the verification of those inclusions for the analytic branches of the switching map.

Chapter summary

A finite geometric partition makes the Fuller Poincaré map continuous branch by branch. A coarse upper-triangular containment table moves every orbit toward D1, while a 3×3 box refinement inside D1 supplies a strict lexicographic descent to the contraction box around qout. Time reversal excludes trapping near qin. Therefore qout attracts every angular point except qin, and the only Fuller trajectories reaching or leaving the singularity are the two

self-similar triangular spirals.

Chapter 19

The cluster-point theorem for the exact Reinhardt map

19.1 The statement

Let $q_0, q_1, q_2, \ldots$

be consecutive switching points of an exact Reinhardt Pontryagin extremal in the blown-up switching section. Assume:

• each q_n has positive radial coordinate $r_n > 0$; • the switching times accumulate at a finite physical time; • a subsequence of q_n converges to the exceptional divisor $r = 0$.

The key theorem is:

Theorem

Theorem 19.1 (Cluster-point theorem; source Theorem 16.4.1). Every such cluster point is q_{in}, and the switching points lie on the one-dimensional stable manifold

---

$W^s(q_{in})$

of the exact Reinhardt Poincaré map.

This upgrades the leading-order Fuller classification to the exact nonlinear system.

19.2 Why exact switching functions are still cubic-like

In rescaled time $s = t/r$, an exact Reinhardt switching function has an analytic expansion

$$\chi_{Rein}(s, r, \xi) = \chi_F(s, \xi) + O(r), \quad r^3$$

where χ_F is a cubic Fuller switching polynomial. The convergence includes derivatives with respect to s on compact sets.

For small r, the third derivative therefore has the same nonzero sign as the Fuller cubic's leading coefficient. Consequently the exact switching function has at most three real zeros and the same qualitative shape as a cubic: at most one inflection point and at most two critical points.

For a precise analytic factorization, the source invokes Weierstrass preparation.

Standard theorem used as a black box

Weierstrass preparation, specialized form. Let f(s, p) be real analytic near (s0, p0), and suppose s 7→f(s, p0) has a zero of multiplicity m at s0. Then near (s0, p0),

f(s, p) = U(s, p)W(s, p),

where U is analytic and nonzero, and

W(s, p) = (s −s0)m + am−1(p)(s −s0)m−1 + · · · + a0(p)

is a monic polynomial whose coefficients are analytic in p.

Apply this near each root of the limiting Fuller cubic and multiply the local Weierstrass polynomials. One obtains a global cubic polynomial in s whose coefficients depend analytically on (r, ξ) and whose real roots are exactly the relevant switching roots near the divisor.

Thus the discriminants, resultants, multiplicities, active-root choices, and cell definitions from the Fuller system extend analytically to the exact Reinhardt system.

19.3 Thickening the cell partition

Let Di be a Fuller cell on the divisor. For sufficiently small r, define a thickened exact cell DRi

> using the same sign conditions on the exact cubic coefficients, discriminants, resultants, and active switching roots. Then DRi ∩{r = 0} = Di. On each DRi , the exact Poincaré map has a continuous branch F iR : DRi →blown-up section

that restricts at r = 0 to the Fuller branch Fi.

The refined boxes D1,ij are thickened similarly. The exact definitions away from r = 0 are not unique; what matters is that they agree on the divisor and preserve the finite containment order for sufficiently small r.

19.4 Finite time converts a cluster point into a limit near a hyperbolic fixed point

Suppose qin is a cluster point. Near qin, the exact map has one contracting eigenvalue transverse to the divisor and three expanding eigenvalues tangent to it. A point whose forward iterates remain near qin and converge to it must lie on the local stable manifold.

Could the orbit repeatedly visit a small neighborhood and then leave? The finite physical-time hypothesis excludes this. The unscaled time between switches is approximately

$\Delta tn = rn\ \Delta sn.$

Near the fixed point, the scale changes by approximately the factor r−1scale. Traveling from outside an ε-neighborhood to inside an ε/rscale-neighborhood consumes a definite fraction of the current radial scale. Infinitely many large excursions would contribute a divergent or nonvanishing amount of time. Hence once a finite-time orbit enters sufficiently deeply, it must remain near the fixed point.

Therefore a cluster at qin is actually convergence to qin, and the orbit lies on

W s(qin).

19.5 Why qout cannot be a cluster point

At qout, the local stable manifold of the exact four-dimensional map is exactly the exceptional divisor. If a positive-radius switching orbit converged to qout, it would have to lie in this stable manifold, forcing rn = 0

for all sufficiently large n. This contradicts the assumption that the switching points themselves are off the divisor.

Thus qout is not a cluster point of an exact non-divisor orbit.

19.6 Descending the set of possible cluster points

Assume for contradiction that a cluster point q exists on the divisor and

$q \neq$ qin, qout.

Assign q to one of the Fuller pieces D in the finite geometric partition, choosing a subsequence qnk that lies in the corresponding thickened piece DR.

Continuity of the branch gives F R(qnk) $\longrightarrow$F(q). The global basin containment relations imply that F(q) lies in a piece eD that is strictly lower in the finite upper-triangular order, unless D is already the outward contraction region. Passing to a further subsequence, the exact images lie in eDR and have a cluster point in eD. Repeat. Because the order is finite and strictly descending, one eventually produces a cluster point in Dout.

Within the outward box, the exact map is analytic and the local structure forces the closed cluster set to contain qout itself. But qout has already been excluded.

This contradiction proves that no cluster point other than qin exists.

19.7 Why the orbit lies on the stable curve

We have shown that the switching points converge to qin. By definition, the stable set is

$$W^s(q_{in}) = \{q : F^n(q) \to q_{in}\} .$$

Near a hyperbolic fixed point, the stable set is the one-dimensional stable manifold. Once a tail of the orbit lies in the local stable manifold, invariance under the diffeomorphism propagates the conclusion backward along all switching points in the considered chattering segment.

Thus the entire switching sequence approaching the singularity lies on the stable curve.

19.8 The logical dependence on the Fuller basin theorem

The cluster theorem uses the global basin theorem only through a finite descent property on the divisor. It does not require exact Reinhardt trajectories to shadow Fuller trajectories for arbitrarily long continuous time. The analytic blowup and cell thickening give enough continuity to transfer the combinatorial order of possible cluster regions.

Geometric intuition

A cluster point is a possible "asymptotic direction" of chattering. The Fuller basin theorem says that every direction except the inward one is eventually carried toward the outward direction. But a true positive-radius orbit cannot converge to the outward fixed point because its stable manifold lies entirely on the divisor. Therefore the only direction available to an exact orbit entering the divisor is the inward stable curve.

Chapter summary

Weierstrass preparation turns exact Reinhardt switching functions near the divisor into analytic cubic polynomials with the same root structure as the Fuller cubics. This thickens the Fuller cell partition to the exact system. Any non-inward cluster direction descends through the finite containment order to the outward box, where positive-radius convergence is impossible because the stable manifold is the divisor. Hence every finite-time chattering sequence converges to qin along its one-dimensional stable manifold.

Chapter 20

The proof of Mahler's First conjecture

20.1 The theorem

Theorem

Theorem 20.1 (Mahler's First conjecture). A global minimizer of the Reinhardt packing problem is a bang-bang solution with finitely many switches. Consequently its boundary is a finite-sided smoothed polygon whose rounded pieces are hyperbolic arcs of Reinhardt–Mahler type.

We now assemble the preceding results in a single proof.

20.2 Step 1: a minimizer is a periodic edge-extremal

Reinhardt's compactness theorem gives a global minimizer. The critical-hexagon reduction and matrix formulation produce an optimal trajectory

$$(g, X, \Lambda 1, \Lambda R, u)$$

satisfying the Pontryagin maximum principle.

Because the trajectory is globally minimizing, it is also minimizing under variations constrained to any edge of the control triangle. Hence it is edge-extremal.

The rotational endpoint conditions give

$$g(tf) = R,\ X(tf) = R{-}1X(0)R,$$

with the same conjugacy conditions for the costates. Extending by rotation makes the lifted trajectory periodic.

20.3 Step 2: if the trajectory avoids the singular locus, we are done

The finite-switching theorem away from the singular locus says:

edge-extremal + avoid Ssing =⇒ finite bang-bang.

Therefore it remains only to prove that the minimizing trajectory does not meet the singular locus.

20.4 Step 3: it cannot remain on the singular locus

A full singular arc is, up to affine transformation, a circular arc. The compactly supported second variation of Chapter 11 lowers its area while preserving endpoint data and convexity. Therefore a minimizer cannot remain on the singular locus during any interval of positive time.

20.5 Step 4: it cannot reach the singular locus with finitely many switches

A constant vertex-control extremal cannot meet the singular locus. Hence a finite concatenation of such segments cannot end there. Any contact with the singular locus must occur through infinitely many switches accumulating in finite time.

Thus the only remaining possibility is chattering.

20.6 Step 5: a chattering sequence has a cluster point on the divisor

Let q0, q1, q2, . . .

be the switching points of a chattering segment approaching a finite time t∗. In the weighted blowup, the exceptional divisor is compact. Therefore some subsequence has a cluster point on the divisor.

The switching points themselves have positive radial coordinate, because the exact trajectory has not yet reached the singular locus.

20.7 Step 6: the cluster theorem forces the inward stable curve

By the cluster-point theorem, the only possible cluster point is qin,

and the switching points lie on the one-dimensional stable manifold

W s(qin).

Thus every exact chattering trajectory entering the singular locus is forced into one specific curve in the four-dimensional blown-up switching section.

20.8 Step 7: the stable curve cannot be part of a periodic orbit

Time reversal identifies W s(qin) with the reverse of the unstable curve leaving qout. The no-return theorem says that the unstable curve from qout hits the boundary of the star domain and never returns to the exceptional divisor. Equivalently:

A trajectory tending to qin did not emanate from the exceptional divisor at any earlier time.

But a periodic trajectory that reaches the singular locus must, when followed one period backward, also have emerged from that same singular locus. This contradicts the no-return theorem.

Therefore the minimizing periodic trajectory cannot chatter into the singular locus.

20.9 Step 8: conclude finite bang-bang behavior

The minimizer neither stays on, reaches by finitely many switches, nor chatters into the singular locus. Hence it avoids the singular locus entirely. The finite-switching theorem applies, and the control is bang-bang with finitely many switches.

Each vertex-control interval gives two straight boundary arcs and one hyperbolic arc. Finitely many intervals give a finite cyclic concatenation of straight segments and hyperbolic arcs. The boundary has no corners and the hyperbolic arcs are tangent to the adjacent straight segments. Therefore the minimizer is a finite-sided smoothed polygon.

20.10 A dependency diagram

20.11 What the theorem does not yet prove

The theorem does not identify the number of sides. There may a priori be other finite smoothed polygons satisfying the Pontryagin conditions and endpoint conditions. To prove the full Reinhardt conjecture, one would need to classify the periodic finite-switching extremals or establish a global area comparison among them.

The source constructs the family of smoothed $(6k + 2)$-gon extremals and shows that among that family the smoothed octagon has the smallest area. It also proves that the smoothed octagon is an isolated extremal. What remains open is the exclusion of all other periodic extremals.

Chapter summary

The conclusion is obtained by exhaustion of possibilities. Regular motion is finite bang-bang. A singular interval is circular and not minimizing. A finite vertex sequence cannot reach the singularity. Any chattering approach must lie on the inward stable curve, but that curve is incompatible with the periodic boundary conditions of a minimizer. Hence the minimizer stays regular and is a finite smoothed polygon.

critical hexagons and balanced multicurves

matrix control system and area cost

Pontryagin extremal; edge-extremal

away from sin- singular locus = circle; gular locus: no singular arc finite bang-bang only remaining obstruction: finite-time chattering

weighted blowup and triangular Fuller map

global basin and cluster theorem

chattering stable curve is nonperiodic

periodic minimizer avoids singularity ⇒finite smoothed polygon

Figure note. *Figure 20.1: Logical dependency of the proof. The long dynamical argument is needed only to exclude chattering at the singular locus.*

Part VI

Proof audit and further orientation

Chapter 21

Proof audit: analytic and computer-assisted components

21.1 Why a proof audit is necessary

The source paper mixes conceptual geometry, exact symbolic calculations, local analytic theorems, and numerical dynamical computations. These have different verification burdens. A reader should know which conclusions follow from hand-checkable formulas and which rely on accompanying computer algebra.

This chapter classifies the main steps.

21.2 Classical external theorems

The following are invoked from established literature rather than reproved in full:

• Fejes Tóth's equality $\delta(K) = \delta_L(K)$ for centrally symmetric planar convex disks; • existence and regularity results of Reinhardt and Mahler; • Filippov's existence theorem for optimal controls; • the Pontryagin maximum principle; • the stable-manifold theorem for hyperbolic fixed points; • Weierstrass preparation; • Gronwall and Rademacher theorems.

The source states or summarizes the needed versions and cites references.

21.3 Exact symbolic derivations

The following parts reduce to finite algebraic or differential calculations and are, in principle, checkable by hand:

• the matrix representation $\sigma_j = g_{s*j}$;

• the star inequalities ρj(X) > 0; • the area formula area(K) = −3 R ⟨J, X⟩; 2 • the control matrix Zu and state equation X′ = [Zu, X]/ ⟨Zu, X⟩; • the Hamiltonian and state–costate equations; • the upper-half-plane parametrization and cost; • constant-control matrix exponentials; • classification of the full singular locus as the circle; • the local second variation excluding circular arcs; • the Fuller truncation and weighted-order estimates; • the triangular Fuller switching cubics and fixed-point equations.

A computer algebra system greatly shortens many of these calculations but is not conceptually essential.

21.4 Symbolic calculations delegated to Mathematica

Several exact assertions are reported as direct Mathematica calculations. Examples include:

• explicit costate formulas on constant-control segments; • switching-function inequalities for the smoothed (6k + 2)-gon family; • the exact Jacobian eigenvalues or numerical enclosures at fixed points; • asymptotic expansions matching Reinhardt and Fuller coordinates; • polynomial discriminants, resultants, and cell boundary equations.

These are finite identities or inequalities. A rigorous audit should inspect the exact arithmetic and simplification rules in the accompanying notebooks.

21.5 The global basin computation

The global basin theorem uses a finite family of set inclusions:

Fi(D) ⊂ [D∗.

The source visualizes and numerically checks images of analytic boundary faces. The logical deduction from the inclusion table is exact, but the inclusions themselves are the core computer-assisted input.

A modern certification could proceed as follows:

1. represent each cell by interval boxes or semialgebraic constraints; 2. isolate the relevant positive root of each switching cubic with interval Newton methods; 3. evaluate the constant-control flow using outward-rounded interval arithmetic; 4. subdivide faces adaptively until each image interval lies in the desired target half-spaces; 5. store a machine-checkable certificate of every subdivision and inequality.

This would convert the visual/boundary computations into a formally auditable interval proof.

21.6 The global unstable curve computation

The source tracks the one-dimensional unstable manifold from qout using 434 numerically generated data points. The dynamics are strongly contracting in the transverse directions, making the numerical continuation stable in practice. The curve is observed to hit the boundary of the star domain and not return to the exceptional divisor.

For a fully certified version one would:

• compute a validated local unstable-manifold enclosure from the parameterization method; • propagate it with interval enclosures under each Poincaré branch; • verify the switching order and star inequalities on each segment; • certify a transverse crossing of the star-domain boundary before any possible return.

Point requiring care

The source itself notes that the unstable-manifold calculation could be repeated with interval arithmetic but does not present such a certificate. Readers interested in formal proof status should distinguish this from the local analytic stable-manifold theorem, which is standard once the Jacobian is known.

21.7 Why the conceptual proof remains valuable

Even when some inclusions are computer-assisted, the proof is not a blind search. The analytic structure determines exactly what the computer must verify:

• the control geometry forces cubic switching functions; • time reversal creates an involution; • discriminants and resultants determine the finite partition; • the upper-triangular containment order provides a discrete Lyapunov function; • hyperbolicity identifies the only stable and unstable channels.

The computation verifies a finite set of geometric inequalities inside a proof architecture dictated by theory.

21.8 A reproducibility checklist

A complete independent audit should answer:

1. Which version of Mathematica and which precision settings were used? 2. Are all fixed-point and root-isolation computations exact or high precision? 3. Are plots merely illustrative, or do they encode containment assertions? 4. For each containment, what analytic inequalities define the source and target cells? 5. How are multiple or nearly multiple switching roots separated?

6. Is every discontinuity surface accounted for by a discriminant or resultant? 7. Can the unstable curve be enclosed rigorously until boundary crossing?

The source provides accompanying code, and these questions indicate how to read it critically.

21.9 The status of these lecture notes

These notes reproduce the mathematical definitions, reductions, theorem statements, and logical implications. They do not re-execute or formally certify the full Mathematica package. The embedded computational figures are explanatory reproductions from the user-provided source PDF.

Chapter summary

Most of the proof architecture is analytic and exact. Two decisive global dynamical inputs rely on extensive computation: the containment relations proving the Fuller global basin and the continuation of the exact Reinhardt unstable curve to the star-domain boundary. The source explains the algorithms and supplies code; a fully formal modern treatment would replace plot-based or high-precision checks with interval certificates.

Chapter 22

What remains open and how to continue studying

22.1 The exact endpoint reached by the source

The 2024 source proves Mahler's First conjecture: a global minimizer is a finite smoothed polygon. It does not prove the full Reinhardt conjecture, which asserts that the minimizer is specifically the smoothed octagon, unique up to affine transformation.

This distinction is easy to miss because the smoothed octagon is present throughout the theory. It is an explicit extremal, it has lower density than the circle, and it is locally isolated among nearby extremals. None of these facts alone proves that no distant periodic extremal has still smaller area.

The remaining global problem can be stated in the language developed in these notes:

Proof roadmap

Classify all periodic, finite-switching Pontryagin extremals satisfying the Reinhardt endpoint conditions, or prove an area inequality that places every such extremal above the smoothed octagon.

22.2 Why Mahler's First is already a major reduction

Before the theorem, a minimizer could in principle have had a boundary with continuously varying curvature, singular arcs, or infinitely many alternations accumulating at a point. After the theorem, only a finite combinatorial object remains:

• a cyclic word in the three vertex controls; • a positive duration attached to each letter; • matrix endpoint equations; • switching inequalities from the maximum principle; • the area computed by a finite sum of elementary integrals.

Thus an infinite-dimensional variational problem has been reduced to a countable union of finite-

dimensional problems.

This reduction does not make the full conjecture automatic. The number of switches is not bounded a priori by the theorem, and the endpoint equations are nonlinear. Still, it changes the nature of the problem decisively.

22.3 A finite-word formulation of the remaining problem

Choose the three vertices e_0, e_1, e_2 of the control triangle. A finite bang-bang trajectory is encoded by $W = ((i_1, t_1), \ldots, (i_N, t_N))$, $i_k \in \{0, 1, 2\}$, $t_k > 0$,

where adjacent indices are different. For a specified initial state X_0, each constant-control piece has an explicit matrix-exponential solution. Multiplying these pieces gives

$$g(T) = G_W(X_0; t_1, \ldots, t_N), \quad T = \sum_{k=1}^{N} t_k.$$

The endpoint equations require

$$g(T) = R, \quad X(T) = R^{-1} X_0 R,$$

and the costate must satisfy the corresponding transversality conditions. The maximum principle adds inequalities saying that the selected vertex really maximizes the Hamiltonian on every interval.

The full Reinhardt conjecture would follow from a statement of the following form.

Theorem

Theorem 22.1 (Desired finite-word theorem). Every periodic finite-switching extremal satisfying the Reinhardt conditions has area at least that of the smoothed octagon, with equality only for its affine orbit.

This is not proved in the source. It is a precise target made possible by Mahler's First.

22.4 Possible routes toward the full Reinhardt conjecture

22.4.1 A priori switch bounds

The strongest simplification would be a universal bound on the number of switches of a minimizing extremal. If one could show, for example, that a minimizer has at most four control pieces during one sixth of the boundary, only finitely many mode words would remain. The endpoint and switching equations could then be analyzed case by case.

What might force such a bound? Possible mechanisms include:

• a lower bound for the cost paid by each additional excursion in the hyperbolic state space; • monotonicity of a rotation or winding number; • a comparison principle that removes short alternating words without increasing area; • a second-order condition excluding high-frequency finite switching.

None of these is presently supplied by the source.

22.4.2 Symbolic dynamics plus coercivity

The control word defines a symbolic itinerary through the three vertex-flow families. One could seek a generating partition and express the endpoint problem through a return map. If long words necessarily move through regions of high cost, a coercive estimate could rule them out for a global minimizer.

The compactification of the star domain is useful here: all relevant trajectories lie in a fixed compact set. Compactness makes uniform estimates on switching times, hyperbolic displacement, and cost more plausible.

22.4.3 A calibration or sharp lower bound

Another route would avoid classifying extremals. One seeks a function or differential form giving a lower bound area(K) ≥area(Koct)

for every admissible trajectory, with equality exactly on the octagonal orbit. In optimal-control language this resembles constructing a global solution or subsolution of the Hamilton–Jacobi–Bellman equation. In geometric language it resembles a calibration.

The difficulty is that the control Hamiltonian is nonsmooth across switching walls, and the boundary condition is periodic modulo rotation. A successful calibration would have to encode both features.

22.4.4 Second variation and conjugate points

The paper proves that the circle fails a second-order necessary condition and that the smoothed octagon is isolated. A more systematic study of second variation along arbitrary bang-bang extremals might identify conjugate points or negative directions whenever the switching pattern is not octagonal.

For bang-bang systems, second variation is delicate because switching times vary. The correct quadratic form includes both continuous state variations and variations of the switching instants. Nevertheless, the finite-dimensional reduction makes this a concrete program.

22.4.5 Certified computation

The endpoint equations and switching inequalities are explicit. A computer-assisted global proof could combine:

1. interval arithmetic for constant-control flows; 2. branch-and-bound over initial states and switching times; 3. symbolic pruning of impossible control words; 4. validated lower bounds for the cost on every surviving box.

This would be substantially larger than the computations in the source, but the theorem proved there supplies the essential finiteness structure.

22.5 Improving the proof of Mahler's First

Even without solving Reinhardt's full conjecture, several aspects of the existing proof invite refinement.

22.5.1 Formal certification of the basin theorem

The global basin argument uses finitely many analytic cells and image-containment checks. Replacing numerical plots by interval certificates would make the computational core independently auditable. The natural output is not a picture but a certificate containing:

• a subdivision of every source face; • an interval enclosure for the relevant switching root; • an interval enclosure for its image; • a verified list of target inequalities.

A small checker could then validate the entire containment table.

22.5.2 Validated unstable-manifold continuation

The no-return result is supported by high-resolution numerical continuation of a one-dimensional unstable curve. A rigorous replacement would use the parameterization method near the hyperbolic fixed point and interval integration thereafter. The main events to certify are the switching sequence, preservation of the star inequalities, and transverse crossing of the star-domain boundary.

22.5.3 A conceptual global Lyapunov function

The finite cell order acts as a discrete Lyapunov function. Finding a single analytic function that decreases under the Fuller–Poincare map would greatly shorten the proof and explain the global basin more transparently. The cell decomposition suggests what level sets of such a function should look like.

22.5.4 Removing auxiliary circular control

The circular-control problem is a powerful model because rotational symmetry yields angular momentum. Logically, however, the final theorem concerns triangular control. A direct triangular derivation of the relevant normal form and global structure might shorten the route, while the circular model would remain valuable intuition.

22.6 Connections worth learning next

The proof sits at the intersection of several subjects. A productive order of further study is:

1. convex geometry and geometry of numbers: support functions, polar bodies, critical lattices;

2. elementary Lie groups: matrix groups, homogeneous spaces, adjoint orbits; 3. optimal control: Pontryagin's principle, bang-bang control, singular arcs; 4. hyperbolic geometry: the action of SL2(R) on the upper half-plane; 5. Hamiltonian dynamics: symplectic forms, Poisson brackets, momentum maps; 6. local dynamical systems: hyperbolic fixed points and stable manifolds; 7. singular perturbation and blowup: weighted projective compactification; 8. validated numerics: interval Newton methods and rigorous ODE integration.

The appendices give a first pass through the background needed for these topics.

22.7 A guided exercise program

The following exercises reproduce the main mechanisms in increasing order of difficulty.

Exercise

Exercise 22.2. Show directly that a centrally symmetric hexagon with edge midpoints s0, . . . , s5 satisfies sj+3 = −sj and s0 + s2 + s4 = 0.

Exercise

Exercise 22.3. For X = ac −ab , verify the formulas for ρ0, ρ1, ρ2 and prove

---

det X = ρ0ρ1 + ρ1ρ2 + ρ2ρ0.

Exercise

Exercise 22.4. Starting from g′ = gX and σj = gs∗j, derive the Green-theorem area formula

---

Z area(K) = −3 tr(JX) dt. 2

Exercise

Exercise 22.5. For a fixed control matrix Z, prove that ⟨X, Z⟩is constant along

[Z, X] X′ = ⟨Z, X⟩.

Then solve the equation by conjugation.

Exercise

Exercise 22.6. Write out the triangular Fuller solution for constant u and verify that every switching function is a real cubic polynomial in time.

Exercise

Exercise 22.7. Check that the weighted dilation

(z1, z2, z3) 7→(rz1, r2z2, r3z3)

preserves the triangular Fuller equations after the time change t 7→rt.

Exercise

Exercise 22.8 (Research-level). Construct an interval-arithmetic certificate for one cell inclusion in the global basin table. State precisely which polynomial root and which target inequalities must be enclosed.

22.8 A compact reading plan for the source

A first pass through the original paper need not be linear. The following sequence is efficient:

1. read the introduction and book summary to understand the theorem hierarchy; 2. study the critical-hexagon and multicurve construction; 3. read the statement of the Reinhardt control problem and the area formula; 4. learn the maximum-principle vocabulary from the source's Chapter 6; 5. read the bang-bang and singular-locus conclusions; 6. jump to the overview at the beginning of Part V; 7. study the triangular Fuller map, blowup, basin theorem, and cluster theorem; 8. return to computational details only after the logical skeleton is clear.

Chapter summary

The 2024 theorem removes the infinite-dimensional pathology: every global minimizer is a finite smoothed polygon. The open task is to determine which finite periodic extremal minimizes area. The most direct targets are a switch bound, a sharp calibration, a global second-variation argument, or a certified exhaustive analysis of finite mode words.

Part VII

Background appendices

Appendix A

Linear algebra, matrix groups, and Lie theory

This appendix develops the matrix language used in the proof. No abstract Lie theory is required beyond differentiating matrix products and exponentials.

A.1 Two-dimensional determinants

For column vectors v = (v1, v2)T and w = (w1, w2)T , write

det(v, w) = v1w2 −v2w1.

Geometrically, det(v, w) is the signed area of the parallelogram spanned by v and w. It is positive when the ordered pair (v, w) is counterclockwise.

Let 0 −1! J = . 1 0

Then Jv is v rotated counterclockwise by π/2, and

det(v, w) = −vT Jw.

Indeed, 0 −1 ! w1 ! − v1 v2 = v1w2 −v2w1. 1 0 w2

If A is any 2 × 2 matrix, then

det(Av, Aw) = det(A) det(v, w).

This identity is the reason determinant-one matrices preserve oriented area.

A.2 The group SL2(R)

Definition

Definition A.1. The special linear group is

$$SL_2(\mathbb{R}) = \{g \in M_2(\mathbb{R}) : \det g = 1\}.$$

It is a group under matrix multiplication.

A general element has the form

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad ad - bc = 1,$$

and $g^{-1} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

The equation $ad-bc = 1$ cuts out a three-dimensional surface in $\mathbb{R}^4$, so $SL_2(\mathbb{R})$ is a three-dimensional smooth manifold.

A.2.1 Tangent vectors at the identity

Let $g(t)$ be a differentiable curve in $SL_2(\mathbb{R})$ with $g(0) = I$. Jacobi's determinant formula gives

$$0 = \frac{d}{dt}\Big|_{t=0} \det g(t) = \operatorname{tr}(g'(0)).$$

Thus a tangent vector at the identity must have trace zero.

Definition

Definition A.2. The Lie algebra of $SL_2(\mathbb{R})$ is

$$\mathfrak{sl}_2(\mathbb{R}) = \left\{ X = \begin{pmatrix} a & b \\ c & -a \end{pmatrix} : a, b, c \in \mathbb{R} \right\}.$$

Its bracket is the commutator $[X, Y] = XY - YX$.

The word "Lie algebra" here simply means that $\mathfrak{sl}_2(\mathbb{R})$ is a vector space closed under commutators. The commutator records the first noncommutative correction in products of small matrix motions.

A.3 The matrix exponential

For any square matrix X, define

$$\exp(X) = \sum_{n=0}^{\infty} \frac{X^n}{n!}.$$

The series converges absolutely. It satisfies

$$\frac{d}{dt} \exp(tX) = X \exp(tX) = \exp(tX)X.$$

If X and Y commute, then $\exp(X + Y) = \exp(X)\exp(Y)$. Without commutativity this formula generally fails. For $X \in \mathfrak{sl}_2(\mathbb{R})$, the Cayley–Hamilton theorem gives an especially simple formula. Since $\operatorname{tr} X = 0$, the characteristic polynomial is $\lambda^2 + \det X$.

Therefore $X2 = -\det(X)I$.

Write $d = \det X$.

• If $d > 0$, then √ √ sin(d t) $\exp(tX) = \cos(d\ t)I + \sqrt{}\ X.$ d • If $d = 0$, then $X2 = 0$ and $\exp(tX) = I + tX$. √ • If $d < 0$, put $\lambda = -d$. Then

sinh(λt) $\exp(tX) = \cosh(\lambda t)I + X.$ λ

These formulas make every constant-control segment in the Reinhardt system elementary.

Example

Example A.3. Since $J2 = -I$,

---

cos t −sin t ! exp(tJ) = . sin t cos t

Thus J is the infinitesimal generator of ordinary rotations.

A.4 Left-trivialized velocity

Let $g(t) \in SL2(R)$. Define $X(t) = g(t)-1g'(t)$. Then $X(t) \in sl2(R)$, and the same equation can be written $g'(t) = g(t)X(t)$.

This is called the left-trivialized velocity because the tangent vector $g'(t)$ is transported back to the identity by left multiplication with $g(t)-1$.

If v is a fixed vector and $\sigma(t) = g(t)v$, then

$\sigma'(t) = g'(t)v = g(t)X(t)v$.

Thus one matrix $X(t)$ describes the simultaneous velocities of all vectors moved by $g(t)$.

If g is reparameterized by $t = t(s)$, then

dt $eX(s) = X(t(s))$ ds. Consequently $\det eX = (dt/ds)2 \det X$. When $\det X > 0$, one may choose the parameter so that $\det X \equiv 1$.

A.5 Adjoint action and Lax equations

For $g \in SL2(R)$ and $X \in sl2(R)$, define $Adg\ X = gXg-1$.

This is conjugation. It preserves trace, determinant, eigenvalues, and commutators:

$Adg[X, Y] = [Adg\ X, Adg\ Y]$.

Differentiating conjugation gives the basic Lax identity. Let P(t) and X(t) be matrix curves. If X(t) = h(t)X0h(t)−1, h′(t) = P(t)h(t),

then X′ = [P, X]. Conversely, a solution of X′ = [P, X] evolves by conjugation. Therefore its characteristic polynomial is constant. Proposition A.4 (Isospectrality). If X′ = [P, X], then tr(Xk) is constant for every positive integer k. In particular, for 2 × 2 trace-zero matrices, det X is constant.

Proof. Proof. By cyclicity of trace, □

d tr(Xk) = k tr(Xk−1[P, X]) = k tr(Xk−1PX −XkP) = 0. dt

The Reinhardt state equation is a control-dependent Lax equation: Zu X′ = ⟨Zu, X⟩, X .

A.6 The trace pairing

On sl2(R), define ⟨X, Y ⟩= tr(XY).

This is a symmetric bilinear form. It is nondegenerate but not positive definite. For a b ! X = , c −a

one has ⟨X, X⟩= 2(a2 + bc) = −2 det X.

Thus matrices with positive determinant have negative squared length in this indefinite geometry.

The pairing is invariant under conjugation: ⟨Adg X, Adg Y ⟩= tr(gXY g−1) = tr(XY).

It also satisfies ⟨X, [Y, Z]⟩= ⟨[X, Y], Z⟩.

This follows from expanding and cyclically permuting traces.

A.6.1 Useful 2 × 2 identities

For X, Y ∈sl2(R), XY + Y X = ⟨X, Y ⟩I. To verify it, write out the entries or polarize X2 = −det(X)I. Several consequences are used repeatedly: [X, Y]2 = ⟨X, Y ⟩2 −⟨X, X⟩⟨Y, Y ⟩ I, ⟨[X, Y], [X, Y]⟩= 2 ⟨X, Y ⟩2 −⟨X, X⟩⟨Y, Y ⟩ , [X, [X, Y]] = 2 ⟨X, X⟩Y −2 ⟨X, Y ⟩X. These are the indefinite analogues of vector triple-product identities in R3.

A.7 A concrete basis for sl2(R)

It is useful to use 1 0 ! 0 1 ! 0 0 ! H = , E = , F = . 0 −1 0 0 1 0

Then [H, E] = 2E, [H, F] = −2F, [E, F] = H. Every X ∈sl2(R) is aH + bE + cF.

Another basis adapted to geometry is

0 −1! 0 1 ! 1 0 ! J = , K = , H = . 1 0 1 0 0 −1

The trace pairing has signs

$\langle J, J\rangle = -2$, $\langle K, K\rangle = 2$, $\langle H, H\rangle = 2$,

and the basis vectors are pairwise orthogonal. Hence sl2(R) is a copy of three-dimensional Minkowski space with signature (−, +, +).

A.8 Adjoint orbits as hyperboloids

Because det J = 1, the adjoint orbit

n o OJ = gJg−1 : g ∈SL2(R)

lies in the quadratic surface det X = 1. Since $\langle X, X\rangle = -2 \det X$, this is one sheet of a two-sheeted hyperboloid in the Minkowski coordinates above. The sign condition $\langle J, X\rangle < 0$ chooses the sheet containing J. The stabilizer of J is the rotation group SO2(R). Therefore OJ ∼= SL2(R)/SO2(R).

The same quotient is the hyperbolic plane. This is the structural reason hyperbolic geometry appears in the control problem.

A.9 The upper-half-plane parametrization

For $z = x + iy$ with $y > 0$, define

x/y −(x2 + y2)/y ! Φ(z) = . 1/y −x/y

A direct calculation gives

+ y2 + 1 tr Φ(z) = 0, det Φ(z) = 1, ⟨J, Φ(z)⟩= −x2 < 0. y

Thus Φ(z) ∈OJ. The group SL2(R) acts on the upper half-plane by Mobius transformations:

a b ! az + b · z = c d cz + d.

The map Φ is equivariant: $\Phi(g \cdot z) = g\Phi(z)g^{-1}$.

One can verify this directly, though the quotient description makes it conceptually inevitable.

A.10 Differentiating determinants: Jacobi's formula

Theorem

Theorem A.5 (Jacobi). For a differentiable invertible matrix curve A(t),

d det A(t) = det A(t) tr A(t)−1A′(t) . dt

For a 2 × 2 matrix, the formula can be checked from coordinates. It implies:

• if g(t) ∈SL2(R), then g−1g′ has trace zero; • if X(t) has constant determinant, then tr(X−1X′) = 0; • the normalization det X = 1 is compatible with the state equation.

A.11 The Cayley transform and su(1, 1)

The source sometimes changes from real matrices to complex matrices because rotations become multiplication by a phase. Set 1 1 −i! C = √ . 2 1 i

Conjugation X 7→CXC−1 identifies sl2(R) with

(it z !) su(1, 1) = : t ∈R, z ∈C . ¯z −it

For a b ! X = , c −a

the transformed coordinates can be written

it z ! b + c b −c CXC−1 = , z = + ia, t = . ¯z −it 2 2

The determinant condition becomes

t2 −|z|2 = det X.

For determinant 1, this is the upper sheet of a hyperboloid when t > 0. These are the hyperboloid coordinates used near the singular locus.

A.12 Matrix product rules used in the proof

The following elementary formulas are worth memorizing:

(AB)′ = A′B + AB′, (A−1)′ = −A−1A′A−1, (ABA−1)′ = A′BA−1 + AB′A−1 −ABA−1A′A−1, d ⟨A, B⟩= A′, B + A, B′ . dt

If A′ = AP, the conjugation formula simplifies to

(ABA−1)′ = A(B′ + [P, B])A−1.

A.13 Exercises

Exercise

Exercise A.6. Prove $XY + Y X = \mathrm{tr}(XY)I$ for all $X, Y \in sl2(R)$ by direct multiplication.

Exercise

Exercise A.7. Show that $X' = [P, X]$ implies $\det X$ is constant without using the general tracepower argument.

Exercise

Exercise A.8. Verify explicitly that $\Phi(i) = J$ and that $\Phi(g \cdot i) = gJg{-}1$.

Exercise

Exercise A.9. Let Z be constant and suppose $X' = [Z, X]/ \langle Z, X\rangle$. Prove that $\langle Z, X\rangle$is constant and obtain $Z\ X(t) = \exp(tP)X(0) \exp(-tP)$, $P = \langle Z, X(0)\rangle$.

Chapter summary

All Lie-theoretic operations in the proof reduce to explicit 2×2 matrix calculus. The essential structures are determinant-one matrices, trace-zero velocities, conjugation, the invariant trace

pairing, and the identification of the determinant-one adjoint orbit with the hyperbolic plane.

Appendix B

Ordinary differential equations and analysis background

This appendix records the analytic facts used in the lecture notes. The goal is not to replace a full course in analysis or differential equations, but to isolate exactly what the proof needs.

B.1 Ordinary differential equations as vector fields

An ordinary differential equation on an open set $U \subset \mathbb{R}^n$ has the form $x'(t) = f(t, x(t))$, $x(t_0) = x_0$.

A solution is a differentiable curve whose tangent vector equals the prescribed vector field. In an autonomous equation $x' = f(x)$, the right-hand side does not depend explicitly on time.

The matrix equations in the Reinhardt problem are ordinary differential equations on finitedimensional vector spaces. For example, after writing all matrix entries as coordinates, the equation

$[Zu, X]\, X' = \langle Zu, X\rangle$ is an ODE in three real variables as long as the denominator does not vanish.

B.2 Existence and uniqueness

Theorem

Theorem B.1 (Picard–Lindelof). Suppose $f(t, x)$ is continuous in (t, x) and locally Lipschitz in x. Then for every initial condition (t_0, x_0) there is a unique local solution of $x' = f(t, x)$, $x(t_0) = x_0$.

The solution depends continuously on the initial data.

A function is locally Lipschitz in x if, on every compact box, there is a constant L such that

$||f(t, x) - f(t, y)|| \leq L\ ||x - y||$.

Every continuously differentiable function is locally Lipschitz. Rational vector fields are locally Lipschitz wherever their denominators stay away from zero.

B.2.1 Why uniqueness matters

Uniqueness permits us to speak of a flow $\varphi t(x0)$: the point reached at time t from initial state x0. It also prevents two distinct trajectories from crossing at an ordinary point. This principle is repeatedly used in the basin and stable-manifold arguments.

The projected Fuller vector field on a quotient space is sometimes not Lipschitz along glued boundaries. In that case the projected equation alone may fail uniqueness, and one must refer to the original smooth system before quotienting.

B.3 Caratheodory solutions and measurable controls

An optimal control u(t) need only be measurable. Then the vector field f(x, u(t)) may be discontinuous in time. Classical differentiability at every time is too restrictive.

A Caratheodory solution is an absolutely continuous curve satisfying

Z t x(t) = x(t0) + f(x(s), u(s)) ds. t0 Its derivative exists almost everywhere and equals the vector field almost everywhere. Under standard boundedness and Lipschitz assumptions, existence and uniqueness still hold.

A piecewise constant bang-bang control gives a familiar special case: solve the smooth ODE on each interval and join the endpoints continuously.

B.4 Absolute continuity

A function x : [a, b] →Rn is absolutely continuous if for every ε > 0 there is δ > 0 such that for any disjoint intervals (ak, bk) with total length less than δ, X ||x(bk) −x(ak)||< ε. k

Equivalently, there is an integrable function v such that Z t x(t) = x(a) + v(s) ds. a Then x′ = v almost everywhere. This is the correct regularity class for controlled trajectories.

B.5 Gronwall's inequality

Theorem

Theorem B.2 (Integral Gronwall inequality). Let $h : [0, T] \to [0, \infty)$ be continuous and suppose

$$h(t) \le A + B\int_0^t h(s)\,ds$$ for constants $A, B \ge 0$. Then $h(t) \le Ae^{Bt}$. More generally, if $A = A(t)$ is nondecreasing, then $h(t) \le A(t)e^{Bt}$.

Proof. Proof. Set $$H(t) = A + B\int_0^t h(s)\,ds.$$ Then $h \le H$ and $H' \le BH$. Hence $(e^{-Bt}H(t))' \le 0$, so $H(t) \le Ae^{Bt}$. □

Gronwall is used for three distinct purposes:

• bounding a state trajectory on a finite time interval; • proving continuous dependence on data; • controlling error terms in asymptotic expansions near the singular locus.

Example

Example B.3 (Matrix growth). If $g' = gX$ and $||X(t)|| \le M$, then

$$||g(t)|| \le ||g(0)|| + M\int_0^t ||g(s)||\,ds,$$ so $$||g(t)|| \le ||g(0)||e^{Mt}.$$

This is one ingredient in compactifying the attainable set.

B.6 Lipschitz functions and Rademacher's theorem

Theorem

Theorem B.4 (Rademacher). Every Lipschitz function $f : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable almost everywhere.

The source proves that the tangent fields of the boundary multicurve are Lipschitz. Rademacher then guarantees the existence of second derivatives almost everywhere, which is enough to define curvature constraints and controls almost everywhere.

B.7 Curvature for a nonsingular plane curve

Let $\gamma(t) = (x(t), y(t))$ be a twice differentiable curve with $\gamma'(t) \neq 0$. Its signed curvature is

$$\kappa(t) = \frac{\det(\gamma'(t), \gamma''(t))}{||\gamma'(t)||^3}.$$

Under an orientation-preserving reparameterization, the sign of curvature is unchanged. A regular simple closed curve oriented counterclockwise bounds a convex region precisely when its signed curvature is nonnegative, interpreted weakly when the curve is only $C^{1,1}$.

In the paper, only the numerator $\det(\gamma', \gamma'')$

is needed because the denominator is positive.

B.8 Landau notation

For functions near $t = 0$, the statement

$f(t) = O(t^k)$

means that there are constants $C, \delta > 0$ such that

$$|f(t)| \le C\,|t|^k \quad (0 < |t| < \delta).$$

Similarly, $f(t) = g(t) + O(t^k)$

means $f - g = O(t^k)$.

The usual algebra rules apply:

$$O(t^a) + O(t^b) = O(t^{\min(a,b)}), \quad O(t^a)O(t^b) = O(t^{a+b}), \quad \int_0^t O(s^a)\,ds = O(t^{a+1}) \ (a > -1).$$

If $h(t) = 1 + O(t)$, then $1/h(t) = 1 + O(t)$. These elementary rules produce the weighted orders

$$w = O(t),\ b = O(t^2),\ c = O(t^3)$$

near the singular locus.

B.9 Taylor's theorem

If f is $k + 1$ times continuously differentiable near 0, then

$$f(t) = \sum_{j=0}^{k} \frac{f^{(j)}(0)}{j!} t^j + O(t^{k+1}).$$

For vector-valued functions the formula holds componentwise. Taylor expansion is used to:

• determine the multiplicity of switching-function zeros; • derive the Fuller truncation; • identify the leading weighted part of the exact Poincare map; • decide how trajectories cross glued boundaries.

B.10 Implicit functions

Theorem

Theorem B.5 (Implicit function theorem). Let $F : R^{n+m} \to R^m$ be continuously differentiable. Suppose $F(x_0, y_0) = 0$

and the $m \times m$ matrix $D_yF(x_0, y_0)$ is invertible. Then near (x_0, y_0) the equation $F(x, y) = 0$ has a unique solution $y = \varphi(x)$,

where φ is continuously differentiable.

The most elementary application in this proof is a transverse switching time. If

$$\chi(t_0, q_0) = 0, \quad \frac{\partial\chi}{\partial t}(t_0, q_0) \neq 0,$$

then for q near q_0 there is a unique nearby switching time $t = t(q)$ depending smoothly on q.

When the switching zero has higher multiplicity, the ordinary implicit function theorem fails; Weierstrass preparation is then used instead.

B.11 Compactness and subsequences

The Bolzano–Weierstrass theorem says every bounded sequence in R^n has a convergent subsequence. In the blown-up system, the exceptional divisor is compact. Therefore any sequence of switching points whose radial coordinate tends to zero has a cluster point on the divisor. This simple compactness observation begins the cluster-point theorem.

A continuous function on a compact set attains its maximum and minimum. Compactification of the star domain therefore permits uniform bounds for denominators, velocities, switching derivatives, and costs.

B.12 Existence of optimal controls: the role of Filippov

The full proof uses a standard result of optimal control.

Filippov existence theorem, simplified form

Consider $x' = f(x, u)$ with u in a compact set U. Suppose the state is confined to a compact set, f is continuous, the velocity set $f(x, U)$ is convex for every x, and the running cost is continuous. Then under appropriate endpoint conditions an optimal measurable control exists.

The Reinhardt problem satisfies these hypotheses after compactification. The triangular control set is compact; the velocity set in upper-half-plane coordinates is a triangle and hence convex; and the relevant state region is compact.

The source also has an independent geometric existence theorem due to Reinhardt. Filippov's theorem ensures that the reformulated control problem is analytically well posed.

B.13 Linearization of a flow

Let $x' = f(x)$ and let $x(t; x0)$ be its solution. A small perturbation $x0 + \varepsilon v$ has first variation

$$\partial x(t; x0) \quad \eta(t) = v. \quad \partial x0$$

Differentiating the ODE with respect to initial data gives the variational equation

$$\eta' = Df(x(t))\eta, \ \eta(0) = v.$$

The matrix solution $\Psi(t)$ with $\Psi(0) = I$ is the fundamental matrix. It is the derivative of the flow with respect to initial conditions.

For a Poincare map, the derivative has an extra correction because the return time varies with initial data. If the return section is $h(x) = 0$ and the return time is $T(x)$, then

$P(x) = \varphi T(x)(x)$.

Differentiating $h(P(x)) = 0$ determines $DT(x)$ and hence $DP(x)$. This is the standard route to the Jacobians used at the fixed points.

B.14 Continuous and discrete time

A flow φt describes continuous motion. A Poincare map records only successive intersections with a section: $qn+1 = P(qn)$.

The map discards time between intersections but retains the recurrence structure. In a chattering problem, switching times are the natural section because the control law changes there.

A fixed point $P(q) = q$ need not mean the continuous trajectory is literally stationary. It can represent a self-similar trajectory: after one switching, the state returns to the same shape after scaling and symmetry reduction.

B.15 Exercises

Exercise

Exercise B.6. Use Gronwall's inequality to prove uniqueness for $x' = f(t, x)$ when f is Lipschitz in x.

Exercise

Exercise B.7. Suppose f(t) = O(t2) and g′(t) = f(t) + O(g(t)) with g(0) = 0. Prove g(t) = O(t3) by Gronwall.

Exercise

Exercise B.8. Let χ(t, q) be smooth with χ(0, q0) = 0 and ∂tχ(0, q0) ≠ 0. Derive the formula

q0) Dt(q0) = −Dqχ(0, . ∂tχ(0, q0)

Exercise

Exercise B.9. Show that a continuous trajectory on a compact state space has at least one cluster point as $t \to \infty$.

Chapter summary

The analytic core uses standard finite-dimensional ODE theory: measurable controls give absolutely continuous trajectories; Lipschitz regularity gives uniqueness; Gronwall controls errors; Taylor and implicit-function arguments control switching; compactness yields cluster points. The genuinely specialized tools, stable manifolds and Weierstrass preparation, are summarized in later appendices.

Appendix C

Hamiltonian mechanics and optimal control

This appendix explains the vocabulary of cotangent bundles, Hamiltonians, costates, and the Pontryagin maximum principle. The exposition begins in Euclidean space and then indicates what changes on a matrix group.

C.1 Tangent and cotangent vectors

For an open set $M \subset \mathbb{R}^n$, a tangent vector at $q \in M$ is an ordinary vector $v \in \mathbb{R}^n$. A cotangent vector is a linear functional on tangent vectors. In coordinates it is represented by a row vector or by a one-form $p = p_1\, dq_1 + \cdots + p_n\, dq_n$.

The pairing is $\langle p, v\rangle_* = \sum_{j=1}^{n} p_j v_j$. The collection of all pairs (q, p) is the cotangent bundle T^*M. A smooth manifold looks locally like an open set in $\mathbb{R}^n$, so tangent and cotangent vectors are defined by changing coordinates consistently. The cotangent bundle always has twice the dimension of M.

C.2 The canonical symplectic form

On $T^*\mathbb{R}^n$ with coordinates $(q_1, \ldots, q_n, p_1, \ldots, p_n)$, define the canonical one-form

$$\theta = \sum_{j=1}^{n} p_j\, dq_j$$

and symplectic form $\omega = -d\theta = \sum_{j=1}^{n} dq_j \wedge dp_j$.

A two-form is symplectic if it is closed and nondegenerate. Nondegeneracy means that for every differential dH there is a unique vector field H satisfying $\omega(H, \cdot) = dH$. This H is the Hamiltonian vector field of H.

In coordinates the equation becomes Hamilton's equations: $q'_j = \frac{\partial H}{\partial p_j}$, $p'_j = -\frac{\partial H}{\partial q_j}$. Example C.1 (Ordinary mechanics). For $H(q, p) = \frac{\|p\|^2}{2m} + V(q)$, Hamilton's equations give $q' = \frac{p}{m}$, $p' = -\nabla V(q)$, which combine to Newton's equation $mq'' = -\nabla V(q)$.

C.3 Poisson brackets

For smooth functions F, G on phase space, define

n ∂F ∂G ∂G ! {F, G} = X −∂F . ∂qj ∂pj ∂pj ∂qj j=1

Along the Hamiltonian flow of H, d dtF(q(t), p(t)) = {F, H}. Therefore F is conserved if {F, H} = 0. The bracket is bilinear, antisymmetric, satisfies a product rule, and obeys the Jacobi identity.

The Fuller system in the source uses a nonstandard but explicit symplectic form, and therefore a nonstandard Poisson bracket. The principle is unchanged.

C.4 From a control problem to a Hamiltonian

Consider q′ = f(q, u), u(t) ∈U,

with running cost Z tf L(q(t), u(t)) dt. 0 Introduce a costate p(t) ∈T q(t)M∗ and a constant multiplier λ0 ≤0. Define the control-dependent Hamiltonian H(q, p, u) = ⟨p, f(q, u)⟩∗+ λ0L(q, u). The first term rewards motion aligned with the costate, while the second incorporates the objective.

The sign convention varies across textbooks. The source uses maximization with λ0 ≤0. In the normal case one rescales to λ0 = −1.

C.5 The Pontryagin maximum principle

Pontryagin maximum principle, free terminal time

Let (q∗, u∗) be an optimal trajectory for a sufficiently regular finite-dimensional control problem. Then there exist an absolutely continuous costate p(t) and a constant λ0 ≤0, not both zero, such that almost everywhere:

∂H q′ = = f(q, u∗), ∂p p′ = −∂H , ∂q H(q, p, u∗) = max H(q, p, u). u∈U

For an autonomous problem with free terminal time, the maximized Hamiltonian is constant and equals zero along the extremal. Endpoint constraints produce transversality conditions on p.

A trajectory satisfying these conditions is a Pontryagin extremal. The principle gives necessary, not sufficient, conditions for optimality.

C.5.1 Normal and abnormal extremals

If λ0 < 0, rescale all multipliers and set λ0 = −1. This is a normal extremal. If λ0 = 0, the cost disappears from the Hamiltonian; the extremal is abnormal. Abnormality can reflect genuine geometry or an artifact of how constraints were encoded.

C.5.2 Why the Hamiltonian is maximized

A short needle variation changes the control from $u*$to v on a tiny interval. The first-order change of endpoint and cost is measured by the Hamiltonian difference

$H(q, p, v) - H(q, p, u*)$.

Optimality forces this difference to be nonpositive for every v, hence maximization by $u*$.

C.6 Bang-bang controls

Suppose U is a compact convex polytope and H depends affinely on u. A nonconstant affine function on a polytope reaches its maximum on a face, usually at a vertex. Thus optimal controls often take extreme values.

In the Reinhardt system the control-dependent term is a ratio of affine functions,

$-\langle \Lambda R, Zu\rangle \langle X, Zu\rangle$,

but along every segment of the control triangle it is monotone. Hence the maximizing set is still a face. On the open regions where a unique vertex wins, the control is bang-bang.

A switching function compares two candidate vertices. If controls i and j give Hamiltonians Hi, Hj, define $\chi ij = Hi - Hj$.

The wall $\chi ij = 0$ is where a switch may occur. A transverse zero gives an isolated switch. An infinite sequence of switches in finite time is chattering.

C.7 Singular controls

A singular interval is one on which the maximum condition fails to determine a unique control. One differentiates switching functions until the control appears, then solves the resulting equations. This procedure may identify a special interior control.

For the Reinhardt problem, full indeterminacy forces

3 $\Lambda R = 0, X = J, \Lambda 1 = 2\lambda 0 J.$

The corresponding control is the center of the triangle,

$u = (1/3, 1/3, 1/3)$,

and the boundary curve is circular. Thus the singular locus has a concrete geometric meaning.

C.8 Transversality

Suppose the final state is constrained to lie on a submanifold $N \subset M$. A permissible endpoint variation v lies in $T_{q(t_f)}N$. The boundary term in the first variation is $\langle p(t_f), v\rangle_*$. Optimality requires $\langle p(t_f), v\rangle_* = 0$ for all $v \in T_{q(t_f)}N$.

Thus the costate is normal to the endpoint manifold.

In the Reinhardt problem, the endpoint is periodic modulo the rotation R. Consequently the state and costate satisfy conjugacy relations such as

$$X(t_f) = R^{-1}X(0)R, \quad \Lambda_R(t_f) = R^{-1}\Lambda_R(0)R.$$

These conditions allow the trajectory to be extended periodically after applying the discrete symmetry.

C.9 Left-invariant systems on a matrix group

For $g \in SL_2(\mathbb{R})$, write the state equation

$g' = gX.$

The vector field is left-invariant because replacing g by hg changes the velocity to hgX. A cotangent vector at g can be transported to the identity and represented by a matrix $\Lambda_1 \in \mathfrak{sl}_2(\mathbb{R})$ using the trace pairing.

The Hamiltonian contribution from the group equation is then

$\langle \Lambda_1, X\rangle.$

Hamilton's equation for the left-trivialized costate becomes

$$\Lambda_1' = [\Lambda_1, X].$$

Its solution is $\Lambda_1(t) = g(t)^{-1}\Lambda_1(0)g(t)$,

so $\det \Lambda_1$ is conserved.

C.10 The Reinhardt Hamiltonian

The state is (g, X) and the reduced costates are (Λ_1, Λ_R). With the trace pairing, the Hamiltonian is $H(\Lambda_1, \Lambda_R, X; u) = \langle \Lambda_1 - \frac{3}{2}\lambda_0 J, X\rangle - \frac{\langle \Lambda_R, Z_u\rangle}{\langle X, Z_u\rangle}$. Set $P = \frac{Z_u}{\langle X, Z_u\rangle}$. The state–costate equations are

$$g' = gX, \quad X' = [P, X], \quad \Lambda_1' = [\Lambda_1, X], \quad \Lambda_R' = [P, \Lambda_R] - \langle \Lambda_R, P\rangle[P, X] + [-\Lambda_1 + \tfrac{3}{2}\lambda_0 J, X].$$

Every term is a 2×2 commutator or trace pairing. The system looks formidable, but it has three simplifying features:

• X remains on a two-dimensional adjoint orbit; • ΛR is constrained by ⟨ΛR, X⟩= 0; • the control triangle has only three vertices and three walls.

C.11 Noether's principle and angular momentum

In mechanics, a continuous symmetry of the Hamiltonian produces a conserved quantity. This is Noether's theorem. A version remains valid for optimal control.

For circular control, the control set is invariant under all rotations. Simultaneous rotation of X, Λ1, ΛR, and the control leaves the Hamiltonian unchanged. The corresponding momentum is

A = ⟨J, Λ1 + ΛR⟩.

It is constant along circular-control extremals.

The triangular control set has only discrete rotational symmetry, so this continuous conservation law is lost. The circular problem is therefore a symmetry-enhanced model used to discover the Fuller normal form.

C.12 Second variation

The maximum principle detects only first-order obstructions. A Pontryagin extremal can fail to minimize because of a negative second variation.

For a one-parameter family qs with q0 = q, write s2 J (qs) = J (q0) + s δJ + δ2J + o(s2). 2 At an extremal, δJ = 0. If one can choose a variation with δ2J < 0, then q is not a local minimizer.

The source applies this to the circular singular arc. A compactly supported matrix perturbation has second-order area term proportional to

Z − (ψ′1ψ2 −ψ1ψ′2) dt.

Choosing the signs of ψ′1, ψ′2 appropriately makes this negative while preserving positive curvature for sufficiently small perturbations. Thus the circle is a Pontryagin extremal but not minimizing.

C.13 Hamilton–Jacobi viewpoint

For intuition, define the value function V (q) to be the least cost from q to the target. Formally, V satisfies a Hamilton–Jacobi–Bellman equation

$$0 = \min_{u \in U} L(q, u) + dV_q(f(q, u)) .$$

The costate along an optimal trajectory is related to dV . A globally smooth value function rarely exists in switching problems, but this viewpoint explains why a sharp global subsolution could prove the full Reinhardt conjecture without classifying every extremal.

C.14 Exercises

Exercise

Exercise C.2. For $H(q, p) = pf(q) - L(q)$ in one dimension, write Hamilton's equations and verify directly that H is constant when it has no explicit time dependence.

Exercise

Exercise C.3. Let $U = [-1, 1]$ and $H = pu$. Determine the maximizing control. What happens when $p = 0$?

Exercise

Exercise C.4. Suppose $H_i - H_j$ has a simple zero at t_0. Explain why the maximizing control switches only once near t_0. Exercise C.5. Verify that $\Lambda_1(t) = g(t)^{-1}\Lambda_1(0)g(t)$ solves $\Lambda_1' = [\Lambda_1, X]$ when $g' = gX$.

Chapter summary

Pontryagin's principle lifts the state trajectory to a Hamiltonian trajectory with costates. The control maximizes a pointwise Hamiltonian, producing vertex controls, switching walls, and singular loci. In the Reinhardt problem, left invariance turns the abstract equations into explicit commutator identities on 2×2 matrices.

Appendix D

Dynamical systems, Poincaré maps, and blowups

This appendix supplies the local dynamical-systems and algebraic-analytic tools used in the final part of the proof.

D.1 Fixed points of a map

Let $F : U \subset R^n \to R^n$ be continuously differentiable. A fixed point is q with $F(q) = q$. The linear approximation is $F(q + h) = q + DF(q)h + o(||h||)$.

The eigenvalues of DF(q) determine the first-order behavior of iterates near q.

Definition

Definition D.1. A fixed point is hyperbolic if no eigenvalue of DF(q) has absolute value 1.

Directions with $|\lambda| < 1$ contract under forward iteration; directions with $|\lambda| > 1$ expand. Hyperbolicity is robust: sufficiently small C1 perturbations retain a nearby hyperbolic fixed point with the same stable and unstable dimensions.

D.2 Stable and unstable sets

For a fixed point q, define

$$W^s(q) = \{x : F^n(x) \to q \text{ as } n \to +\infty\}, \quad W^u(q) = x : F^{-n}(x) \to q \text{ as } n \to +\infty,$$

where the inverse iterates are defined. These sets need not be smooth globally, but near a hyperbolic fixed point they are manifolds.

Stable-manifold theorem for a diffeomorphism

Let F be a Cr local diffeomorphism, r ≥1, with a hyperbolic fixed point q. Then there are local Cr manifolds W loc(q)s and W loc(q)u tangent at q to the stable and unstable eigenspaces

of DF(q). Points on the local stable manifold converge to q under forward iteration; points on the local unstable manifold converge under backward iteration.

In the blown-up Reinhardt map, each relevant fixed point has one radial eigenvalue and several angular eigenvalues. The dimensions of its stable and unstable manifolds follow from the absolute values of those eigenvalues.

D.3 Poincare first-return maps

Let x′ = f(x) be a smooth flow and let Σ be a hypersurface transverse to the flow. For x ∈Σ, let T(x) > 0 be the first time the trajectory returns to Σ. The Poincare map is

P(x) = φT(x)(x).

It converts a continuous-time problem to a discrete one.

In a bang-bang system, switching walls are natural sections. The first recurrence map starts at a switching point, follows the constant-control flow until the next admissible switch, applies symmetry normalization, and records the new switching state.

The map is generally only piecewise analytic because different switching roots or control branches can be selected in different regions. The discontinuity surfaces are determined by root collisions and simultaneous wall events.

D.4 Self-similarity and fixed points after quotienting

Suppose a system admits scaling

(z1, z2, z3, t) 7→(rz1, r2z2, r3z3, rt).

A trajectory can reproduce itself after one switching up to this scaling and a discrete rotation. Once these symmetries are quotiented out, the trajectory becomes a fixed point of the recurrence map.

Thus the inward and outward Fuller spirals are not fixed in the original phase space; they are fixed shapes under renormalization.

D.5 Basins of attraction

The basin of a stable fixed point q is

B(q) = {x : F n(x) →q} .

Local stability only proves that a neighborhood of q lies in the basin. A global basin theorem must control every other initial condition.

The source constructs a finite partition into cells D1, . . . , Dm and proves image relations of the form

F(Di) ⊂ [Dj. j≤i

If the partial order is strict away from a small neighborhood of q, then the cell index is a discrete Lyapunov function: trajectories can move only downward and eventually enter the stable region.

D.5.1 Lyapunov functions

For a map, a Lyapunov function is a scalar V satisfying

V (F(x)) < V (x)

away from the target. A finite cell order is a coarse Lyapunov function. Instead of a real number, it assigns a rank in a finite partially ordered set.

D.6 Time-reversing involutions

An involution τ satisfies τ 2 = I. It reverses time for a map F if

τFτ = F −1.

Then a stable fixed point is carried to an unstable fixed point, and

τ(W s(q)) = W u(τq).

The triangular Fuller system has such an involution from complex conjugation and sign changes. This symmetry reduces many arguments to one time direction.

D.7 Ordinary blowup of the origin

The origin is a singular point when the dynamics depends on direction but not smoothly on magnitude. Blowup separates directions. In R2, write (x, y) = r(cos θ, sin θ), r ≥0.

For r > 0 this is ordinary polar coordinates. Instead of collapsing all points with r = 0 to the origin, blowup retains the circle of directions θ ∈S1. That circle is the exceptional divisor.

A vector field singular at the origin may become regular after multiplication by a suitable power of r and extension to r = 0. The induced vector field on the exceptional divisor describes the leading directional dynamics.

D.8 Weighted blowup

The Fuller asymptotics have different orders:

z1 = O(t), z2 = O(t2), z3 = O(t3).

Ordinary polar coordinates ignore this structure. A weighted blowup uses z1 = rez1, z2 = r2ez2, z3 = r3ez3.

The variable r measures distance with weights (1, 2, 3); the tilded variables record weighted direction. The exceptional divisor is r = 0 modulo the residual scaling relation.

The triangular Fuller equations are homogeneous for these weights. Therefore their recurrence map is exactly the map induced on the exceptional divisor. The exact Reinhardt equations equal the Fuller leading term plus higher weighted-order corrections.

D.9 Blowup charts

A projective blowup is covered by coordinate charts. In a simple weighted chart one chooses a nonzero component, for example z1, and sets

z2 z3 r = |z1| , ez2 = r2 , ez3 = r3 ,

with a phase normalization for z1/r. Other charts cover directions where z1 = 0 but another weighted component is nonzero.

The source chooses coordinates adapted to switching walls and symmetries, reducing the divisor to a three-dimensional section. The exact coordinate formulas are specialized, but the underlying idea is standard weighted projective geometry.

D.10 Analytic extension across the divisor

Suppose the exact recurrence map in blown-up coordinates has the form

R(r, q) = r a(r, q), F(q) + rG(r, q) .

If a, G extend analytically to r = 0, then

R(0, q) = (0, F(q)).

Thus the exceptional divisor is invariant and carries the Fuller map F. Fixed points of F become fixed points of the exact map on the divisor.

The difficult part is that the return time is defined as a root of a switching equation. Near a multiple root, ordinary implicit-function theory does not apply. Weierstrass preparation supplies a polynomial factor whose roots capture the return times.

D.11 Weierstrass preparation

Weierstrass preparation theorem, one distinguished variable

Let $f(t, x)$ be real-analytic near $(0, 0)$, where $t \in \mathbb{R}$ and $x \in \mathbb{R}^n$. Suppose

$$f(0, 0) = \partial_t f(0, 0) = \cdots = \partial_t^{m-1} f(0, 0) = 0, \quad \partial_t^m f(0, 0) \neq 0.$$

Then near $(0, 0)$,

$$f(t, x) = U(t, x)\, \left(t^m + a_{m-1}(x)t^{m-1} + \cdots + a_0(x)\right),$$

where U is analytic and nonzero, and the coefficients a_j are analytic with $a_j(0) = 0$.

The nonvanishing factor U has no zeros, so the zeros of f are exactly the roots of the monic degree-m polynomial. Root selection can then be analyzed algebraically.

For the Fuller switching problem, the switching functions are already cubic. For the exact Reinhardt map, preparation shows that the relevant switching equation is an analytic perturbation of such a cubic.

D.12 Discriminants

For a polynomial $p(t)$, the discriminant vanishes exactly when p has a repeated complex root. For

$$p(t) = at^2 + bt + c,$$

the discriminant is $b^2 - 4ac$. For a cubic

$$p(t) = at^3 + bt^2 + ct + d,$$

the discriminant is $\Delta = b^2c^2 - 4ac^3 - 4b^3d - 27a^2d^2 + 18abcd$.

The sign of the cubic discriminant determines whether there are three distinct real roots or one real root and a complex-conjugate pair.

Discriminant surfaces partition parameter space into regions with a fixed root pattern. This is one source of the cells used for the Fuller recurrence map.

D.13 Resultants

The resultant $\mathrm{Res}_t(p, q)$ is a polynomial in the coefficients of p and q that vanishes exactly when p and q share a complex root. It can be computed as the determinant of the Sylvester matrix.

In a switching problem, two candidate switching functions vanish simultaneously precisely on a resultant surface. Such surfaces mark changes in which wall is reached first. Together, discriminants and resultants provide an algebraic skeleton for the piecewise analytic partition.

D.14 Cluster points and invariant manifolds

Suppose qn is an orbit of the exact recurrence map with radial coordinate rn > 0 and rn →0. Compactness of the divisor gives a cluster point q∗. The leading map on the divisor constrains q∗.

A typical argument has three stages:

1. prove every divisor cluster point lies in a compact invariant set of the Fuller map;
2. use the global basin theorem to force the cluster point to the inward fixed point;
3. use local hyperbolicity and analytic extension to show the entire exact orbit lies on the stable manifold of that fixed point.

The final step is stronger than mere convergence: it identifies a unique low-dimensional channel through which chattering can occur.

D.15 No-return arguments

A local unstable manifold says how trajectories leave a fixed point. To rule out periodicity one needs global information. The source continues the unstable curve from the outward fixed point and shows it reaches the boundary of the admissible star domain without returning to the divisor.

Time reversal then implies that a trajectory entering through the inward stable curve cannot have emerged from the divisor at an earlier time. This one-sided history contradicts the two-sided recurrence required by a periodic minimizer.

D.16 Validated numerics for maps

A numerical picture is not by itself a proof of a set inclusion. Rigorous computation replaces points by sets and ordinary arithmetic by outward-rounded interval arithmetic.

For a map defined by a switching root, a validated step has the following structure:

1. enclose the initial cell by interval boxes; 2. isolate the desired positive root using interval Newton or Krawczyk methods; 3. evaluate the flow and normalization on the root interval; 4. prove every resulting interval satisfies the target cell inequalities; 5. subdivide if an interval is too wide.

The finite containment graph can then be checked by a small independent verifier.

D.17 Exercises

Exercise

Exercise D.2. For $F(x, y) = (x/2, 2y)$, find the stable and unstable manifolds of the origin.

Exercise

Exercise D.3. Show that the map $F(r, \theta) = (r/2, \theta + \alpha)$ has the entire circle $r = 0$ as an invariant exceptional divisor, even though the unblown origin is a single point.

Exercise

Exercise D.4. Compute the discriminant of t3 + pt + q and describe the regions with one or three real roots.

Exercise

Exercise D.5. Let $F(r, q) = (ra(q), G(q) + rH(r, q))$. Show that $r = 0$ is invariant and that a fixed point of G gives a fixed point of F on the divisor.

Chapter summary

The last part of the proof replaces a singular finite-time limit by regular dynamics on a weighted exceptional divisor. The triangular Fuller system is the induced leading map. Hyperbolic fixed points provide stable and unstable channels; a global basin theorem selects the only possible channel; and a no-return computation makes that channel incompatible with periodicity.

Appendix E

Notation, theorem index, and formula sheet

This appendix is designed for repeated consultation. The notation follows the source as closely as possible, with occasional simplification explained in the main text.

E.1 Geometric objects

Symbol or term Meaning

K A centrally symmetric convex disk in R2, centered at the origin. Kmin A global minimizer of greatest packing density, equivalently area after normalization. Kccs The class of centrally symmetric convex disks. Kbal The class of balanced disks admitting the six-curve structure used in the control reduction.

δ(K) Greatest packing density of K. For centrally symmetric planar disks, this equals greatest lattice-packing density. L A lattice in R2. det(L) Covolume of the lattice: area of a fundamental parallelogram. HK A least-area centrally symmetric hexagon circumscribing K; a critical hexagon. hK The centrally symmetric hexagon formed by joining six critical boundary points. sj Midpoints of the six edges of a critical hexagon, indexed modulo 6. s∗j The normalized reference multipoint, usually the sixth roots of unity. Tj The equilateral triangular sector containing the jth boundary arc after normalization. σj(t) The six synchronized boundary curves; all have the same image after periodic extension.

Symbol or term Meaning

multicurve The indexed family (σ0, . . . , σ5) satisfying the multipoint relations at every time. star condition The tangent σ′j lies in the open cone prescribed by neighboring critical points.

smoothed polygon A finite cyclic concatenation of straight segments and tangent hyperbolic arcs of Reinhardt–Mahler type.

E.2 Matrix and Lie-theoretic notation

Symbol Meaning

SL2(R) Real 2 × 2 matrices of determinant 1. sl2(R) Real 2 × 2 matrices of trace 0. J 01 −10 , the infinitesimal rotation. R exp(πJ/3), rotation by π/3. g(t) Matrix curve carrying the reference multipoint to the current one: σj(t) = g(t)s∗j. X(t) Left-trivialized velocity X = g−1g′, normalized by det X = 1. [X, Y] Matrix commutator XY −Y X. ⟨X, Y ⟩ Trace pairing tr(XY). Adg X Adjoint action gXg−1. OJ Adjoint orbit of J; the determinant-one sheet containing J. Φ(z) Parametrization of OJ by the upper half-plane. ρ0, ρ1, ρ2 Linear functions of X encoding the star inequalities and edge directions. Zu Control matrix obtained affinely from u = (u0, u1, u2). P Normalized control matrix P = Zu/ ⟨Zu, X⟩.

E.3 Control and Hamiltonian notation

Symbol Meaning

u = (u0, u1, u2) Normalized curvature control. UT Triangular control set ui ≥0, u0 + u1 + u2 = 1. e0, e1, e2 Vertices of UT ; bang-bang control values. UC, UI, Ur Circular control sets used in the symmetry-enhanced model.

Symbol Meaning

κj State-dependent signed curvature numerator for the even-indexed boundary curves. uj κj/(κ0 + κ1 + κ2). Λ1 Costate paired with the group variable g. Λ2 Original costate paired with X. ΛR Reduced costate [Λ2, X], constrained by ⟨ΛR, X⟩= 0. λ0 Pontryagin cost multiplier; λ0 = −1 in the normal convention. H Control-dependent Hamiltonian. H+ Maximized Hamiltonian maxu∈U H. χij Switching function comparing control vertices i and j. wall A codimension-one set on which two control vertices tie. singular locus The set X = J, ΛR = 0, Λ1 = 32λ0J; geometrically the circle. edge-extremal A trajectory that is also extremal for every restricted edge-control subproblem.

chattering Infinitely many switches accumulating in finite time.

E.4 Hyperbolic coordinates

For z = x + iy with y > 0, x/y −(x2 + y2)/y ! Φ(z) = . 1/y −x/y

The upper-half-plane star domain is

1 1 h∗= x + iy : −1√ < x < √ x2 + y2 > . 3 3, 3

The area integrand becomes −3 ⟨J, X⟩= 3 x2 + y2 + 1 . 2 2 y The center z = i corresponds to X = J and hence to the circular singular state.

E.5 Core equations

E.5.1 Multipoint relations

For j modulo 6, sj+3 = −sj, sj + sj+2 + sj+4 = 0, √ 3 det(sj, sj+1) = . 2

E.5.2 Star coordinates

For a b ! X = , c −a √ c + 3a ρ0(X) = √ , 3 √ c − 3a ρ1(X) = √ , 3 + c ρ2(X) = −3b√ . 2 3 The star condition is ρj(X) > 0 for j = 0, 1, 2, and

det X = ρ0ρ1 + ρ1ρ2 + ρ2ρ0.

E.5.3 Control matrix

For u0 + u1 + u2 = 1, √ (u1 −u2)/ 3 (u0 −2u1 −2u2)/3 ! Zu = √ . u0 (u2 −u1)/ 3

It satisfies uj = det(s∗2j, Zus∗2j).

E.5.4 State and cost

g′ = gX, [Zu, X] X′ = ⟨Zu, X⟩, Z tf area(K) = −3 ⟨J, X⟩dt. 2 0 Endpoint conditions: g(0) = I, g(tf) = R, X(tf) = R−1X(0)R.

E.5.5 Hamiltonian system

With P = Zu/ ⟨Zu, X⟩, Zu⟩ H = Λ1 −3 X −⟨ΛR, 2λ0J, ⟨X, Zu⟩, and g′ = gX, X′ = [P, X], Λ′1 = [Λ1, X], Λ′R = [P, ΛR] −⟨ΛR, P⟩[P, X] + [−Λ1 + 2λ0J,3 X].

E.6 Fuller notation

Symbol Meaning

w, b, c Complex hyperboloid coordinates centered at the singular locus. z1, z2, z3 Rescaled leading variables of weighted degrees 1, 2, 3. ζ Primitive cube root exp(2πi/3). VT Triangular control values {1, ζ, ζ2}. HF Fuller Hamiltonian. AF Fuller angular-momentum analogue. E Exceptional divisor of the weighted blowup. PF Fuller Poincare first-recurrence map on the divisor. PR Exact Reinhardt Poincare map after analytic extension. qin Fixed point representing the inward self-similar chattering trajectory. qout Fixed point representing the outward self-similar trajectory. W s, W u Stable and unstable manifolds of a fixed point.

The triangular Fuller system is

z′3 = z2, z′2 = z1, z′1 = −iu, u ∈{1, ζ, ζ2}.

For constant u and initial values z0j ,

z1(t) = z01 −itu, z2(t) = z02 + z01t −iut2 , 2 z01t2 z3(t) = z03 + z02t + −iut3 . 2 6

Therefore every wall comparison is a real cubic in t.

E.7 The main theorem chain

Result Role in the proof

Fejes Toth equality Replaces arbitrary packings by lattice packings for centrally symmetric planar disks. Critical-hexagon theorem Expresses density as area(K)/ area(HK) and supplies six synchronized boundary points.

Balanced-multicurve Converts the minimizer into six regular curves satisfying reduction algebraic and convexity constraints.

Result Role in the proof

Matrix representation Writes all six curves as σj = gs∗j with g ∈SL2(R). State-control theorem Encodes curvature by u ∈UT and obtains X′ = [Zu, X]/ ⟨Zu, X⟩. Area formula Converts the geometric objective to an integral cost.

Compactification Confines every possible minimizer to a compact subset of the star domain.

Pontryagin maximum Produces costates, switching walls, and necessary principle Hamiltonian dynamics.

Finite switching away from Shows every edge-extremal avoiding the singular locus is a singularity finite smoothed polygon.

Singular-locus classification Identifies full singular arcs with circular arcs.

Second variation Excludes a circular singular interval from a global minimizer.

Fuller truncation Gives the universal leading chattering system near the singular locus.

Weighted blowup Replaces the singular point by the exceptional divisor carrying triangular Fuller dynamics.

Fixed-point theorem Finds the inward and outward self-similar chattering states.

Global basin theorem Forces all nonexceptional divisor trajectories toward the outward fixed point.

Analytic extension Transfers the fixed points and local invariant manifolds to the exact Reinhardt map. Cluster-point theorem Forces every exact finite-time chattering orbit onto W s(qin). No-return theorem Shows the time-reversed unstable curve cannot close into a periodic orbit.

Mahler's First A periodic global minimizer cannot chatter; hence it is finite bang-bang and a smoothed polygon.

E.8 What is analytic and what is computer-assisted

Category Examples

Classical theorem Fejes Toth, Filippov, Pontryagin, stable-manifold theorem, Weierstrass preparation.

Direct exact calculation Matrix reduction, area formula, state and costate equations, singular-locus classification, Fuller truncation.

Symbolic algebra Switching inequalities, fixed-point polynomials, Jacobians, discriminants, resultants, asymptotic coordinate changes.

Global computer-assisted Fuller cell-image containments and numerical continuation of the step exact unstable curve.

E.9 Source-to-notes reading crosswalk

The notes reorganize rather than duplicate the source. A rough crosswalk is:

Lecture-note material Main source location

Packing geometry and critical Source Chapters 1–2. hexagons

Matrix and control reduction Source Chapters 3–5.

Maximum principle Source Chapter 6.

Bang-bang polygons and Source Chapters 7–8. singularity

Circular model and Fuller Source Chapters 9–12. system

Triangular Fuller map and Source Chapters 13–15. blowup

Global basin and final proof Source Chapters 16 and the concluding part of Part V.

Background structures Source Appendices A–B, supplemented here at a more elementary level.

E.10 A one-page proof mnemonic

Remember the sequence

> packing →critical hexagon →six curves →SL2(R) →curvature control →PMP →bang-bang except at the circle
>
> →chattering model →weighted blowup →global basin
>
> →unique stable channel →nonperiodic →finite smoothed polygon.

Chapter summary

The proof uses many coordinate systems, but each has one job. Planar geometry produces the critical hexagon; SL2(R) synchronizes six curves; the upper half-plane exposes the compact star domain; costates expose switching; complex Fuller coordinates expose chattering; the weighted blowup exposes its limiting directions. The final conclusion is coordinate-independent: the minimizer has only finitely many straight and hyperbolic pieces.

Bibliography and primary source

The principal source for every theorem specific to the Reinhardt problem in these notes is the first item below. The remaining references are selected background works cited by that source or useful for the prerequisites developed here.

Bibliography

[1] Thomas Hales and Koundinya Vajjha, Packings of Smoothed Polygons, arXiv:2405.04331v1, 7 May 2024.

[2] Karl Reinhardt, “Uber die dichteste gitterformige Lagerung kongruenter Bereiche in der Ebene und eine besondere Art konvexer Kurven,” Abhandlungen aus dem Mathematischen Seminar der Universitat Hamburg 10 (1934), 216–230.

[3] Kurt Mahler, “On the Area and the Densest Packing of Convex Domains,” Proceedings of the Koninklijke Nederlandse Akademie van Wetenschappen 50 (1947), 108 ff.

[4] Kurt Mahler, “On the Minimum Determinant and the Circumscribed Hexagons of a Convex Domain,” Proceedings of the Koninklijke Nederlandse Akademie van Wetenschappen 50 (1947), 692–703.

[5] Walter Ledermann and Kurt Mahler, “On Lattice Points in a Convex Decagon,” Acta Mathematica 81 (1949), 319–351.

[6] Laszlo Fejes Toth, “Some Packing and Covering Theorems,” Acta Scientiarum Mathematicarum (Szeged) 12A (1950), 62–67.

[7] Laszlo Fejes Toth, Gabor Fejes Toth, and Wlodzimierz Kuperberg, Lagerungen: Arrangements in the Plane, on the Sphere, and in Space, Springer, 2023.

[8] Peter M. Gruber, Convex and Discrete Geometry, Springer, 2007.

[9] Janos Pach and Pankaj K. Agarwal, Combinatorial Geometry, Wiley, 2011.

[10] Thomas C. Hales, “On the Reinhardt Conjecture,” Vietnam Journal of Mathematics 39 (2011), 287–307.

[11] Koundinya Vajjha, On the Reinhardt Conjecture and Formal Foundations of Optimal Control, Ph.D. thesis, University of Pittsburgh, 2022; arXiv:2208.04443.

[12] Velimir Jurdjevic, Geometric Control Theory, Cambridge University Press, 1996.

[13] Velimir Jurdjevic, Optimal Control and Geometry: Integrable Systems, Cambridge University Press, 2016.

[14] Andrei A. Agrachev and Yuri L. Sachkov, Control Theory from the Geometric Viewpoint, Springer, 2004.

Bibliography Bibliography

[15] Francis Clarke, Functional Analysis, Calculus of Variations and Optimal Control, Springer, 2013.

[16] Anthony T. Fuller, "Study of an Optimum Non-linear Control System," International Journal of Electronics 15 (1963), 63–71.

[17] Mikhail Zelikin and Vladimir F. Borisov, Theory of Chattering Control: With Applications to Astronautics, Robotics, Economics, and Engineering, Springer, 2012.

[18] Larisa Manita and Mariya Ronzhina, "Optimal Spiral-like Solutions near a Singular Extremal in a Two-input Control Problem," Discrete and Continuous Dynamical Systems B 27 (2022), 3325–3353.

[19] Ralph Abraham and Jerrold E. Marsden, Foundations of Mechanics, AMS Chelsea, 2008.

[20] Jerrold E. Marsden and Tudor S. Ratiu, Introduction to Mechanics and Symmetry, Springer, 2013.

[21] Hector J. Sussmann, "Symmetries and Integrals of Motion in Optimal Control," Banach Center Publications 32 (1996), 379–393.

[22] Michael C. Irwin, Smooth Dynamical Systems, World Scientific, 2001.

[23] Herbert Amann, Ordinary Differential Equations: An Introduction to Nonlinear Analysis, De Gruyter, 2011.

[24] Askold M. Perelomov, Integrable Systems of Classical Mechanics and Lie Algebras, Birkhauser, 1990.

Relation to the source paper

The citation data above follow the bibliography of the user-provided source where possible. These lecture notes are explanatory and should not be cited as the authority for the theorem; cite Hales–Vajjha and the original historical sources.